

Desarrollo de aplicaciones para la visualización de datos

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Agenda

1. Introducción asignatura
2. Introducción a los datos y su analítica
3. Arquitecturas de procesamiento de datos
4. Datos y modelos matemáticos
5. Introducción al aprendizaje automático
6. Introducción al aprendizaje profundo
7. Casos de uso en la realidad
8. Productos basados en datos

Programa de la asignatura

1. **Introducción a python**
 - a. **Python básico**
 - b. **Pandas**
 - c. **Modelos de machine learning**

2. **Desarrollo de aplicaciones para visualización de datos**
 - a. **Diseño de data stories**
 - b. **Gráficos con Mathplotlib, Seaborn**
 - c. **Web app con Dash**

3. **Despliegue de aplicaciones**
 - a. **APIs, CI, aws**

<https://github.com/dmartincc/davd-26-27>

Evaluación asignatura

Propuesta de idea (10%)

Prueba intertrimestral (10%)

Desarrollo aplicación de datos y presentación (50%)

Examen final (30%)

Planificación semanal

Fecha	Clase
8 de septiembre 2025	Presentación
15 de septiembre 2025	Estructuras básicas + Funciones y clases + Ejercicios
22 de septiembre 2025	Lectura de datos (Pandas) + Ejercicios
29 de septiembre 2025	Modelos de machine learning + Ejercicios
6 de octubre 2025	Visualizaciones + Ejercicios / Entrega propuesta
13 de octubre 2025	Ejercicios de repaso
20 de octubre 2025	Prueba intertrimestral
27 de octubre 2025	Visualizaciones interactivas + Trabajo app
3 de noviembre 2025	Dash + Trabajo app
10 de noviembre 2025	Dash Producción + API + Trabajo app
17 de noviembre 2025	Trabajo app
24 de noviembre 2025	Repaso para examen final + Trabajo app
1 de diciembre 2025	Presentación app (12)
8 de diciembre 2025	Presentación app (12) + Entrega final app
TBD	Exámen final
* Sujeto a cambios	

Lunes de 15:00 a 16:50

Propuesta, entrar a las 15:15 hasta 16:50

85% Asistencia obligatoria

Motivación profesional de la asignatura

Motivación. Hay datos de sobra; lo que falta es convertir un problema de negocio en algo que se pueda ver, explorar y compartir (no un Excel ni un gráfico muerto).

Perfil egresado. Alguien que prototipa y despliega cuadros de mando / apps de datos con criterio de producto: qué mostrar, a quién y cómo dejarlo usable en una URL.

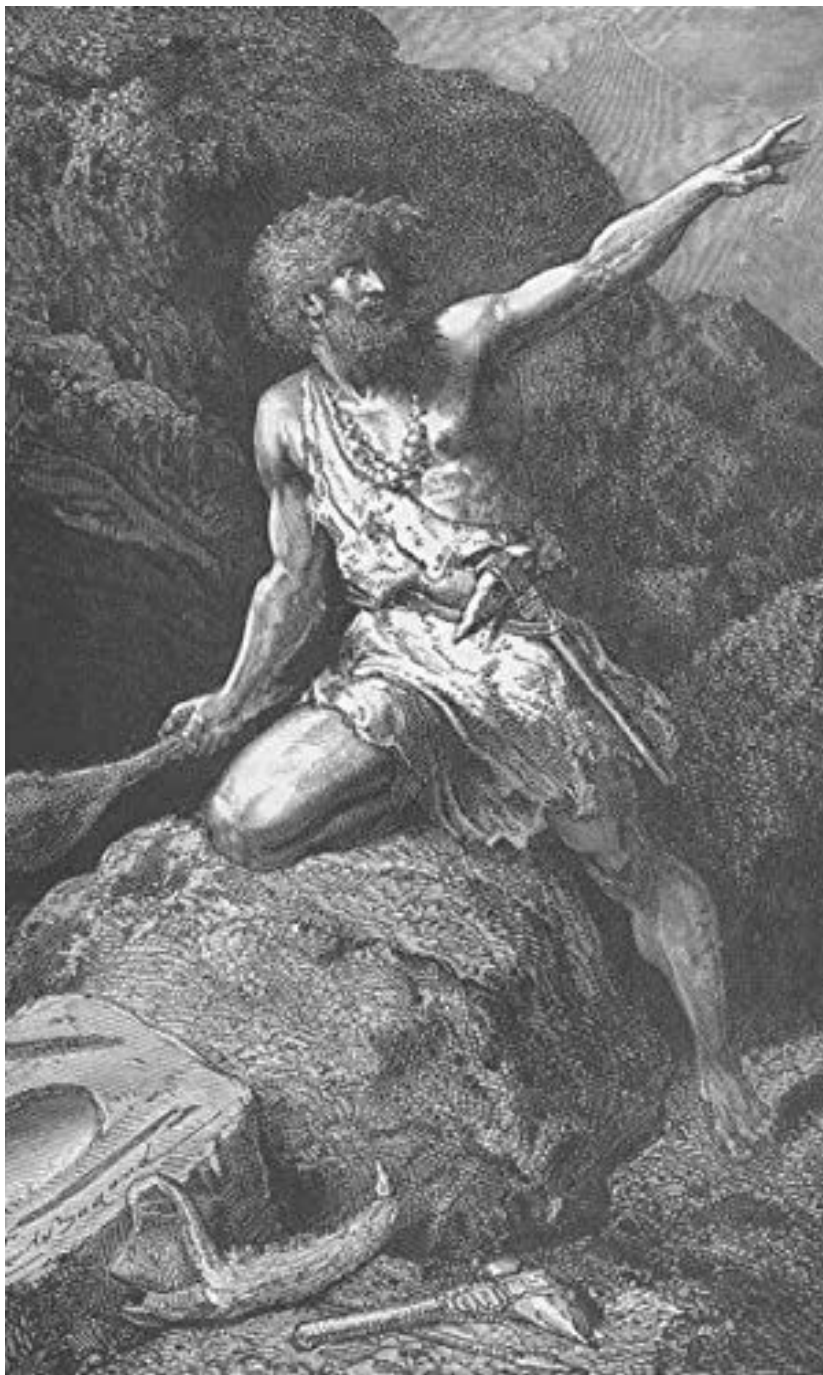
Data Scientist, Product Managers, AI Engineers, Business analysts, ...

Casos de uso. Panel de ventas o churn; KPIs operativos; indicadores públicos/ESG; explorador interno de un dataset o de un modelo.

Sobre David Martín-Corral

- ▶ Cofundador de **D3LF**, High Frequency Trading Fund
- ▶ Cofundador de **Zensei**, **Politibot** y **Souly**
- ▶ He trabajado en **IBM**, **CaixaBank**, **Bankinter**, **BBVA**, **Smartvel** y **Sanitas** en las primeras plataformas de Big Data de España y como data scientist
- ▶ **Ingeniero Eléctrico** por la UCLM
- ▶ **Ingeniero Industrial** por la UPM
- ▶ **Máster en Investigación Operativa** por la Universidad de Strathclyde (UK)
- ▶ **Doctor en Ingeniería Matemática** por la UC3M
- ▶ Imparte o ha impartido clases en **Instituto de Empresa**, **Universidad Complutense de Madrid**, **The Valley**, **ThePowerMBA**, **Universidad Carlos III de Madrid** y **Universidad de Alcalá**. David ha publicado en revistas científicas de alto nivel como **Nature** y **PNAS** y ha sido investigador invitado en el laboratorio de dinámica humana del **MIT Media Lab** (USA).

Introducción a los datos y su analítica



Situación de los datos y social

Datos

Conexión social



Neolítico

No existían

Muy baja



Principios s. XX

Escasos

Media alta

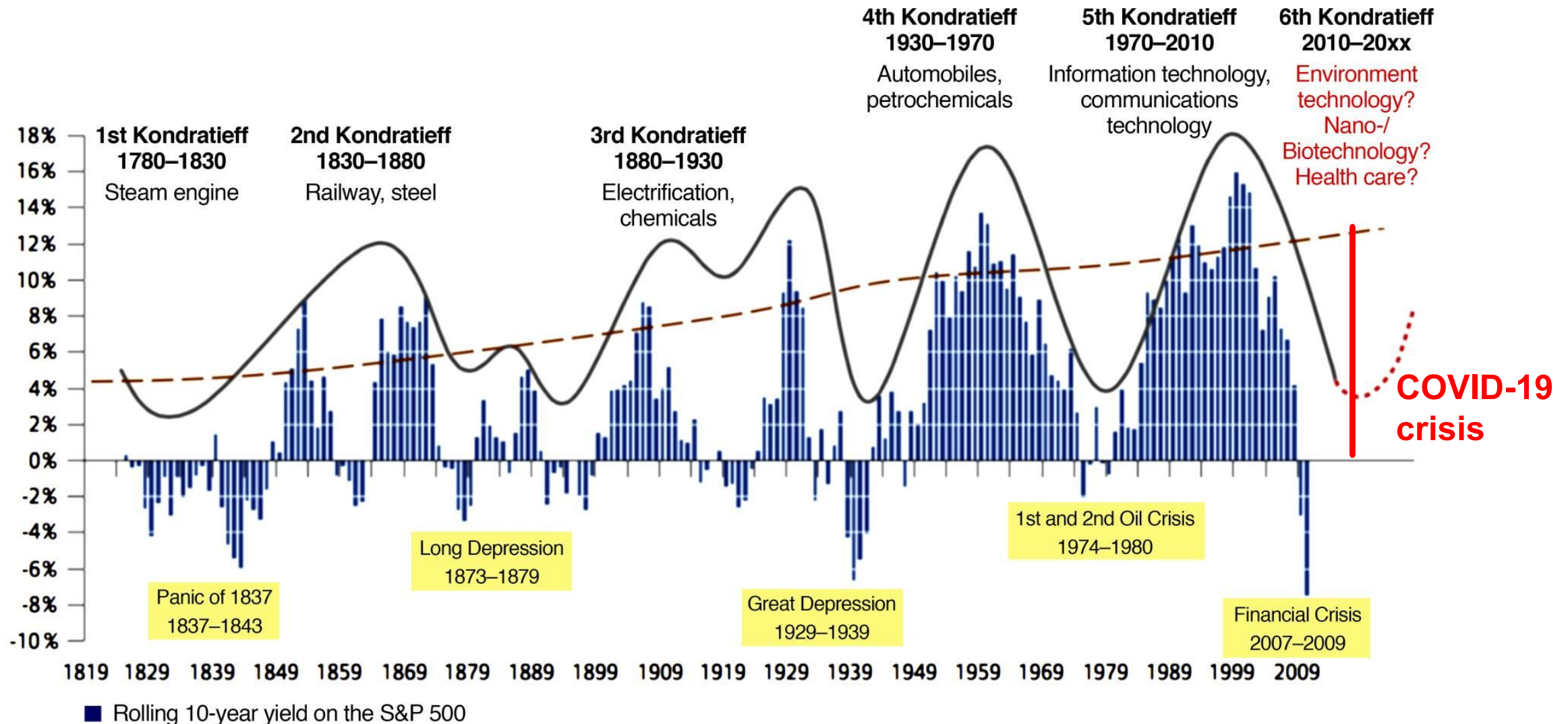


Principios s. XXI

Exponencial

Hiperconectados

Estamos en el inicio de la sexta Kondratieff (ola de innovación), donde existen oportunidades de grandes retornos en salud y clima. Pero tenemos que cambiar.



Fuente: [The ownership paradox: nurturing continuity and change for the future ASCI. 2019. Marr](#)

¿De donde podemos extraer los datos?

► Fuentes de datos propias

Bases de datos

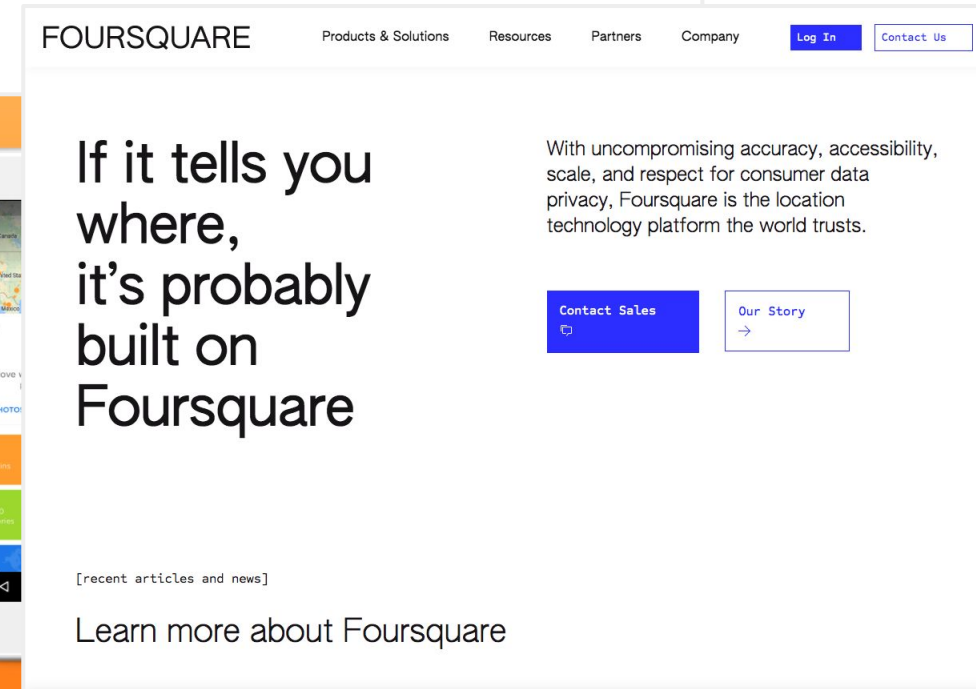
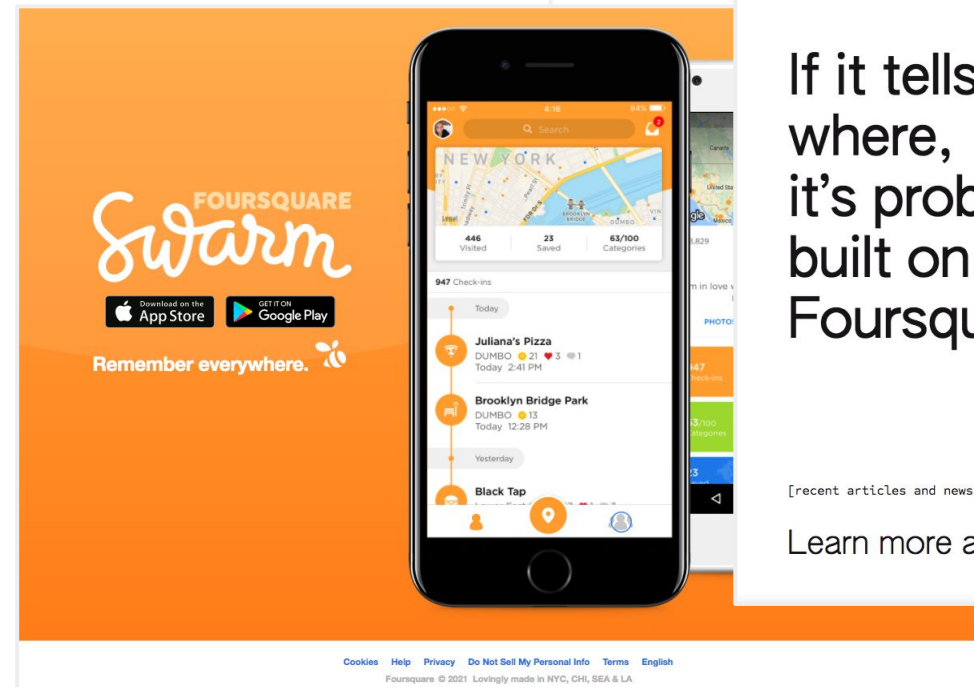
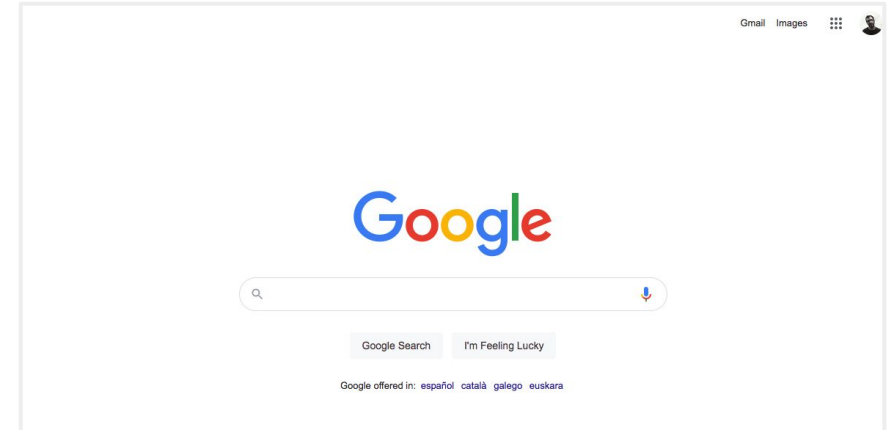
Datos de comportamiento

Sensores

► Proveedores de datos

► Datos abiertos

► Scrapeado de internet



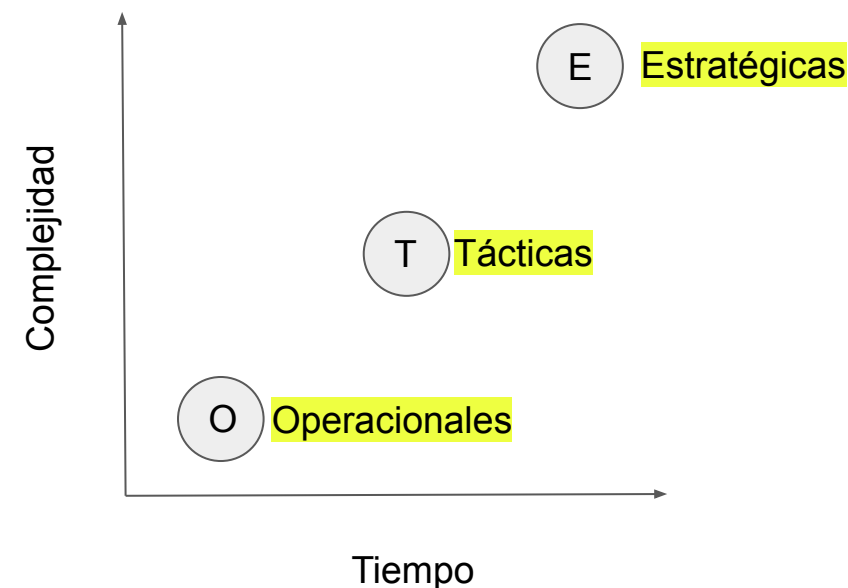
¿Para qué podemos utilizar los datos?

Toma de decisiones

- ▶ Operacionales
- ▶ Tácticas
- ▶ Estratégicas

	Operativa	Táctica	Estratégica
Opciones	Binarias o Bajas	Múltiples	Miles o millones
Incertidumbre	Baja	Media	Alta
Automatización	Alta	Media	Baja

En términos de escala temporal y complejidad podemos dividirlos en:

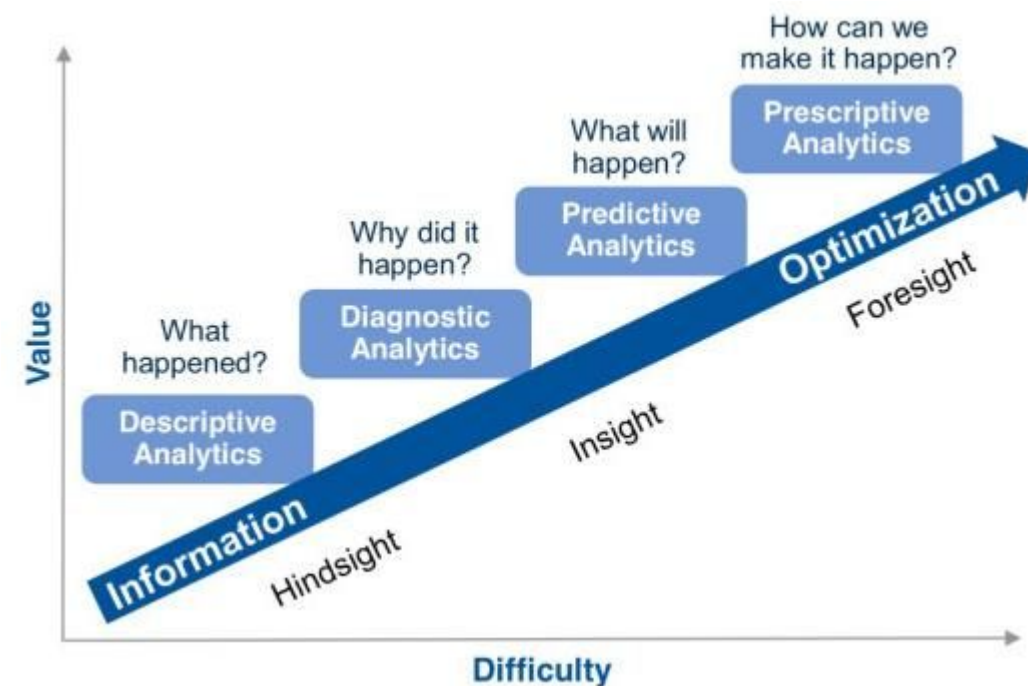


Análisis que podemos realizar sobre los datos

Analítica

- ▶ Descriptiva
 - ¿Qué sucedió?
- ▶ Diagnóstica
 - ¿Por qué sucedió?
- ▶ Predictiva
 - ¿Qué sucederá?
- ▶ Prescriptiva
 - ¿Cómo hacemos que suceda?

En términos de valor y dificultad podemos dividir las en:



Tipos de datos

► Estructurados

Tabulares (csv, excel,...)

Diccionarios (json, xml,...)

► No estructurados

Texto

Imágenes

Sonido

Vídeo

dreams.csv

1

subreddit,date,num_comments,score,title,description,replies

2

DreamInterpretation,2018-03-09 11:42:21,0,1,Pulling Worms and bugs out of my brea

3

DreamInterpretation,2018-03-09 15:58:52,0,1,Doing barrel rolls in a older airplane

4

DreamInterpretation,2018-03-09 17:53:18,1,3,"Accidentally taking car that wasn't m

5

DreamInterpretation,2018-03-10 15:36:41,0,2,Friend Dropped His Phone In The Harbor

6

DreamInterpretation,2018-03-10 17:46:20,1,3,Glen Campbell imparts cryptic life les

7

DreamInterpretation,2018-03-10 22:00:16,0,2,My mom dies in my dreams.,[deleted],

8

DreamInterpretation,2018-03-10 22:09:42,0,3,"When I realize I am dreaming, I know

9

DreamInterpretation,2018-03-10 22:24:41,0,2,The guest house and black hooded figur

10

DreamInterpretation,2018-03-11 02:45:59,1,3,Need help filling in the blanks.,[dele

11

DreamInterpretation,2018-03-11 05:25:23,0,2,Aztec Ritual,"Hi, I had a weird dream

12

DreamInterpretation,2018-03-11 05:43:46,2,3,An indoor beach,"I went walking in a p

13

DreamInterpretation,2018-03-12 01:12:13,0,2,Family Members torturing people,"I ha

14

DreamInterpretation,2018-03-12 04:36:26,0,3,NSFW. I (female) often dream of having

15

DreamInterpretation,2018-03-12 07:45:56,0,1,Being treated like a child,[deleted],

16

DreamInterpretation,2018-03-12 09:59:46,0,2,Dreamt of a long time crush and being

17

DreamInterpretation,2018-03-12 10:16:57,0,1,"Cemetery, crypt, butchered bodies,,[

18

DreamInterpretation,2018-03-12 12:24:30,0,1,Random Ex-Coworker messaged me this -

19

DreamInterpretation,2018-03-12 18:34:51,0,1,transforming assassin,"I took some Va

20

DreamInterpretation,2018-03-12 21:39:03,0,1,Lucid dreaming Discord x),,[

21

DreamInterpretation,2018-03-12 22:21:18,1,1,Floating.. a recurring dream,"So, I ke

22

DreamInterpretation,2018-03-13 00:20:03,2,1,"In my dream, I saw myself watching my

23

DreamInterpretation,2018-03-13 06:04:33,3,1,BLACK CAT DREAM,[deleted],[{}

24

DreamInterpretation,2018-03-13 06:21:38,0,1,Another to interpret,[deleted],[]

25

DreamInterpretation,2018-03-13 18:18:10,4,2,"I had the most awful dream last night

26

DreamInterpretation,2018-03-14 12:49:34,0,1,I can never get through to 911/Nobody

27

DreamInterpretation,2018-03-14 01:35:25,0,1,Small black scorpion stinging very lar

28

DreamInterpretation,2018-03-14 05:08:35,0,1,Recurring dream?,"Just curious if anyo

29

DreamInterpretation,2018-03-14 13:39:10,2,1,Reoccurring Dream about Classes,[deleted],[{}

30

DreamInterpretation,2018-03-14 13:47:53,2,2,"Saw my ""ancestors""?,"[deleted],[{}

31

DreamInterpretation,2018-03-14 23:12:09,1,2,idek?,[deleted],[{}

32

Period

Saxo

001-102 I:XX To

001-008 I:Enfer

009-041 II:Turbo

042-043 III:Enf.

044-045 IV:Enfer

046-049 V:Trast

050-052 VI-VIII:I

053-061 IX:Enfe

062-067 X:En

068-071 XI:Enfe

072-075 XII:Enfe

076-079 XIII:Enfe

080-083 XIV:Enfe

084-087 XV:Enfe

088-091 XVI:Enfe

092-095 XVII:Enfe

096-099 XVIII:Enfe

100-103 XIX:Enfe

104-107 XX:Enfe

108-111 XXI:Enfe

112-115 XXII:Enfe

116-119 XXIII:Enfe

120-123 XXIV:Enfe

124-127 XXV:Enfe

128-131 XXVI:Enfe

132-135 XXVII:Enfe

136-139 XXVIII:Enfe

140-143 XXIX:Enfe

144-147 XXX:Enfe

148-151 XXXI:Enfe

152-155 XXXII:Enfe

156-159 XXXIII:Enfe

160-163 XXXIV:Enfe

164-167 XXXV:Enfe

168-171 XXXVI:Enfe

172-175 XXXVII:Enfe

176-179 XXXVIII:Enfe

180-183 XXXIX:Enfe

184-187 XL:Enfe

188-191 XLI:Enfe

192-195 XLII:Enfe

196-199 XLIII:Enfe

200-203 XLIV:Enfe

204-207 XLV:Enfe

208-211 XLVI:Enfe

212-215 XLVII:Enfe

216-219 XLVIII:Enfe

220-223 XLIX:Enfe

224-227 L:Enfe

228-231 L I:Enfe

232-235 L II:Enfe

236-239 L III:Enfe

240-243 L IV:Enfe

244-247 L V:Enfe

248-251 L VI:Enfe

252-255 L VII:Enfe

256-259 L VIII:Enfe

260-263 L IX:Enfe

264-267 L X:Enfe

268-271 L XI:Enfe

272-275 L XII:Enfe

276-279 L XIII:Enfe

280-283 L XIV:Enfe

284-287 L XV:Enfe

288-291 L XVI:Enfe

292-295 L XVII:Enfe

296-299 L XVIII:Enfe

300-303 L XIX:Enfe

304-307 L XX:Enfe

308-311 L XXI:Enfe

312-315 L XXII:Enfe

316-319 L XXIII:Enfe

320-323 L XXIV:Enfe

324-327 L XXV:Enfe

328-331 L XXVI:Enfe

332-335 L XXVII:Enfe

336-339 L XXVIII:Enfe

340-343 L XXIX:Enfe

344-347 L XXX:Enfe

348-351 L XXXI:Enfe

352-355 L XXXII:Enfe

356-359 L XXXIII:Enfe

360-363 L XXXIV:Enfe

364-367 L XXXV:Enfe

368-371 L XXXVI:Enfe

372-375 L XXXVII:Enfe

376-379 L XXXVIII:Enfe

380-383 L XXXIX:Enfe

384-387 L XL:Enfe

388-391 L XLI:Enfe

392-395 L XLII:Enfe

396-399 L XLIII:Enfe

400-403 L XLIV:Enfe

404-407 L XLV:Enfe

408-411 L XLVI:Enfe

412-415 L XLVII:Enfe

416-419 L XLVIII:Enfe

420-423 L XLIX:Enfe

424-427 L L:Enfe

428-431 L I:Enfe

432-435 L II:Enfe

436-439 L III:Enfe

440-443 L IV:Enfe

444-447 L V:Enfe

448-451 L VI:Enfe

452-455 L VII:Enfe

456-459 L VIII:Enfe

460-463 L IX:Enfe

464-467 L X:Enfe

468-471 L XI:Enfe

472-475 L XII:Enfe

476-479 L XIII:Enfe

480-483 L XIV:Enfe

484-487 L XV:Enfe

488-491 L XVI:Enfe

492-495 L XVII:Enfe

496-499 L XVIII:Enfe

500-503 L XIX:Enfe

504-507 L XX:Enfe

508-511 L XXI:Enfe

512-515 L XXII:Enfe

516-519 L XXIII:Enfe

520-523 L XXIV:Enfe

524-527 L XXV:Enfe

528-531 L XXVI:Enfe

532-535 L XXVII:Enfe

536-539 L XXVIII:Enfe

540-543 L XXIX:Enfe

544-547 L XXX:Enfe

548-551 L XXXI:Enfe

552-555 L XXXII:Enfe

556-559 L XXXIII:Enfe

560-563 L XXXIV:Enfe

564-567 L XXXV:Enfe

568-571 L XXXVI:Enfe

572-575 L XXXVII:Enfe

576-579 L XXXVIII:Enfe

580-583 L XXXIX:Enfe

584-587 L XL:Enfe

588-591 L XLI:Enfe

592-595 L XLII:Enfe

596-599 L XLIII:Enfe

600-603 L XLIV:Enfe

604-607 L XLV:Enfe

608-611 L XLVI:Enfe

612-615 L XLVII:Enfe

616-619 L XLVIII:Enfe

620-623 L XLIX:Enfe

624-627 L L:Enfe

628-631 L I:Enfe

632-635 L II:Enfe

636-639 L III:Enfe

640-643 L IV:Enfe

644-647 L V:Enfe

648-651 L VI:Enfe

652-655 L VII:Enfe

656-659 L VIII:Enfe

660-663 L IX:Enfe

664-667 L X:Enfe

668-671 L XI:Enfe

672-675 L XII:Enfe

676-679 L XIII:Enfe

680-683 L XIV:Enfe

684-687 L XV:Enfe

688-691 L XVI:Enfe

692-695 L XVII:Enfe

696-699 L XVIII:Enfe

700-703 L XIX:Enfe

704-707 L XX:Enfe

708-711 L XXI:Enfe

712-715 L XXII:Enfe

716-719 L XXIII:Enfe

720-723 L XXIV:Enfe

724-727 L XXV:Enfe

728-731 L XXVI:Enfe

732-735 L XXVII:Enfe

736-739 L XXVIII:Enfe

740-743 L XXIX:Enfe

744-747 L XXX:Enfe

748-751 L XXXI:Enfe

752-755 L XXXII:Enfe

756-759 L XXXIII:Enfe

760-763 L XXXIV:Enfe

764-767 L XXXV:Enfe

768-771 L XXXVI:Enfe

772-775 L XXXVII:Enfe

776-779 L XXXVIII:Enfe

780-783 L XXXIX:Enfe

784-787 L XL:Enfe

788-791 L XLI:Enfe

792-795 L XLII:Enfe

796-799 L XLIII:Enfe

800-803 L XLIV:Enfe

804-807 L XLV:Enfe

808-811 L XLVI:Enfe

812-815 L XLVII:Enfe

816-819 L XLVIII:Enfe

820-823 L XLIX:Enfe

824-827 L L:Enfe

828-831 L I:Enfe

832-835 L II:Enfe

836-839 L III:Enfe

840-843 L IV:Enfe

844-847 L V:Enfe

848-851 L VI:Enfe

852-855 L VII:Enfe

856-859 L VIII:Enfe

860-863 L IX:Enfe

864-867 L X:Enfe

868-871 L XI:Enfe

872-875 L XII:Enfe

876-879 L XIII:Enfe

880-883 L XIV:Enfe

884-887 L XV:Enfe

888-891 L XVI:Enfe

892-895 L XVII:Enfe

896-899 L XVIII:Enfe

900-903 L XIX:Enfe

904-907 L XX:Enfe

908-911 L XXI:Enfe

912-915 L XXII:Enfe

916-919 L XXIII:Enfe

920-923 L XXIV:Enfe

924-927 L XXV:Enfe

928-931 L XXVI:Enfe

932-935 L XXVII:Enfe

936-939 L XXVIII:Enfe

940-943 L XXIX:Enfe

944-947 L XXX:Enfe

948-951 L XXXI:Enfe

952-955 L XXXII:Enfe

956-959 L XXXIII:Enfe

960-963 L XXXIV:Enfe

964-967 L XXXV:Enfe

968-971 L XXXVI:Enfe

972-975 L XXXVII:Enfe

976-979 L XXXVIII:Enfe

980-983 L XXXIX:Enfe

984-987 L XL:Enfe

988-991 L XLI:Enfe

992-995 L XLII:Enfe

996-999 L XLIII:Enfe

untitled

1

<?xml version="1.0" encoding="utf-8"?>

2

3

<root_element>

4

5

<child_element_1>

6

<child_element_2>Content</child_element_2>

7

</child_element_1>

8

9

<child_element_1>

10

<child_element_2>Content</child_element_2>

11

</child_element_1>

12

13

<child_element_1>

14

<child_element_2>Content</child_element_2>

15

<child_element_2>Content</child_element_2>

16

</child_element_1>

17

18

</root_element>

thunder: {} s3-event.json: ...

1

{

2

"Records": [

3

{

4

"eventVersion": "2.0",

5

"eventSource": "aws:s3",

6

"awsRegion": "eu-west-1",

7

"eventTime": "2018-04-09T13:39:25.298Z",

8

"eventName": "ObjectCreated:Put",

9

"userIdentity": {

10

"principalId": "AWS:AROAI4UHMFLMCF60KUC:thunder-dev-refinery"

11

},

12

"requestParameters": {

13

"sourceIPAddress": "34.244.58.59"

14

},

15

"responseElements": {

16

"x-amz-request-id": "5F87B1DF8246EAA4",

17

"x-amz-id-2": "0uLqe9041By99GCTGr+ArtKsGJ561kv/NYMBB1Xr0XmXodJre5jKdkZsus8VgG+csi"

18

},

19

"s3": {

20

"s3SchemaVersion": "1.0",

21

"configurationId": "28baf0a1-170b-4530-8d0b-8bdbab865bf7",

22

"bucket": {

23

"name": "dev.refinery.eu-west-1.zenseiapp.com",

24

"ownerIdentity": {

25

"principalId": "AN61RV6AN93VA"

26

},

27

"arn": "arn:aws:s3:::dev.refinery.eu-west-1.zenseiapp.com"

28

},

29

"object": {

30

"key": "refinery/6359304/2020-09-21/airquality.json",

31

"size": 208,

32

"eTag": "075902cdc66242cc9628286156701303",

33

"sequencer": "005ACB608036801D97"

34

}

35

}

36

}

37

}

38

}

Tipos de datos

► Estructurados

Tabulares (csv, excel,...)

Diccionarios (json, xml,...)

► No estructurados

Texto

Imágenes

Sonido

Vídeo

Cualitativas (No numéricos)

Cuantitativos (Numéricos)

Nombre	Color de pelo	Estatura
Pedro	Rubio	1,86m
Ana	Morena	1,76m
Cualitativo	Cualitativo	Cuantitativo

Tipos de datos

► Estructurados

Tabulares (csv, excel,...)

Diccionarios (json, xml)

► No estructurados

Texto

Imágenes

Sonido

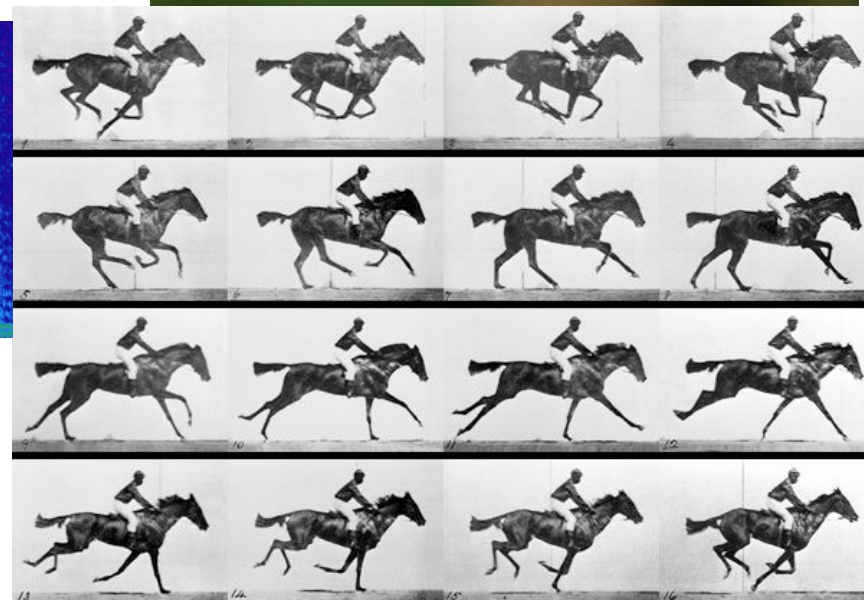
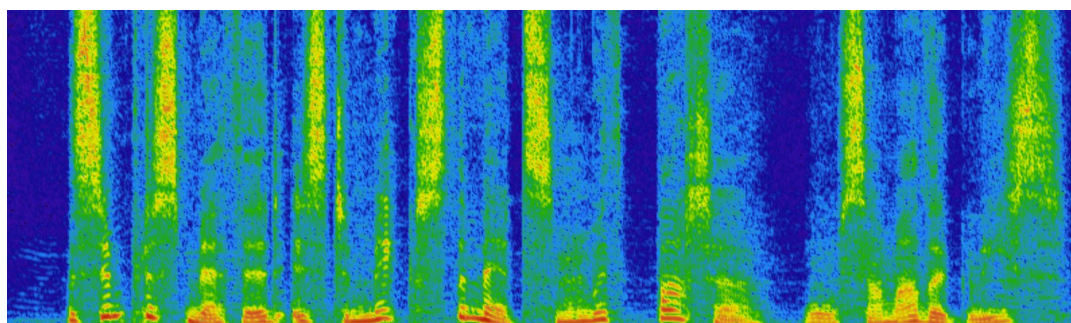
Vídeo



David Perell @david_perell · 1m

Talented kids are more capable than we think:

- Leonardo da Vinci was an apprentice to Verrocchio at 14
- Andrew Carnegie finished schooling at 12. At 13, he became a office boy. By 16, he earned most of the family income.



Tipos de variables

► Contenido

1. Texto
 - a. Cadenas de caracteres
 - b. Mensajes
2. Categóricas
 - a. Color, marca, producto, ranking
3. Numérica

► Origen

1. Dummy
2. Latente o sintética

► Función

1. Independientes
 - a. Precio
 - b. Marketing
 - c. Leads
2. Dependientes
 - a. Ventas

¿Cómo podemos trabajar con datos?

► No estructurados

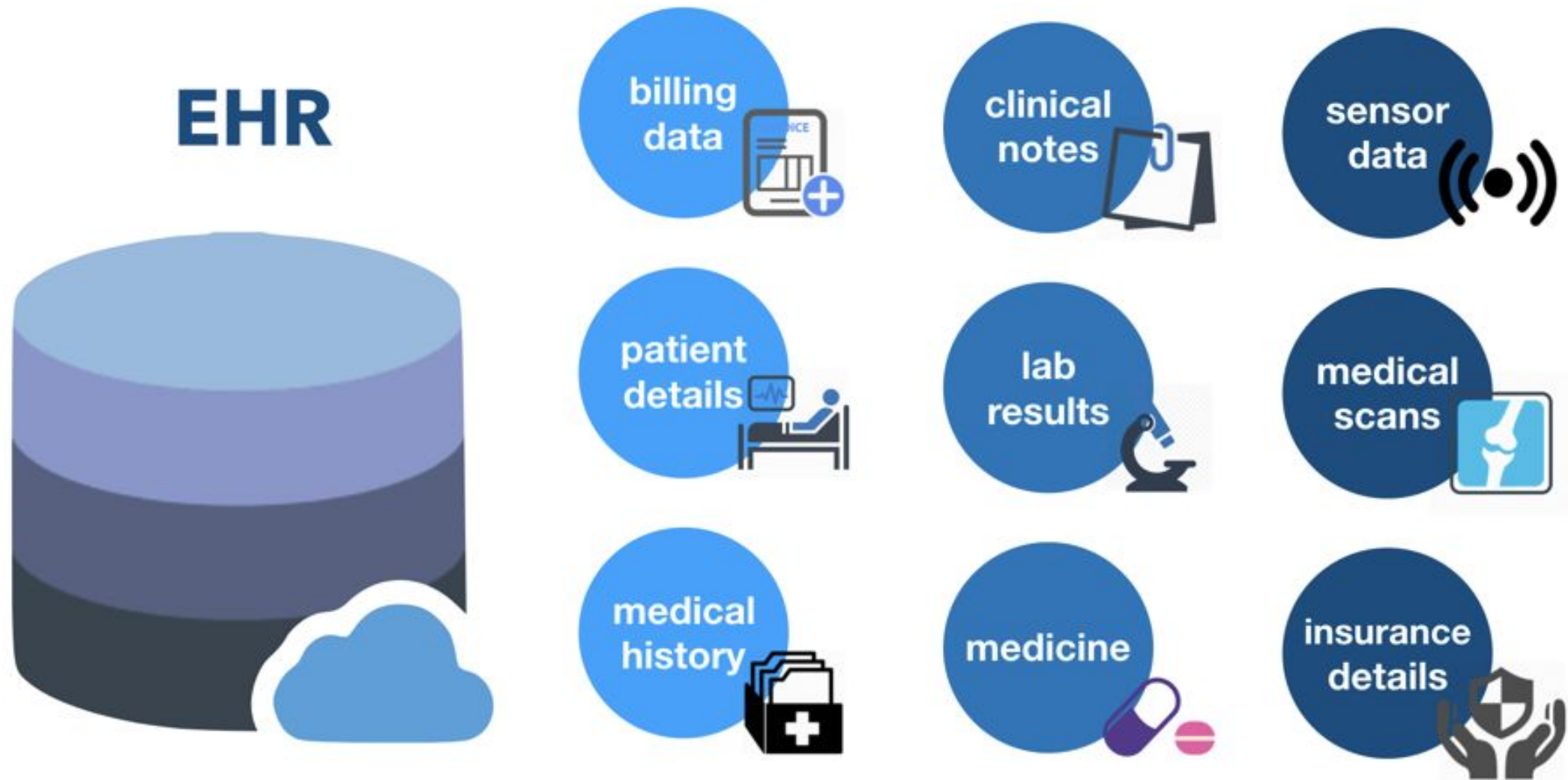
1. Técnicas de procesamiento del...
 - a. Lenguaje
 - b. Visión
 - c. Sonidos

► Estructurados

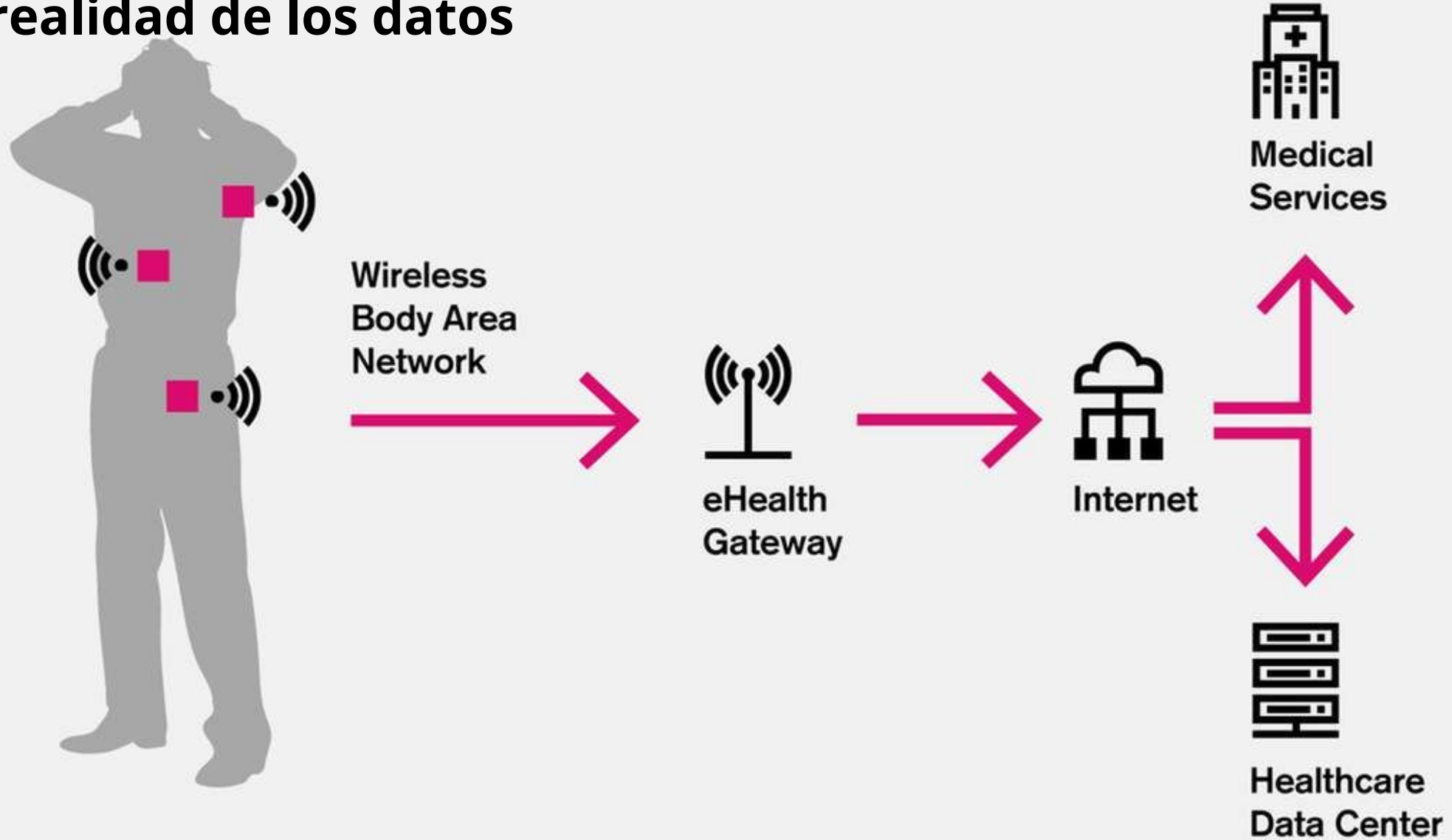
1. Técnicas estadísticas y matemáticas

► Plataformas de procesamiento Big Data

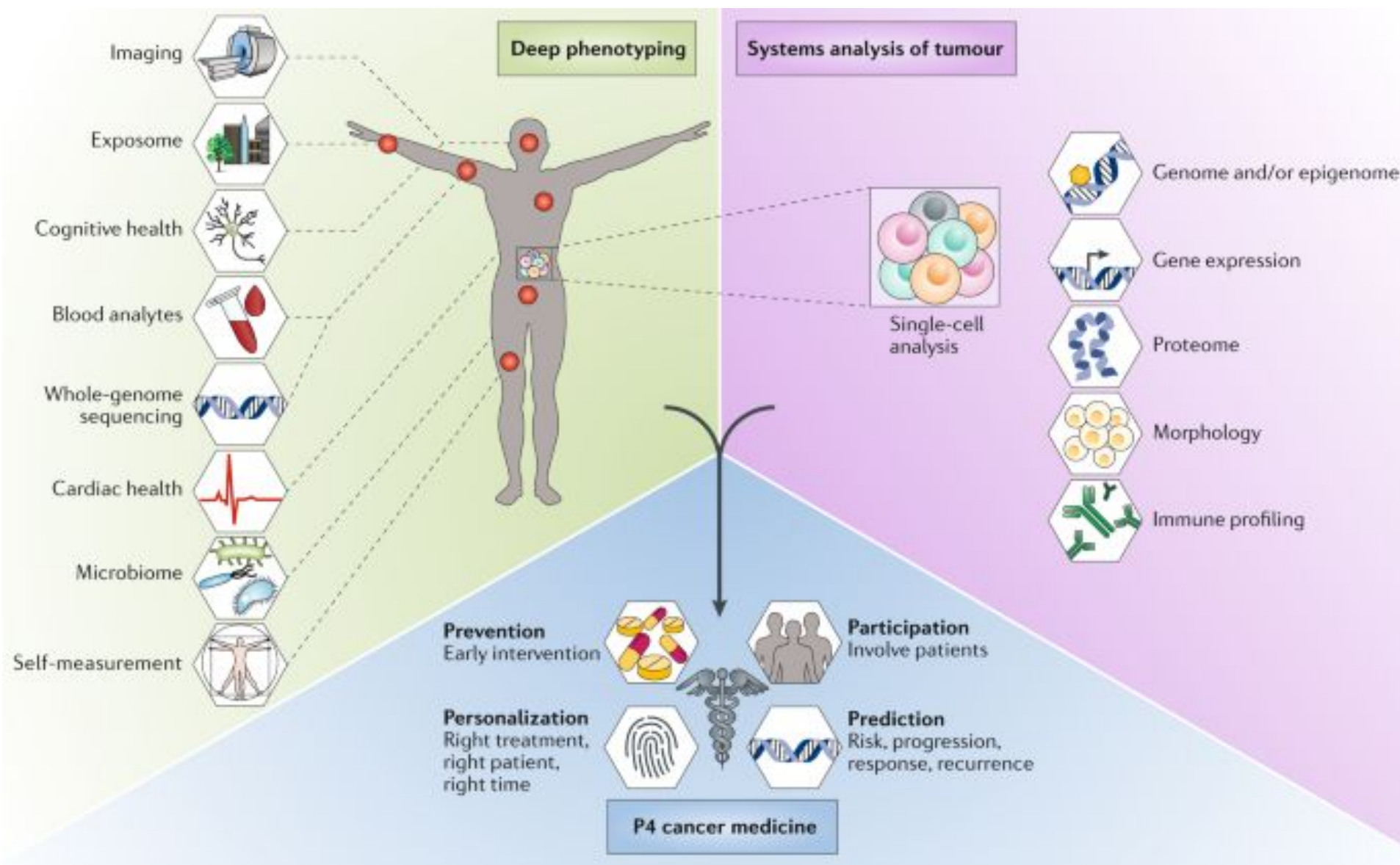
La realidad de los datos



La realidad de los datos



La realidad de los datos



Arquitecturas de procesamiento de datos

Tipos de tareas de una plataforma de Big data

- ▶ Procesamiento batch
- ▶ Procesamiento en tiempo real
- ▶ Exploración interactiva de datos
- ▶ Analítica

Qué es el famoso data lake

► *Es una base de datos distribuida donde todas las fuentes de datos de una organización están integradas y en su formato original para posteriores procesamientos . Permitiendo eliminar silos de información.*

Tipos de arquitecturas

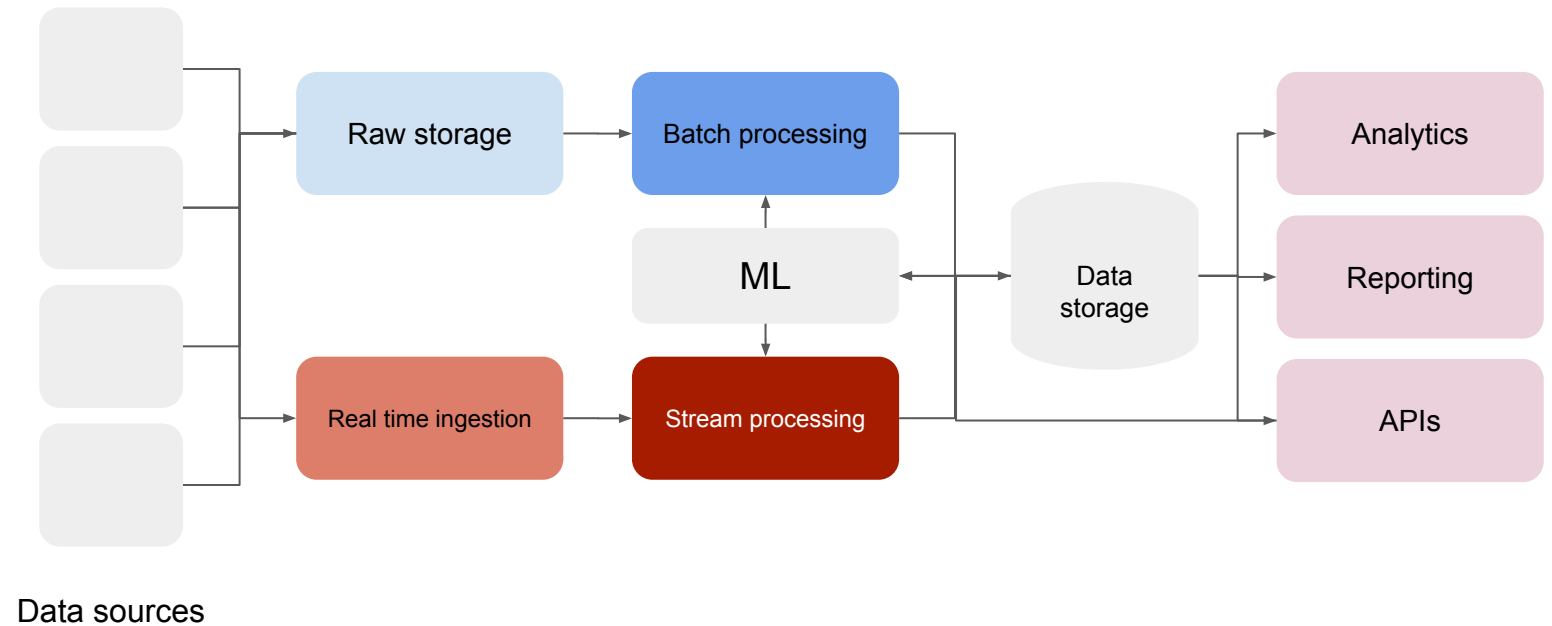
- ▶ Tradicional
- ▶ Lambda
- ▶ Kappa

Tipos de arquitecturas

► Tradicional

► Lambda

► Kappa

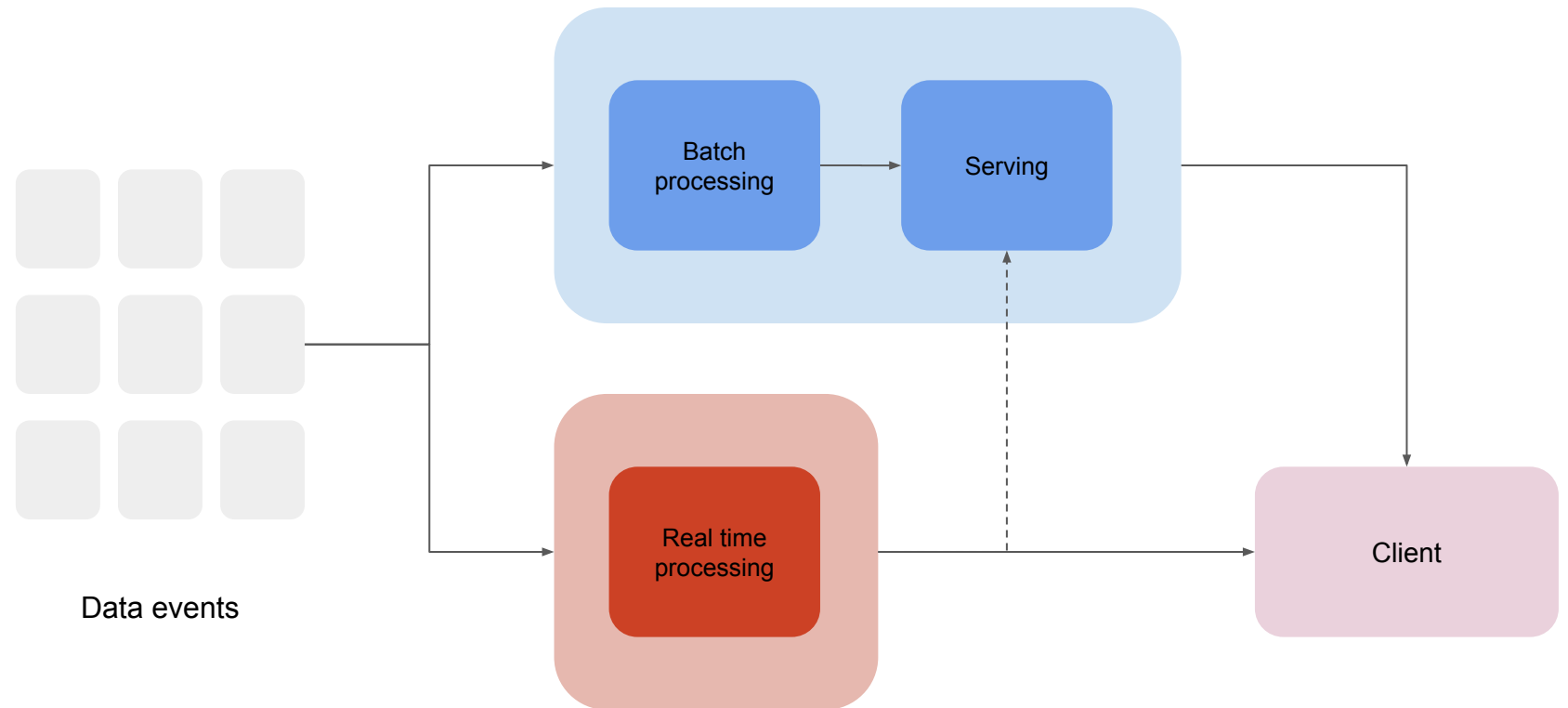


Tipos de arquitecturas

► Tradicional

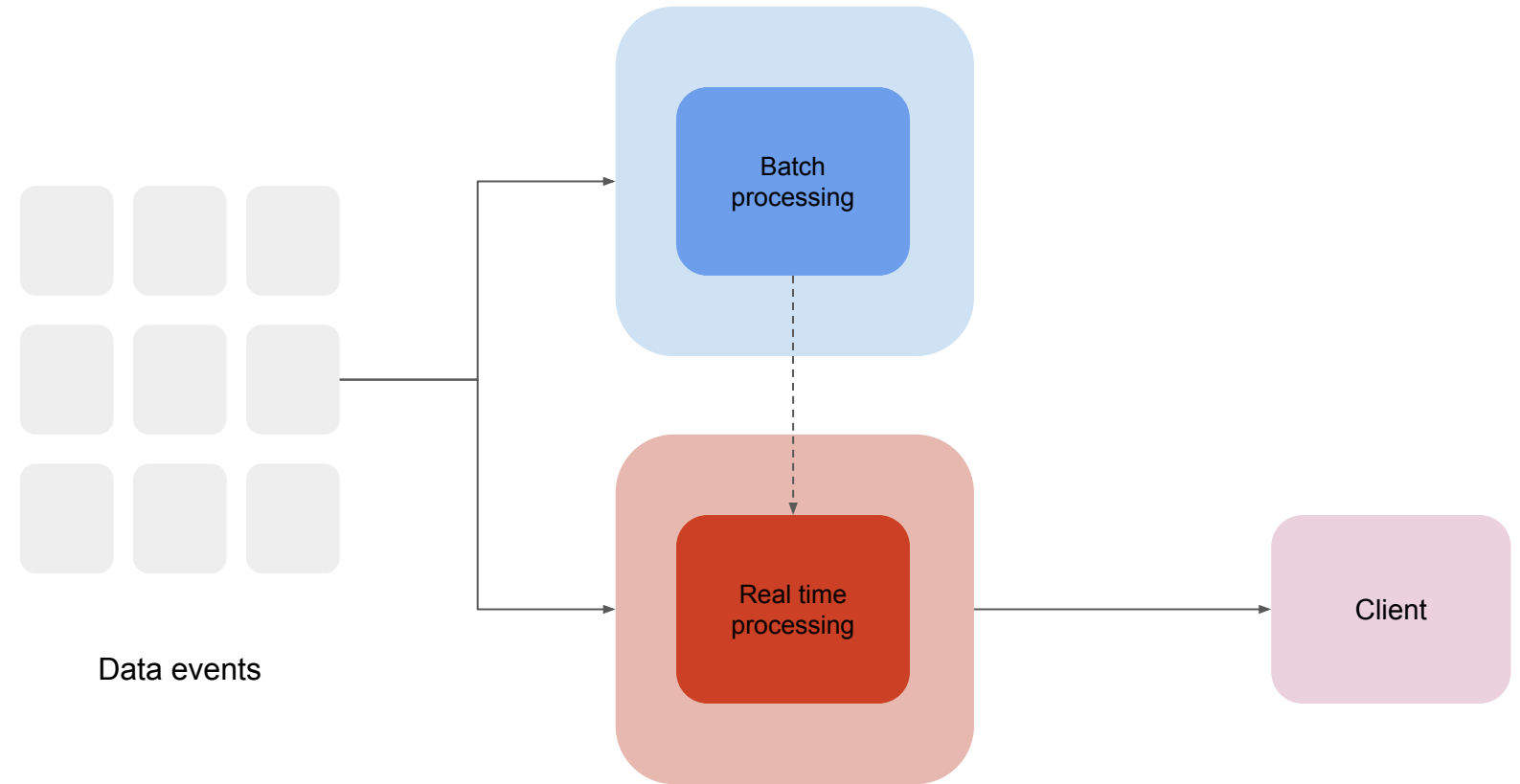
► Lambda

► Kappa



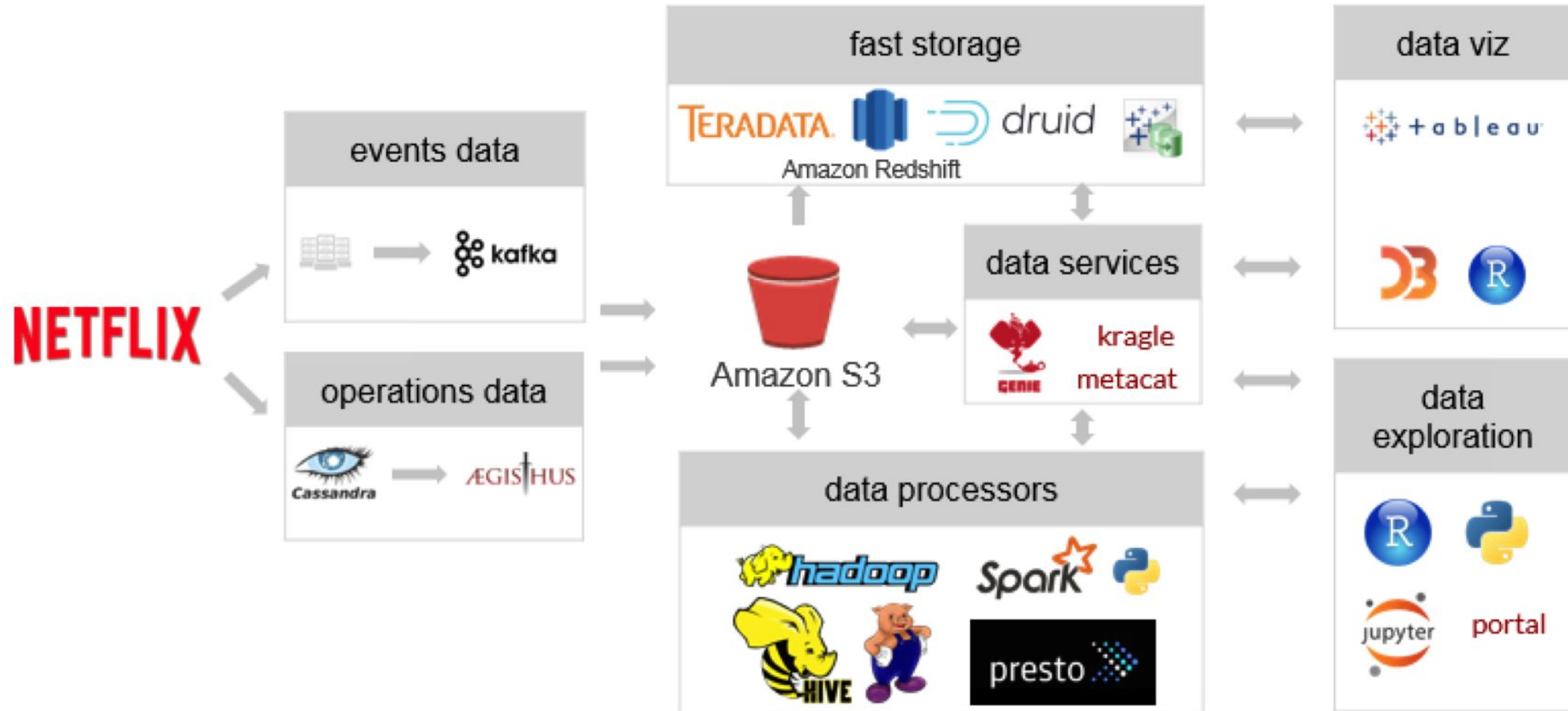
Tipos de arquitecturas

- ▶ Tradicional
- ▶ Lambda
- ▶ **Kappa**



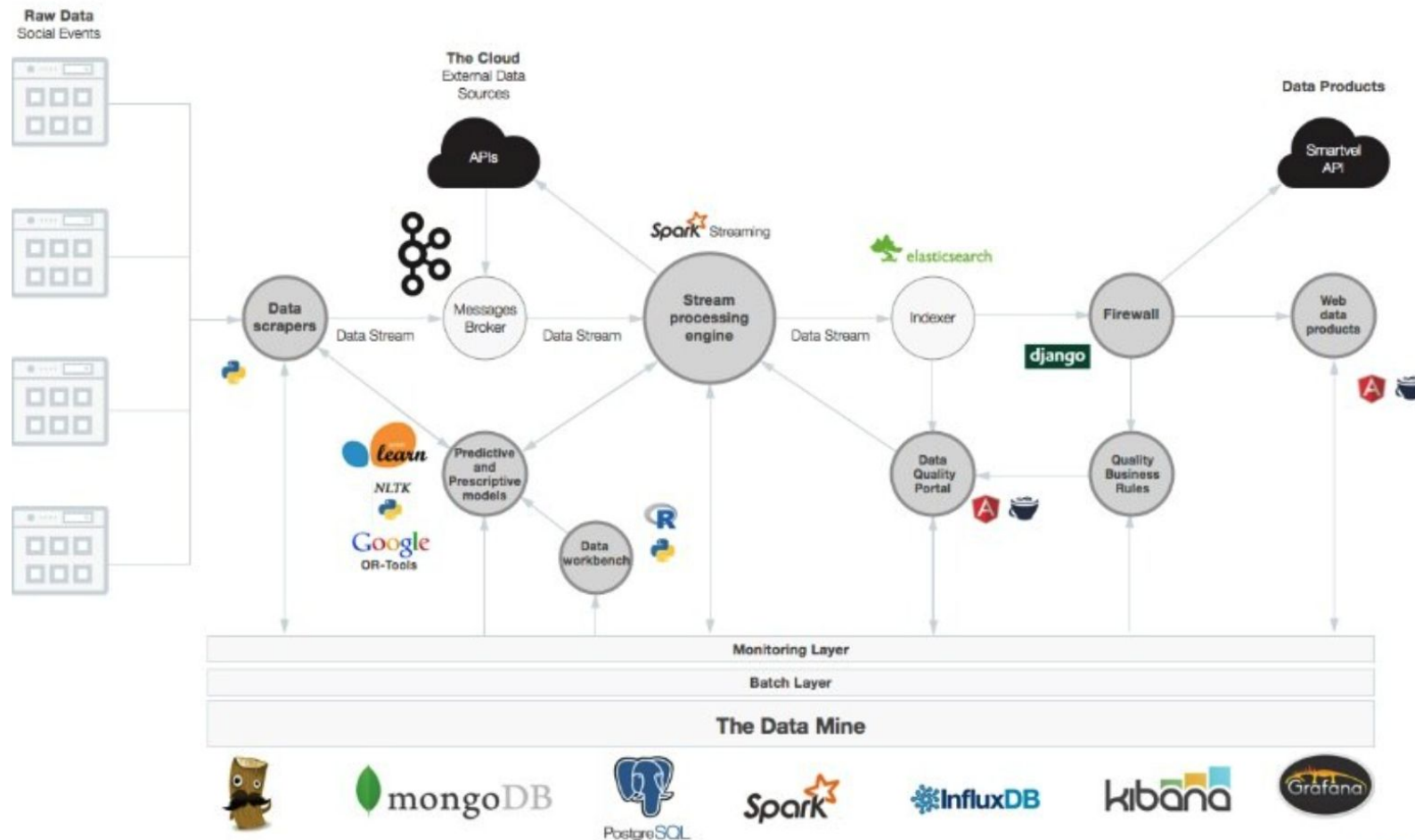
Tipos de arquitecturas

► Arquitectura de microservicios de Netflix



Tipos de arquitecturas

► Arquitectura de microservicios (Otro ejemplo)



Trabajo previo a la implantación

- ▶ Casos de uso y de negocio
- ▶ Análisis de fuentes de datos

Problema - Qué problemas tiene el negocio.

Solución - Qué solución tiene impacto.

Tecnología - Cómo podemos implementarla.

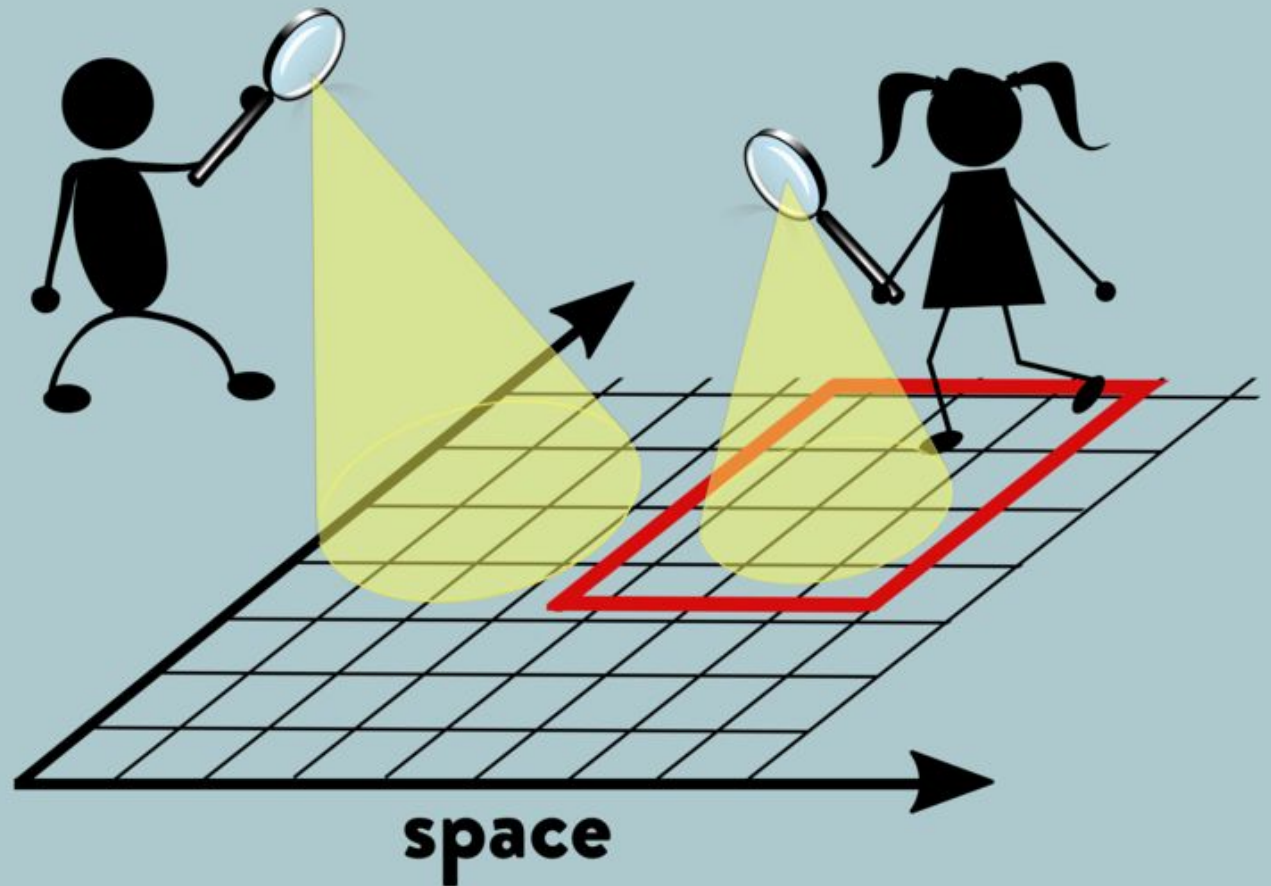
Cuidado con los arcos de iglesia y las culturas de técnicos tiranos.

Datos y modelos matemáticos

Datos

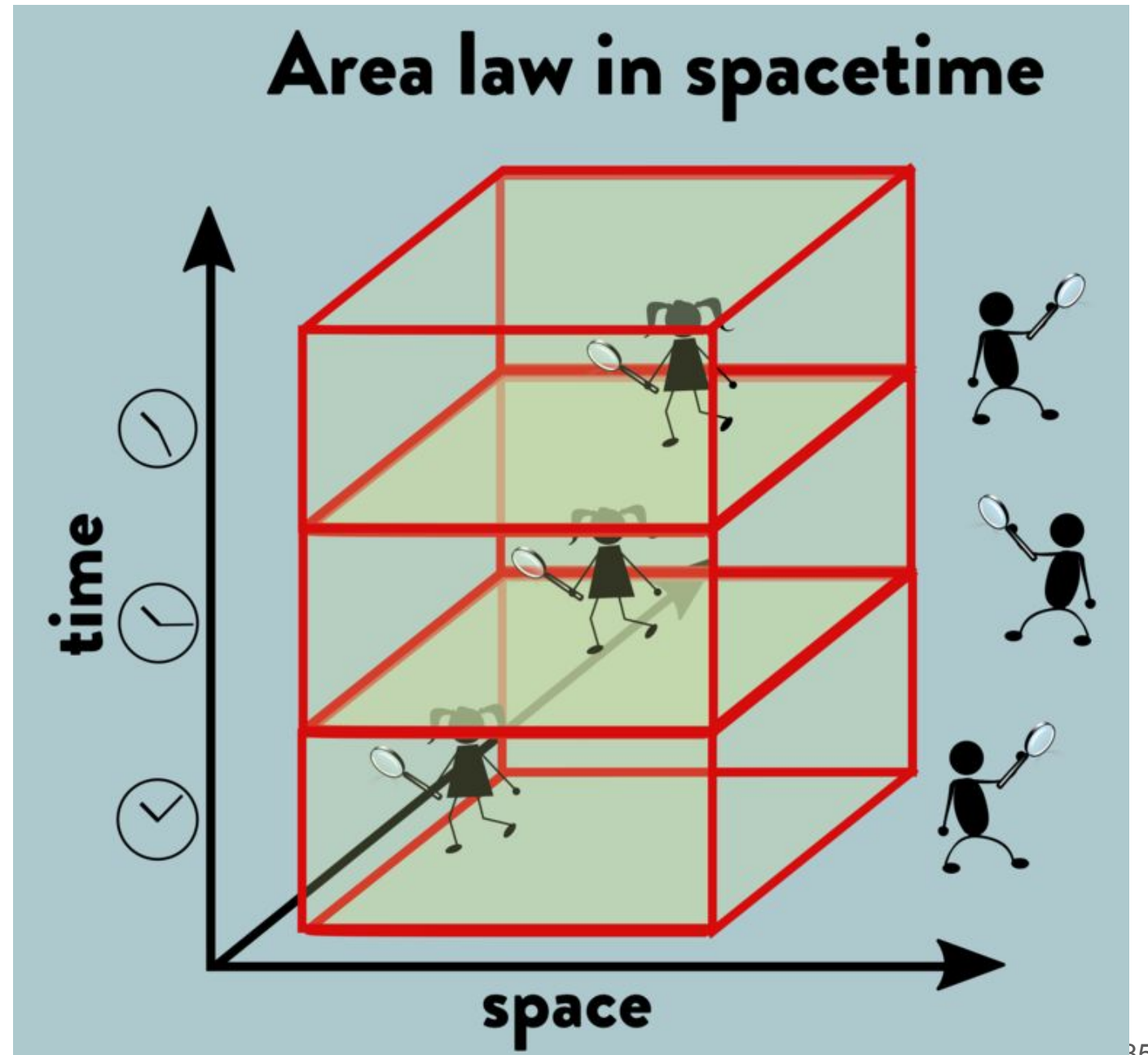
Espacio

Area law in space



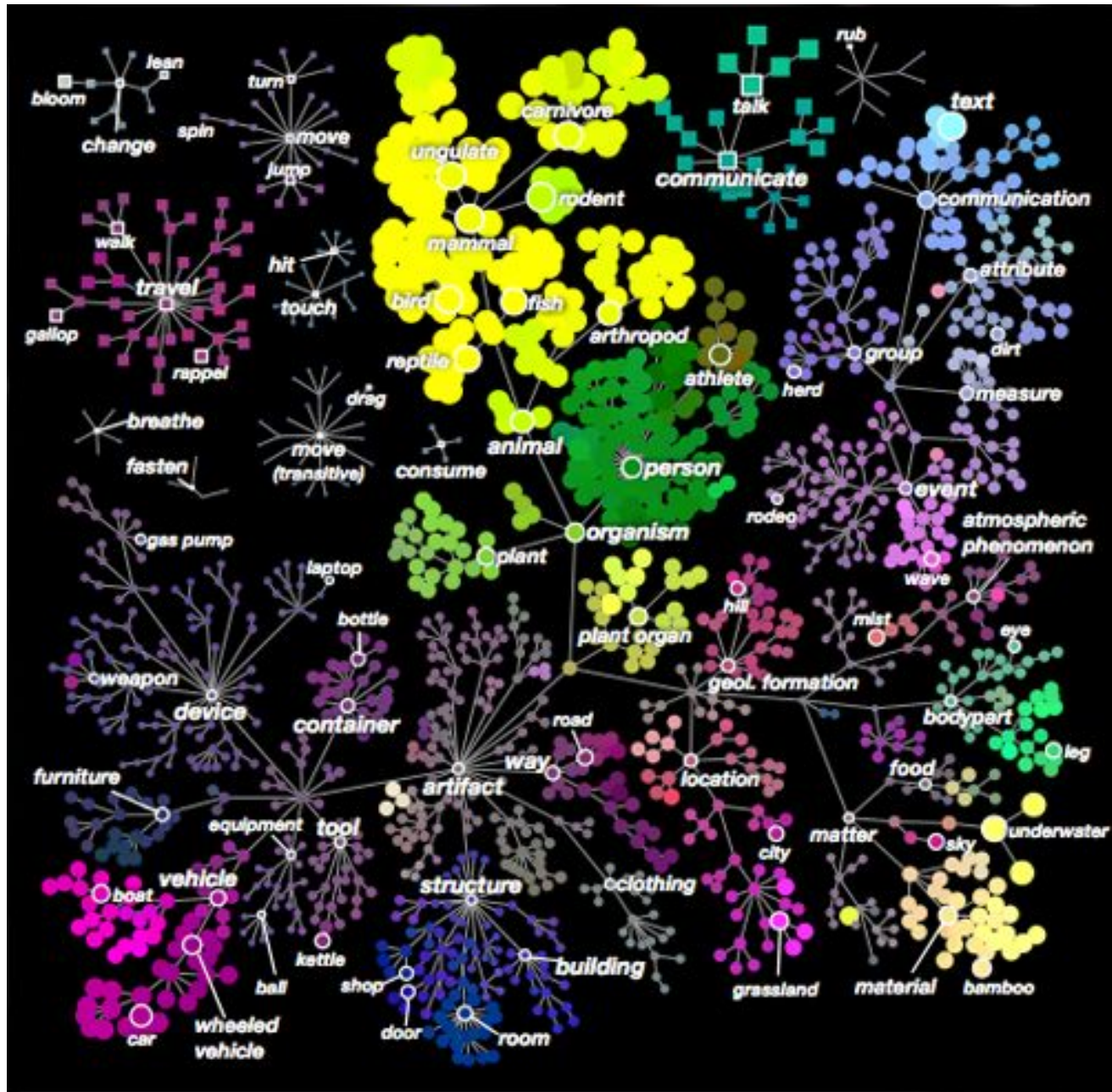
Datos

Espacio-tiempo



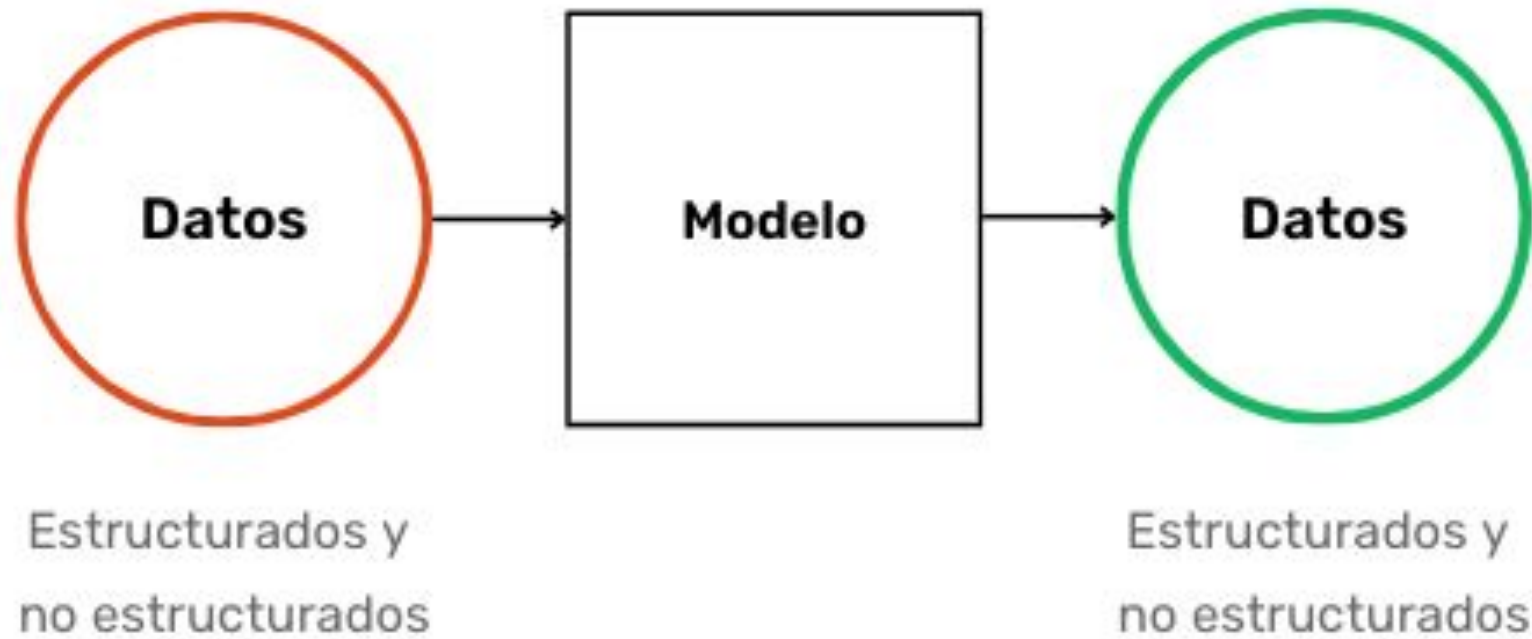
Datos

Representación simbólica,
fonológica, semántica y
gramatical = Lenguaje



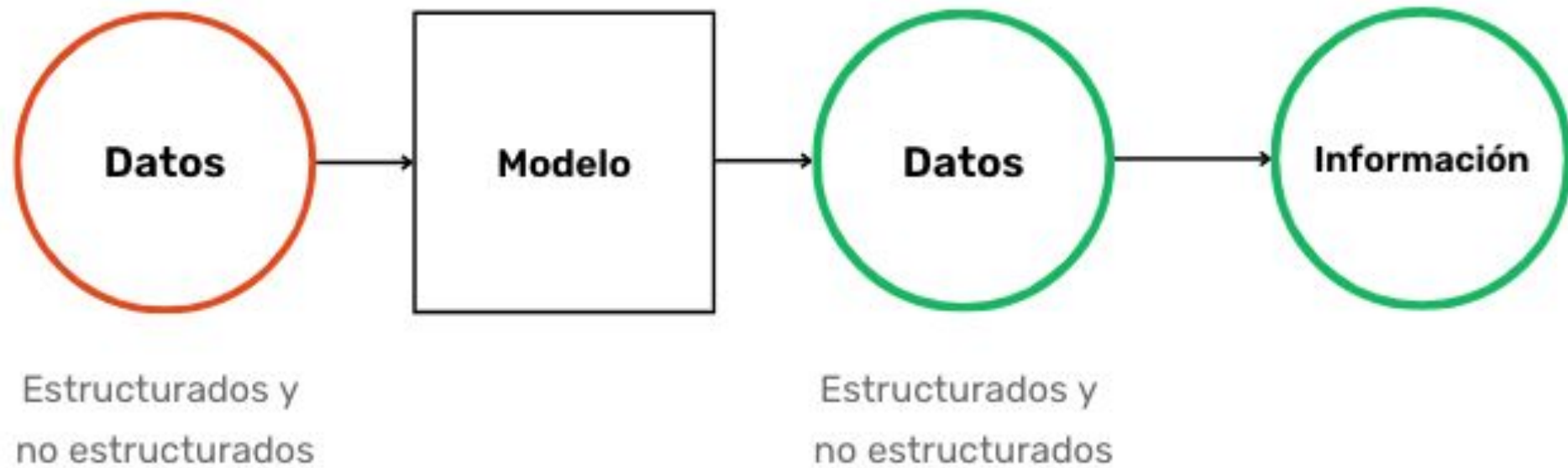
Modelos

Datos + Modelo = Datos



Modelos

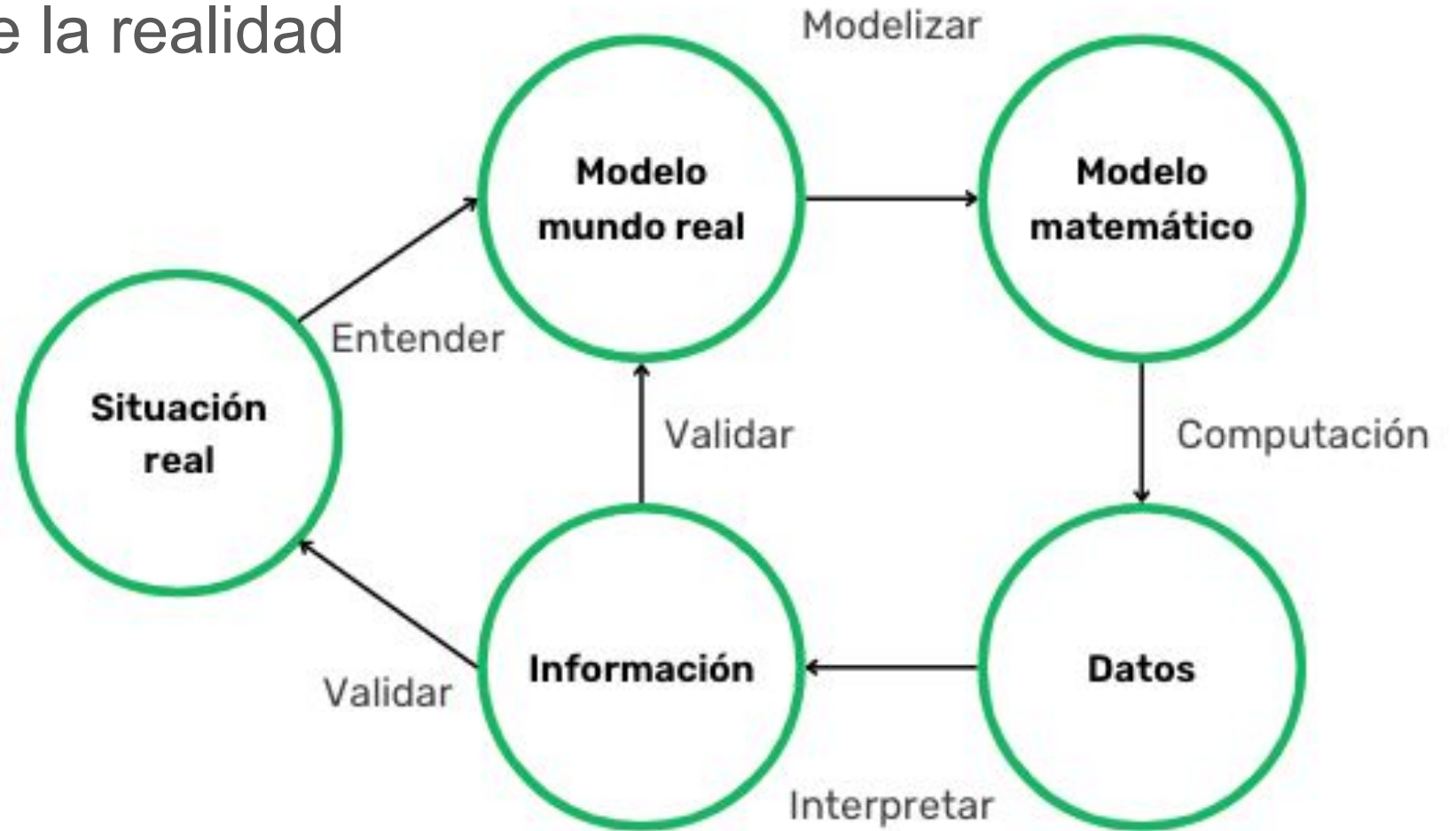
Datos + Modelo = Datos



Modelos

¿Qué es un modelo?

Una representación de la realidad



Introducción al aprendizaje automático

Cognición

"Proceso mental que incluye el pensamiento, el aprendizaje, la memorización, la conciencia del entorno y el sentido de realidad o juicio."

Cognición

Nos creemos dioses y queremos crear inteligencias artificiales que sean capaces de ...

- ▶ Percibir = Sensores = Datos
- ▶ Aprender = Representación de la información de los datos
- ▶ Memorizar = Recordar en el corto y largo plazo
- ▶ Razonar = Lógica, asociación, inducción, deducción
- ▶ Sentir emociones
- ▶ Ser conscientes

Aprendizaje

El aprendizaje es el proceso de adquirir nuevos conocimientos, comportamientos, habilidades, valores, actitudes y preferencias.

- ▶ Supervisado
- ▶ Semi supervisado
- ▶ No supervisado
- ▶ Refuerzo

Aprendizaje

Supervised Learning

Data: (x, y)

x is data, y is label

Goal: Learn function to map
 $x \rightarrow y$

Examples: Classification, regression, object detection, semantic segmentation, etc.

Unsupervised Learning

Data: x

x is data, no labels!

Goal: Learn some *hidden* or *underlying structure* of the data

Examples: Clustering, feature or dimensionality reduction, etc.

Modelos matemáticos para resolver problemas

- ▶ Regresión = Variable numérica
- ▶ Clasificación = Variable categórica
- ▶ Clusterización = Similitud
- ▶ Series temporales = Variables temporales
- ▶ Análisis de redes = Interdependencia
- ▶ Reducción de la dimensionalidad = Condensación de la información
- ▶ Optimización = Resolución óptima o factible de problemas
- ▶ Simulaciones = Solo tenemos datos de distribuciones estadísticas
- ▶ Agentes = Toma de acciones en mundos virtuales
- ▶ Recomendadores = Busca elementos similares

TIPOS DE APRENDIZAJE

Tareas



Introducción al aprendizaje profundo

El despertar de la inteligencia artificial

'Deep Voice' Software Can Clone Anyone's Voice With Just 3.7 Seconds of Audio

Using snippets of voices, Baidu's 'Deep Voice' can generate new speech, accents, and tones.

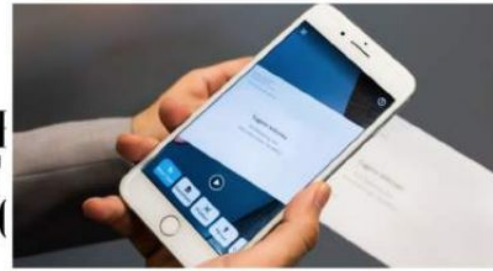
SHARE 7 FAVORITE

AI
Nv
pr
DEAN1



t with
**DEEPMIND
STARCRAFT
TRIUMPH**

Let There Be Sight: How Deep Learning Is Helping the Blind 'See'



Technology outpacing security measures

| Facial Recognition | Features and Interviews

AI beats docs in cancer spotting

A new study provides a fresh example of machine learning as an important diagnostic tool. Paul Biegler reports.

AI Can Help In Predicting Cryptocurrency Value

By Rickards Last updated Jan 25, 2018



'Creative' AlphaZero leads way for chess computers and, maybe, science

Former chess world champion Garry Kasparov likes what he sees of computer that could be used to find cures for diseases



How an A.I. 'Cat-and-Mouse Game' Generates Believable Fake Photos

By GIDE MEYER and KEITH COLLINS JAN. 3, 2018

Stock Predictions Based On AI: Is the Market Truly Predictable?

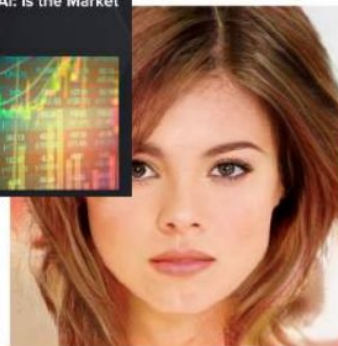
By David Higgins, Contributor and CEO, iStockphoto



Complex of bacteria-infecting viral proteins modeled at CASP-13. The complex consists of 10 subunits that were modeled individually. PDB: 5A8K

Google's DeepMind acers protein folding

By Robert E. Service | Dec. 6, 2018, 12:05 PM



To create the final image in this set, the system generated 10 million revisions over 18 days.

After Millions of Trials, These Simulated Humans Learned to Do Perfect Backflips and Cartwheels

By George Dvorsky 4/10/18 11:34am • Filed to: AI



On, Faked Data
The new AI model can generate realistic data for the purpose of self-driving.
JANUARY 24, 2018
SELF-DRIVING STARTUP



Neural networks everywhere

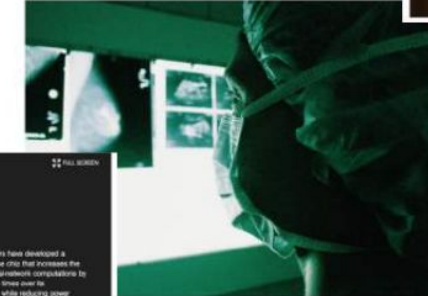
New chip reduces neural networks' power consumption by up to 95 percent, making them practical for battery-powered devices.

Wed, 01/16/2018 - 8:00am 1 Comment by Kenny Walter - Digital Reporter - @RandiMagazine



Researchers introduce a deep learning method that converts mono audio recordings into 3D sounds using video scenes

By Michael Muller | December 26, 2018 10:22 am



The faces show how far AI image generation has come in just four years

The faces on the right aren't real; they're the product of machine learning



Automation And Algorithms: De-Risking Manufacturing With Artificial Intelligence

Sarah Goehrke Contributor
Manufacturing
I focus on the industrialization of additive manufacturing.

TWEET THIS

The two key applications of AI in manufacturing are pricing and manufacturability feedback

ARTIFICIAL INTELLIGENCE

Any technique that enables computers to mimic human behavior



MACHINE LEARNING

Ability to learn without explicitly being programmed



DEEP LEARNING

Extract patterns from data using neural networks

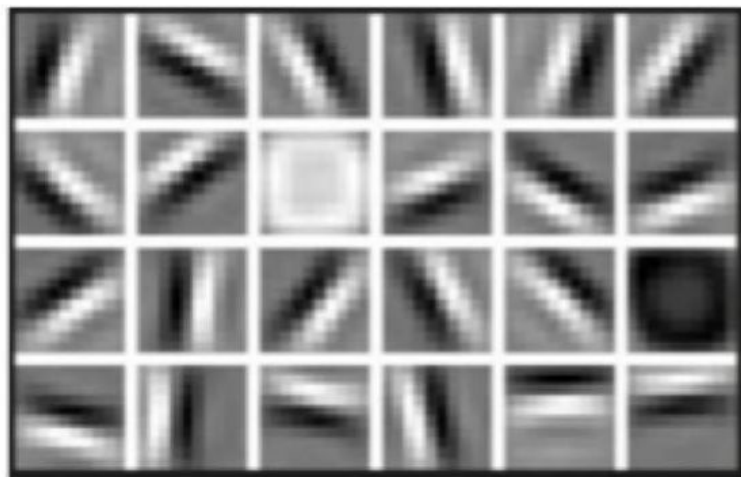


Por qué el aprendizaje profundo

Hand engineered features are time consuming, brittle and not scalable in practice

Can we learn the **underlying features** directly from data?

Low Level Features



Lines & Edges

Mid Level Features



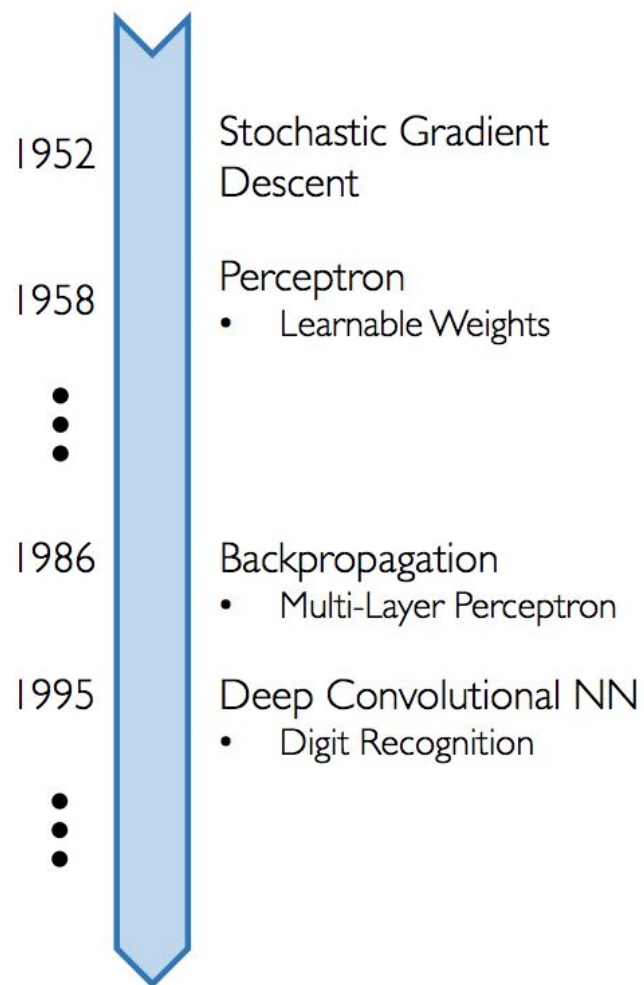
Eyes & Nose & Ears

High Level Features



Facial Structure

Por qué ahora



Neural Networks date back decades, so why the resurgence?

1. Big Data

- Larger Datasets
- Easier Collection & Storage

IMAGENET



2. Hardware

- Graphics Processing Units (GPUs)
- Massively Parallelizable



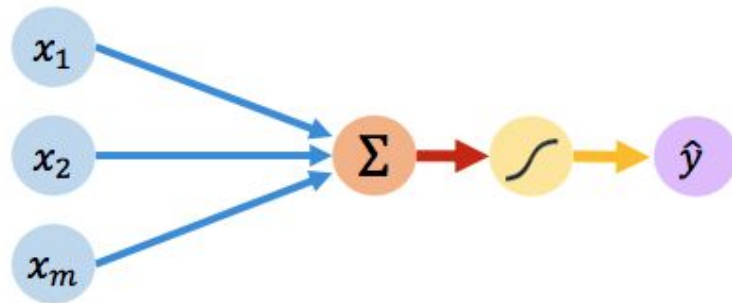
3. Software

- Improved Techniques
- New Models
- Toolboxes



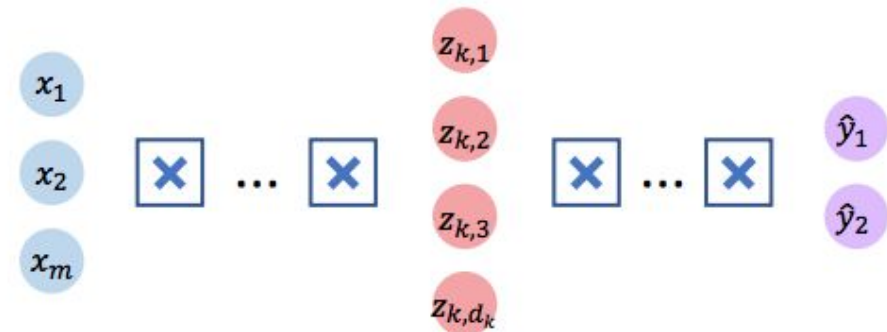
The Perceptron

- Structural building blocks
- Nonlinear activation functions



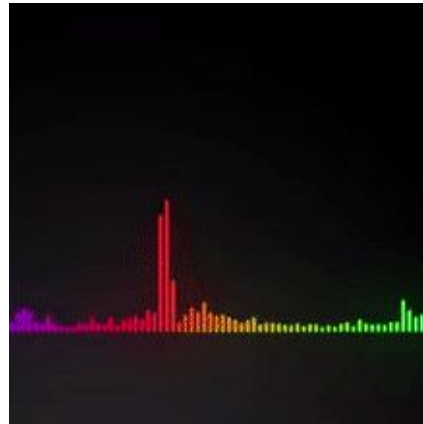
Neural Networks

- Stacking Perceptrons to form neural networks
- Optimization through backpropagation





Vision



Secuencias



Generativa



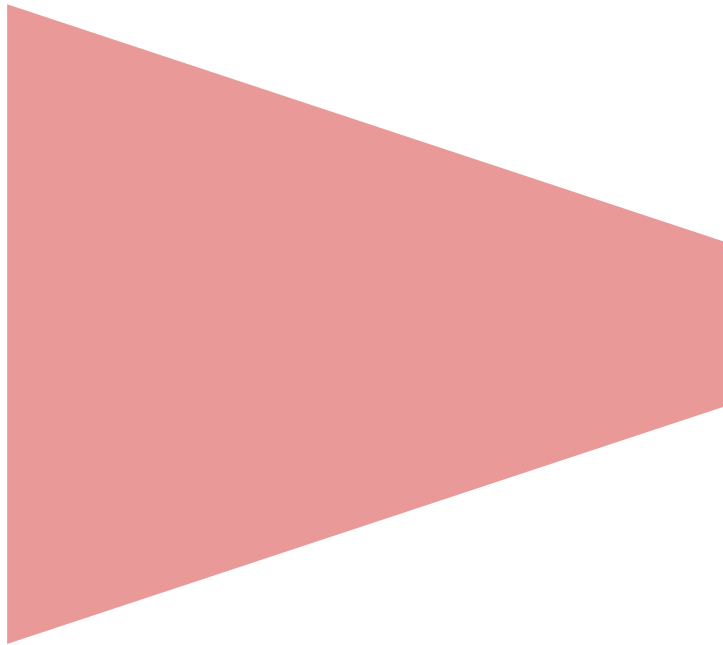
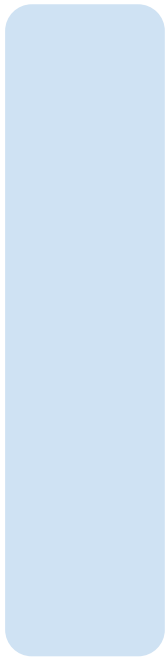
Refuerzo

Convolución

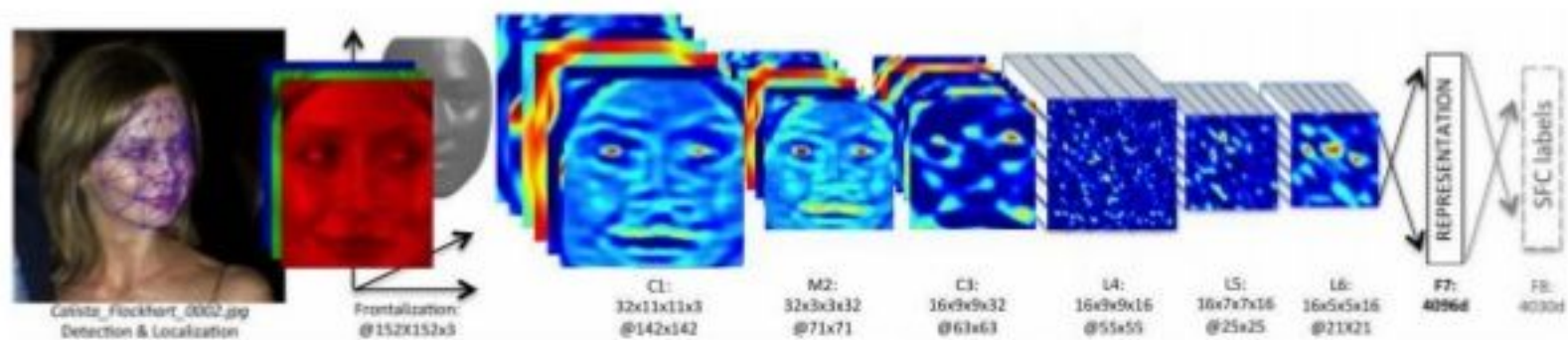
Entrada
X

Aprende
patrones

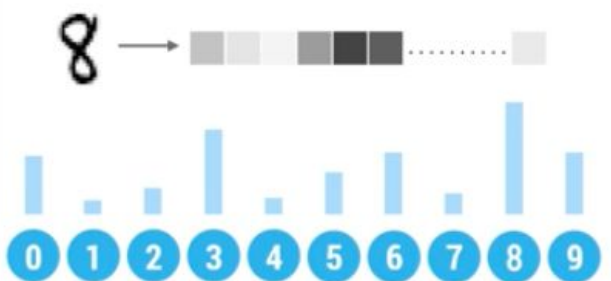
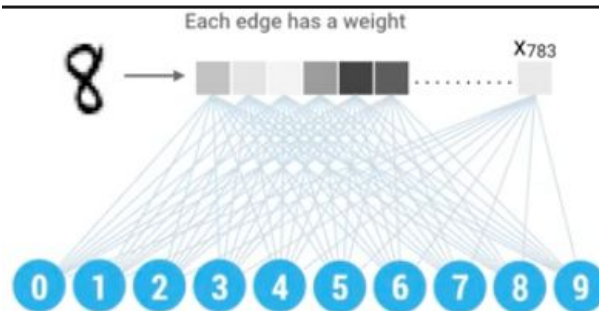
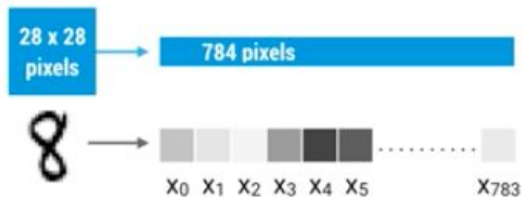
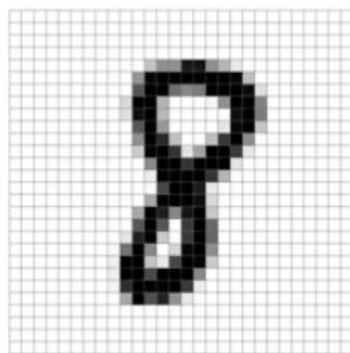
Predicción
y



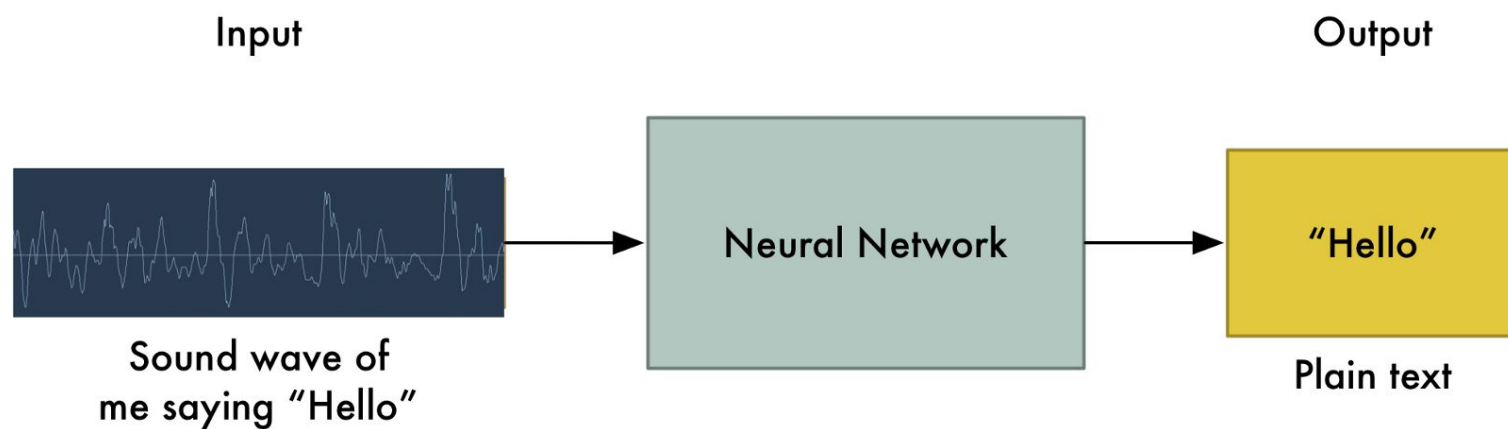
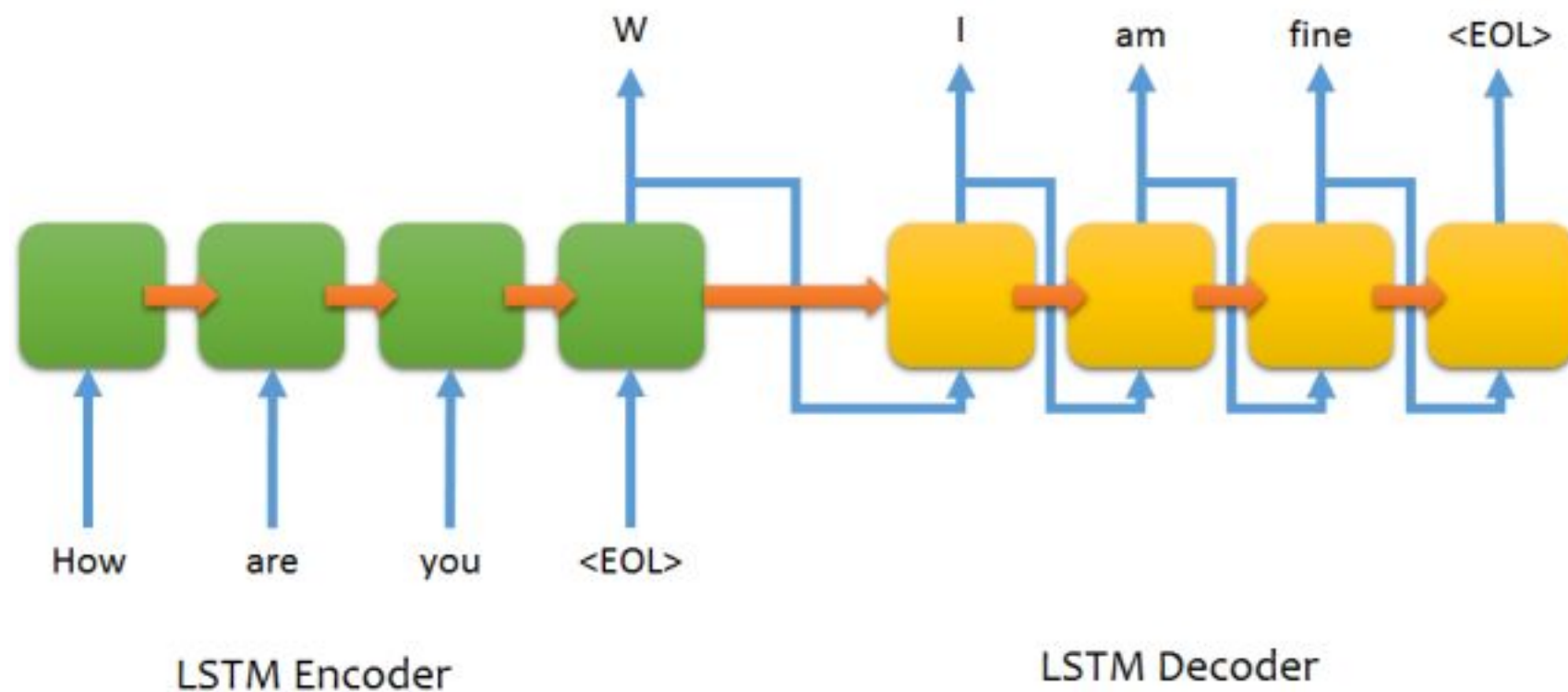
DeepFace Architecture



Yaniv Taigman, etc (Facebook) . [DeepFace: Closing the Gap to Human-Level Performance in Face Verification](#), CVPR 2014

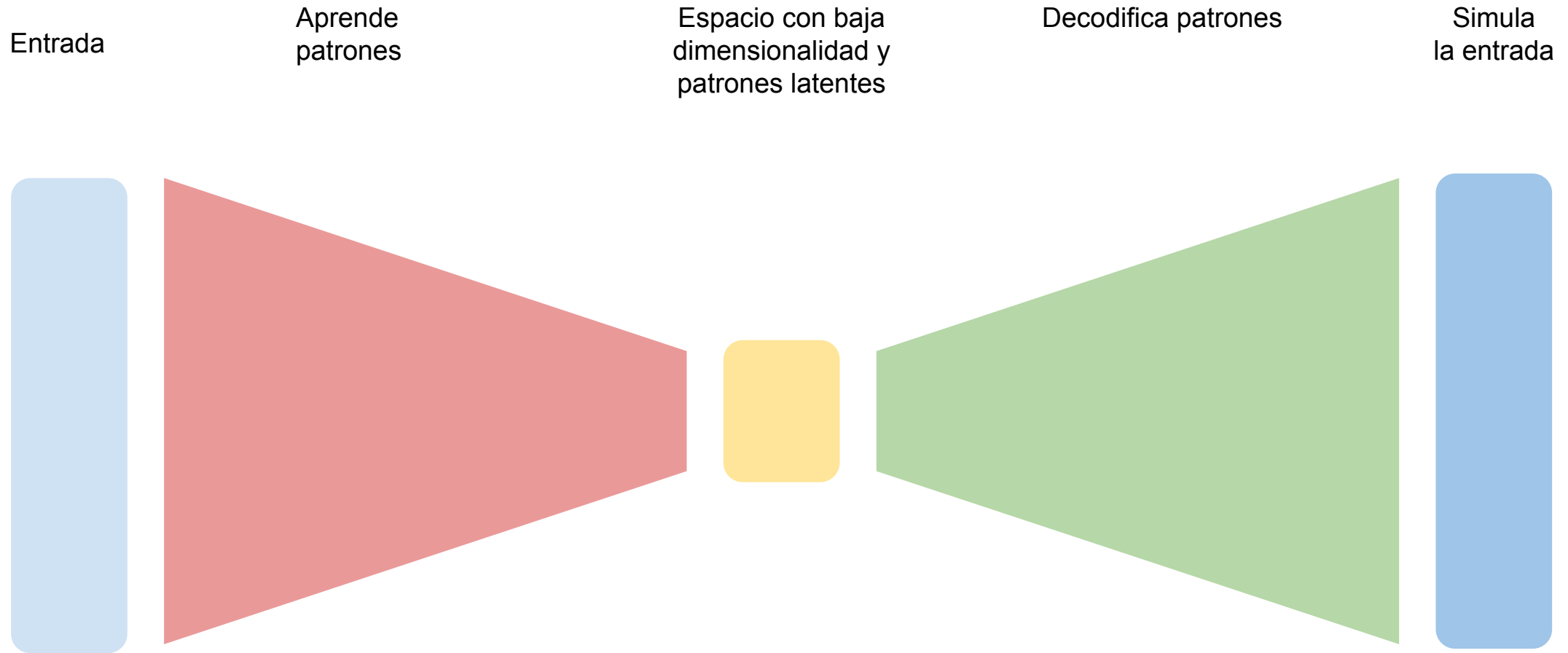


$$\text{evidence } y_i = \sum_j W_{i,j} x_j$$

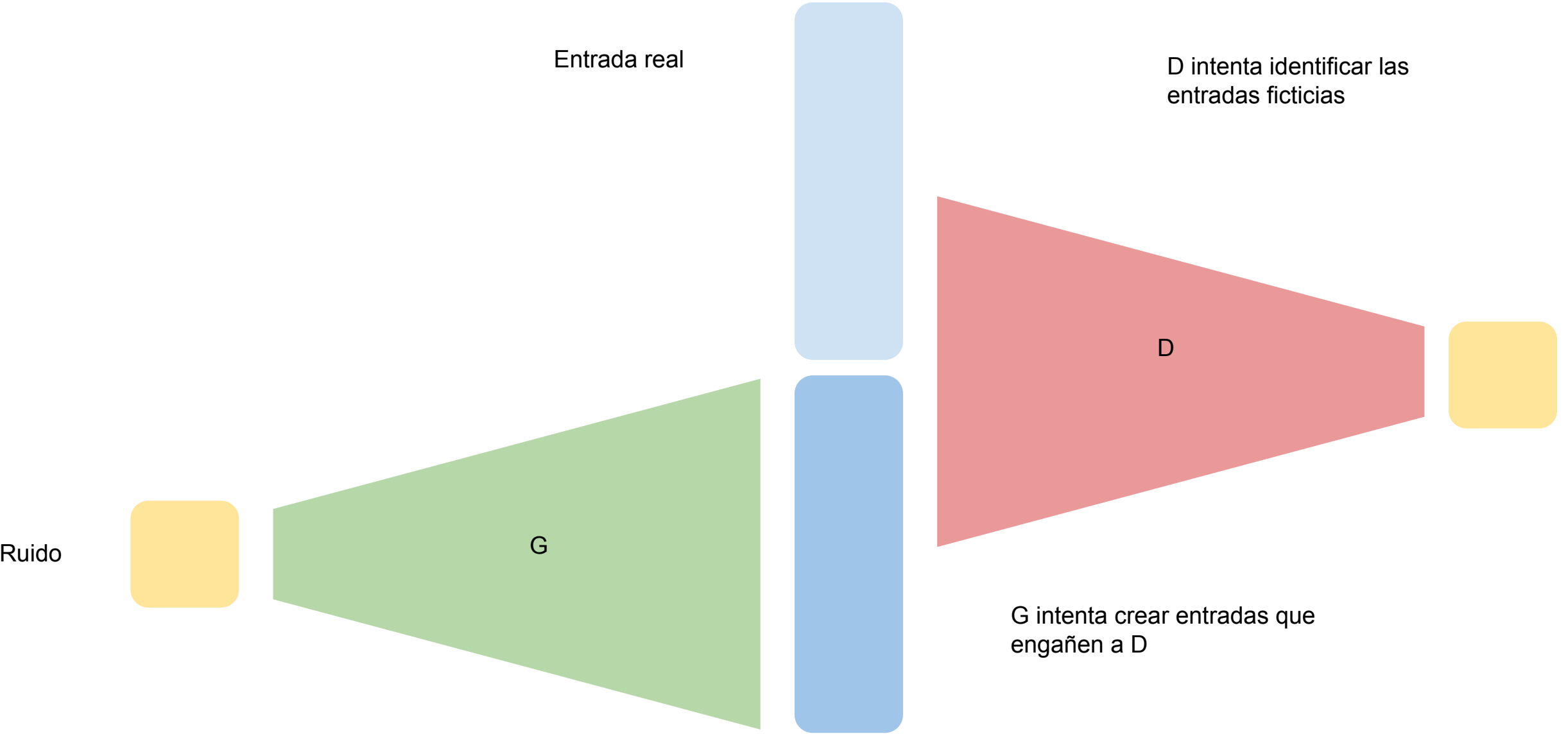


Autoencoders

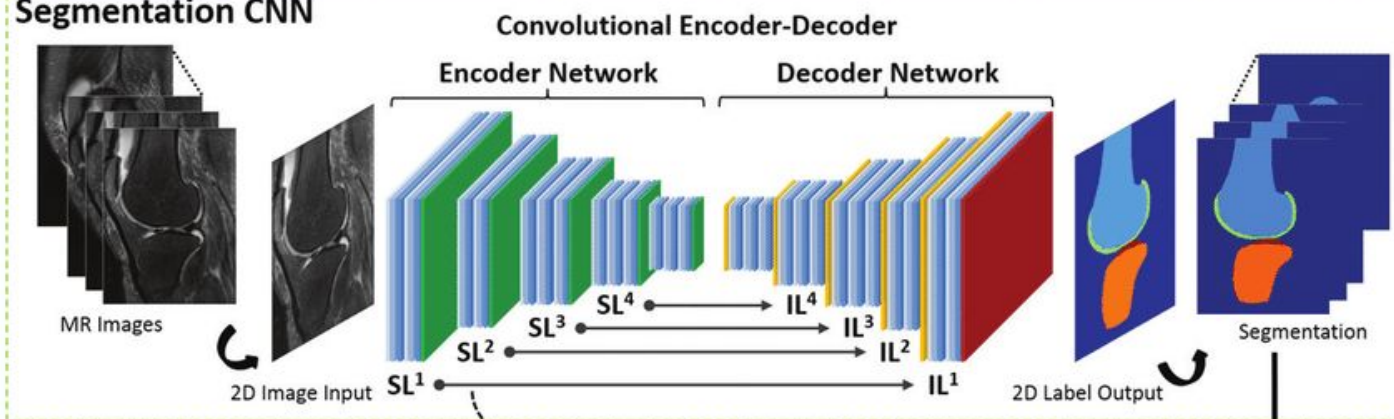
Permite reconstruir datos originales, es como una comprensión de los datos.



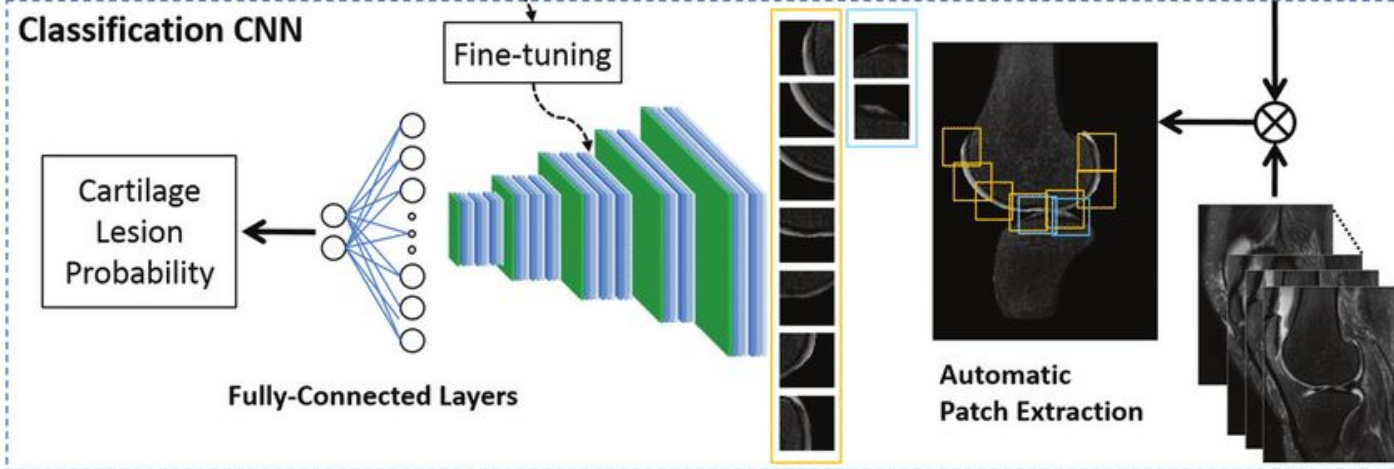
Generative Adversarial Networks



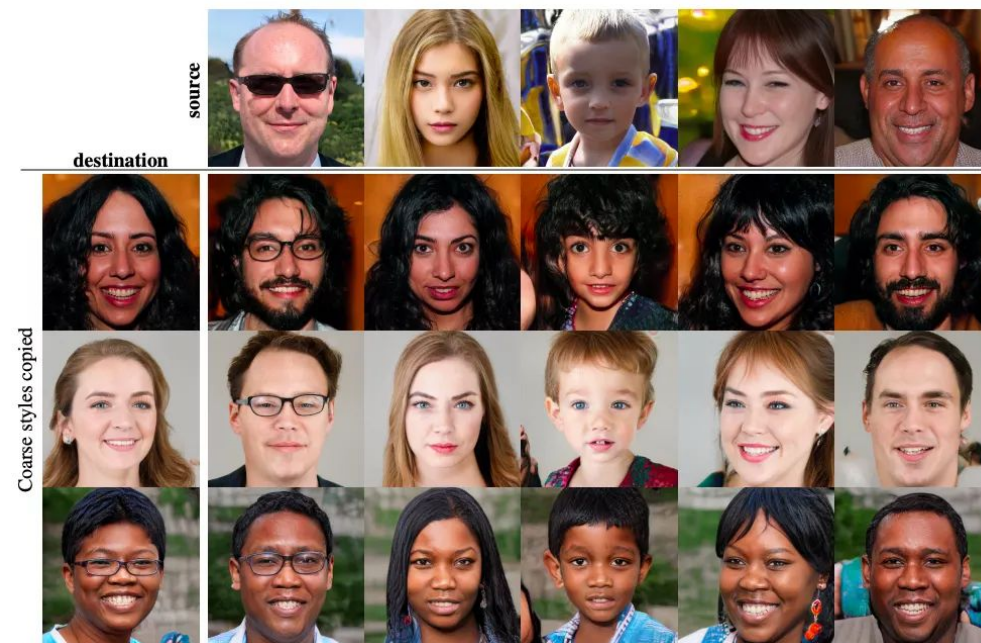
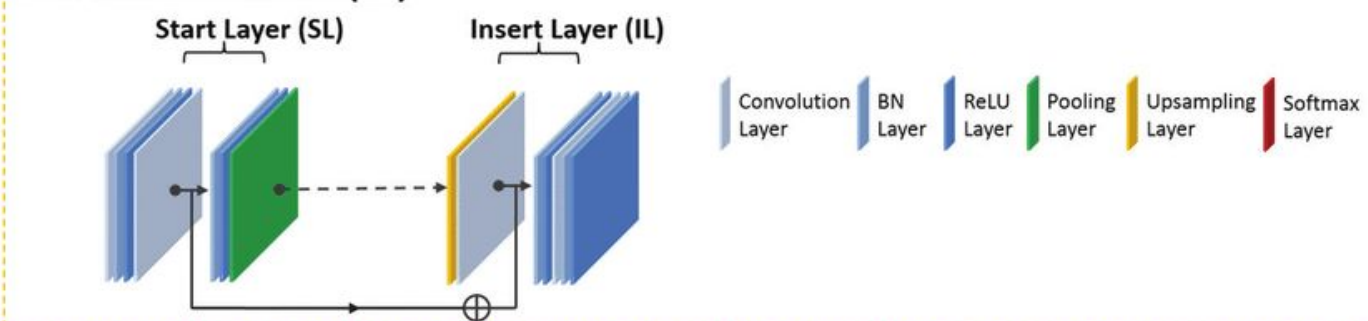
Segmentation CNN



Classification CNN



Shortcut Connection (SC)



Monet ↔ Photos



Monet → photo



photo → Monet

Zebras ↔ Horses



zebra → horse

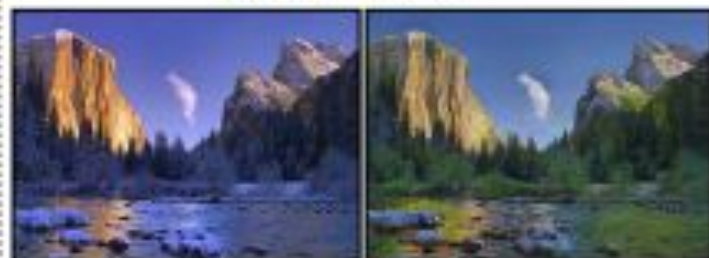


horse → zebra

Summer ↔ Winter



summer → winter



winter → summer



Photograph



Monet



Van Gogh



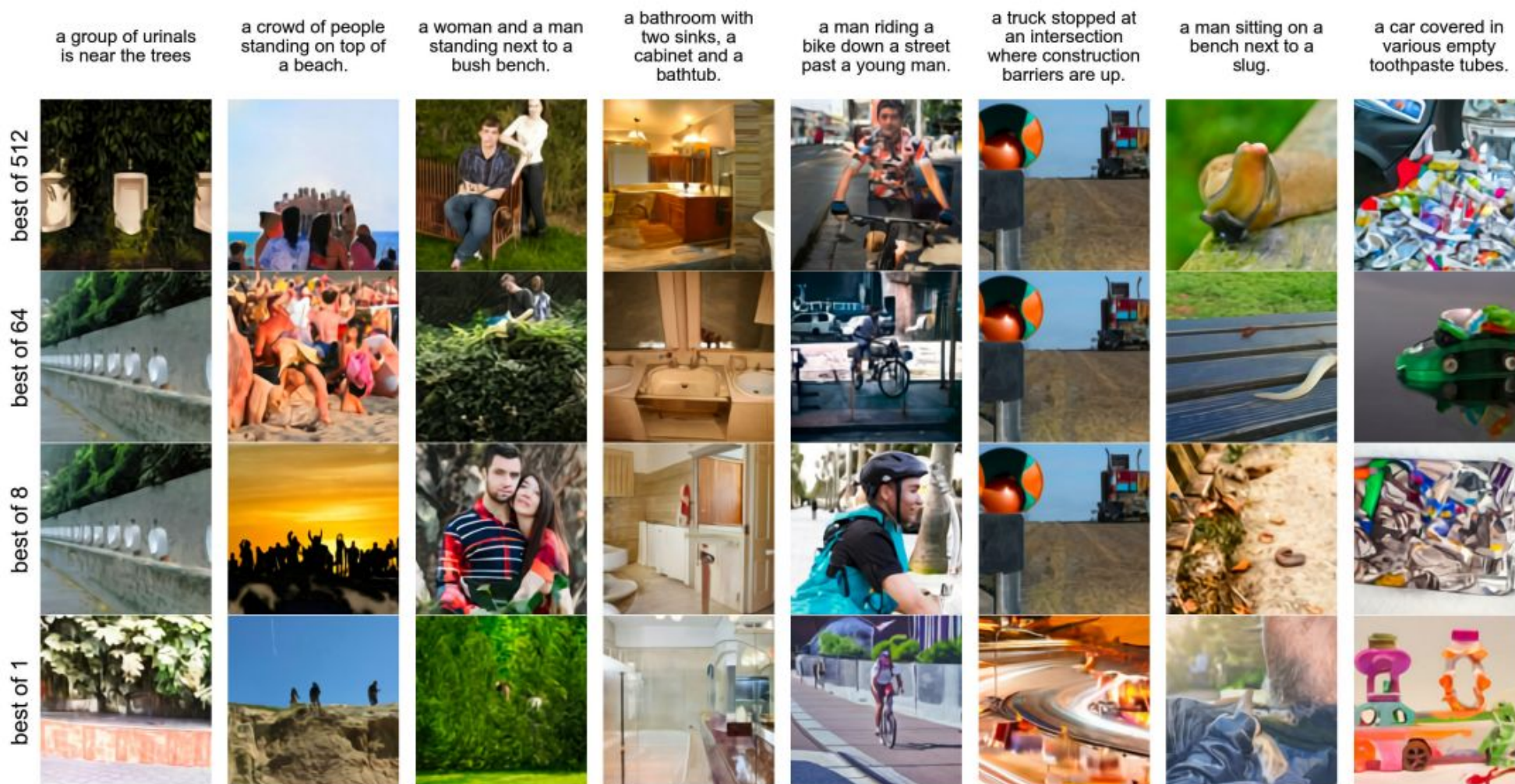
Cezanne



Ukiyo-e

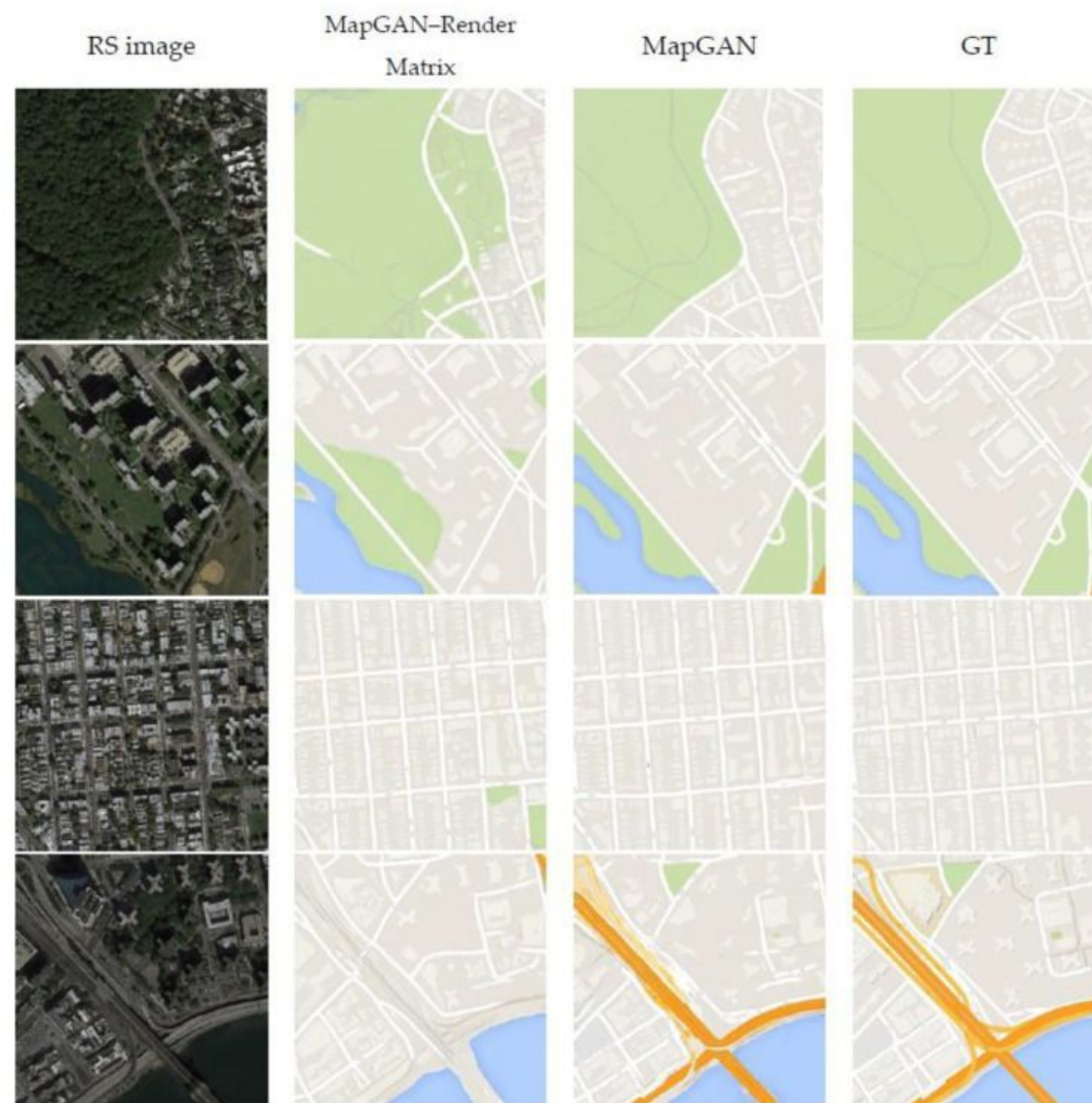
Text to image

Unpaired Image-to-Image Translation
using Cycle-Consistent Adversarial Network

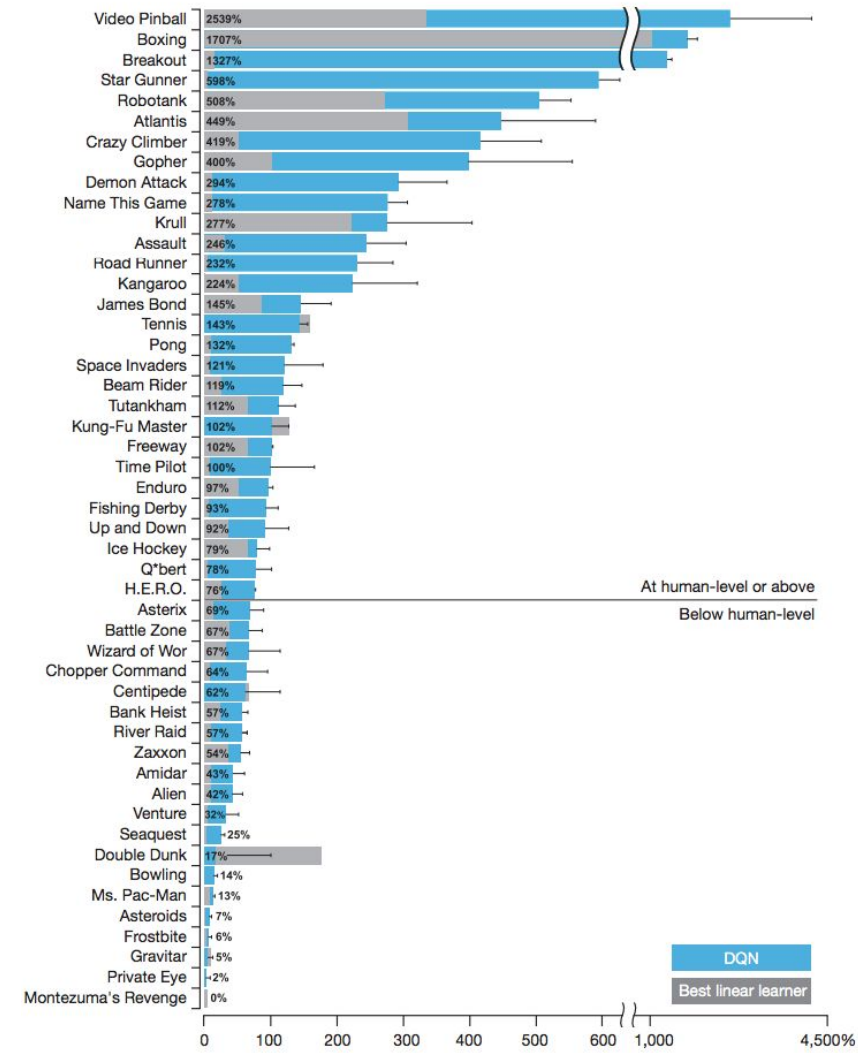
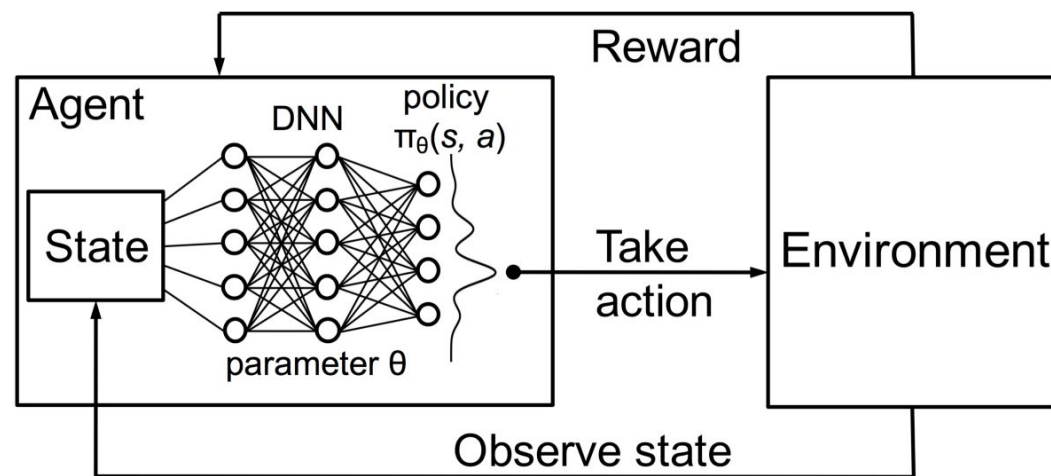
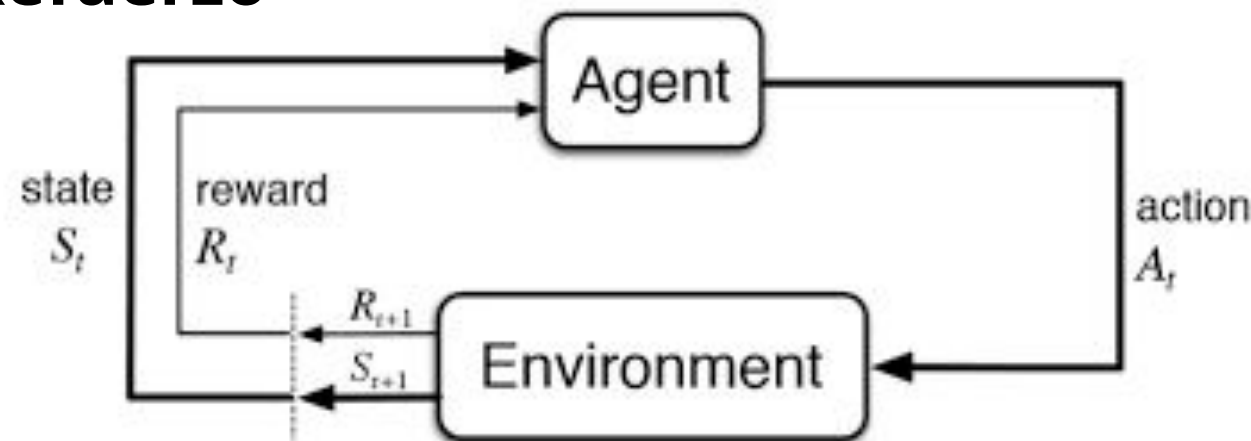


Creación de mapas

MapGAN: An Intelligent Generation Model for Network Tile Maps



Refuerzo



Atención

Efecto de realzar algunos patrones de los datos de entrada y permite transferir información entre redes neuronales

Attention Is All You Need

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Niki Parmar*
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Jakob Uszkoreit*
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Abstract

The dominant sequence transduction models are based on convolutional neural networks that include an encoder performing models also connect the encoder and decoder mechanism. We propose a new simple network architecture based solely on attention mechanisms, dispensing with recurrently. Experiments on two machine translation tasks are superior in quality while being more parallelizable and less time to train. Our model achieves 28.4 BLEU on the to-German translation task, improving over the existing ensembles, by over 2 BLEU. On the WMT 2014 English-to-French translation task, our model establishes a new single-model state-of-the-art training for 3.5 days on eight GPUs, a small fraction of the best models from the literature. We show that the Transformer can be applied to other tasks by applying it successfully to English constituency parsing and limited training data.

1 Introduction

Recurrent neural networks, long short-term memory [13] and generative models, have been firmly established as state of the art approaches for sequence transduction.

*Equal contribution. Listing order is random. Jakob proposed replacing the effort to evaluate this idea. Ashish, with Illia, designed and implemented the model. Illia has been crucially involved in every aspect of this work. Noam proposed the attention mechanism and the parameter-free position representation and became the main contributor to the implementation.

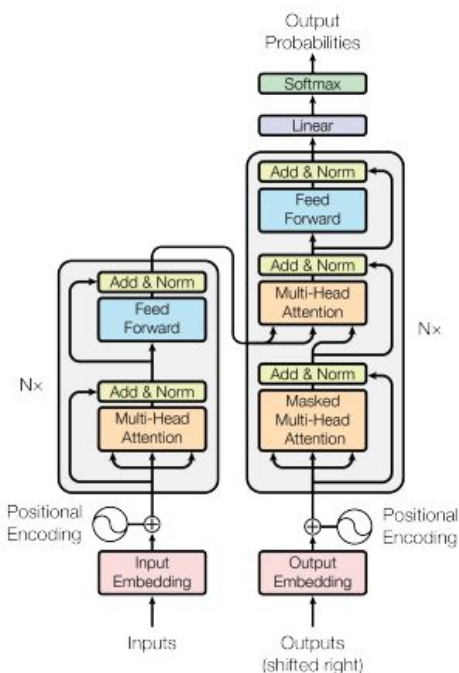
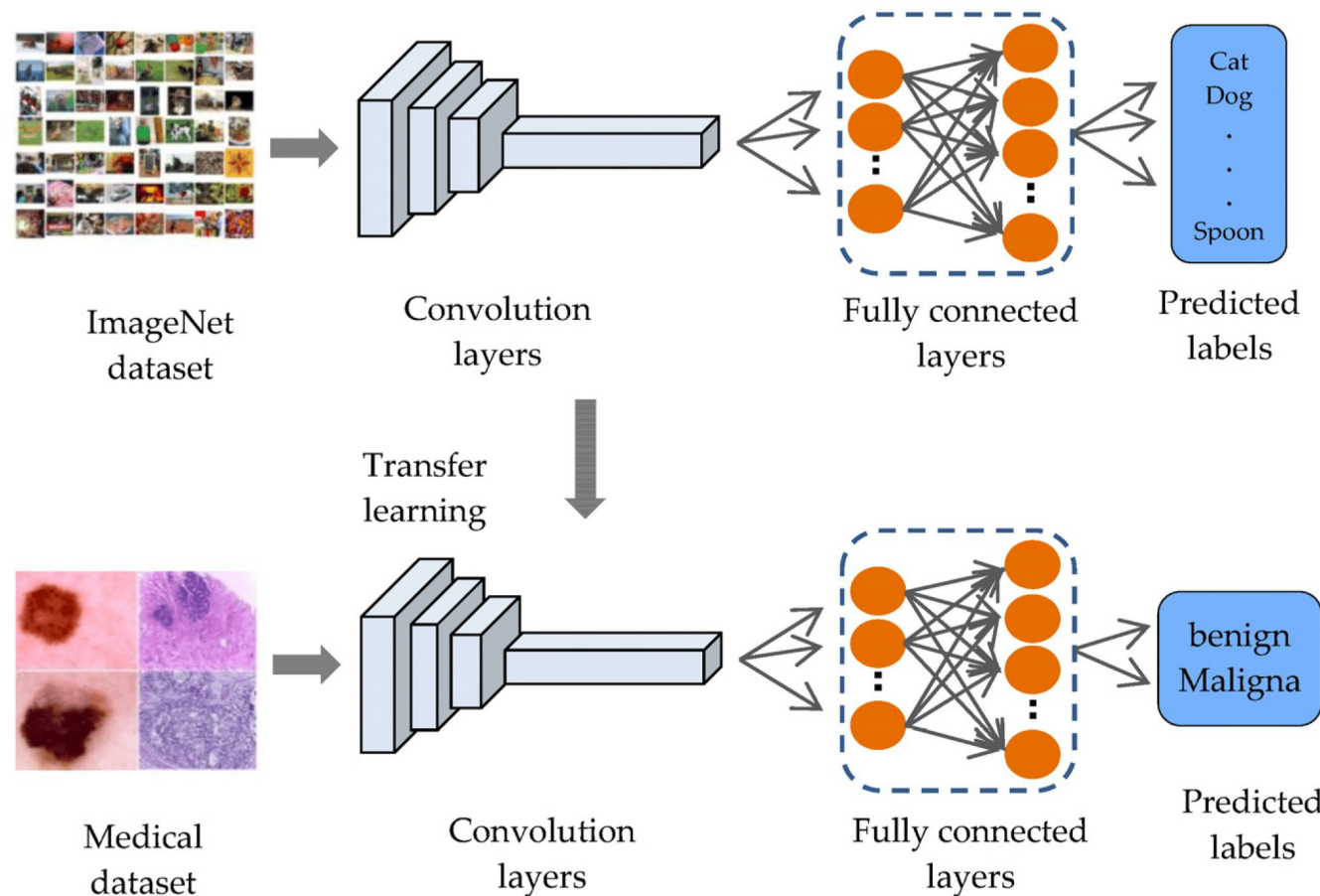
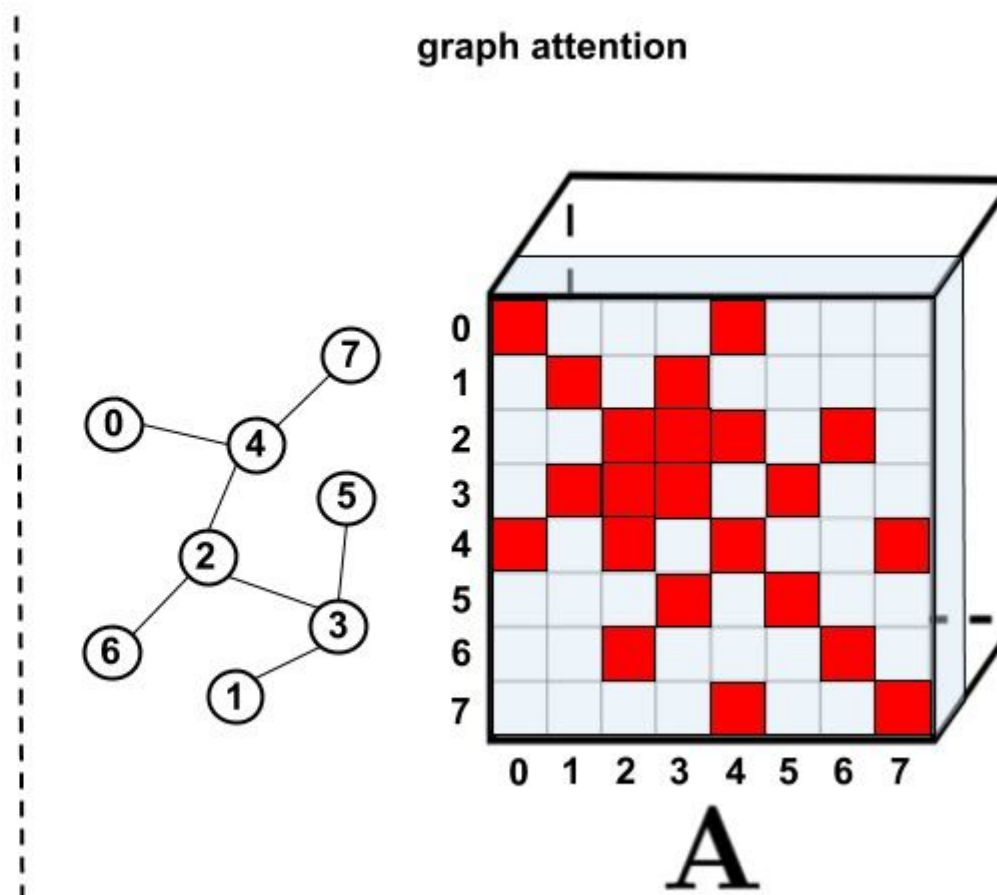
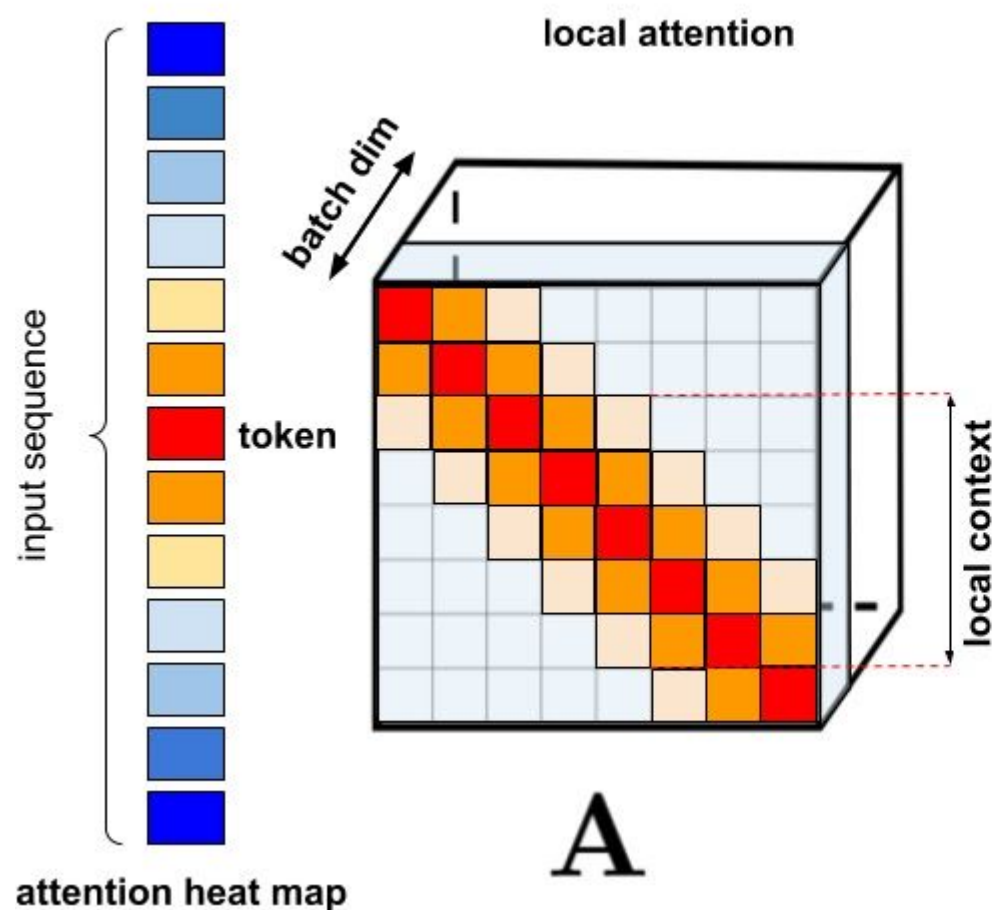


Figure 1: The Transformer - model architecture.



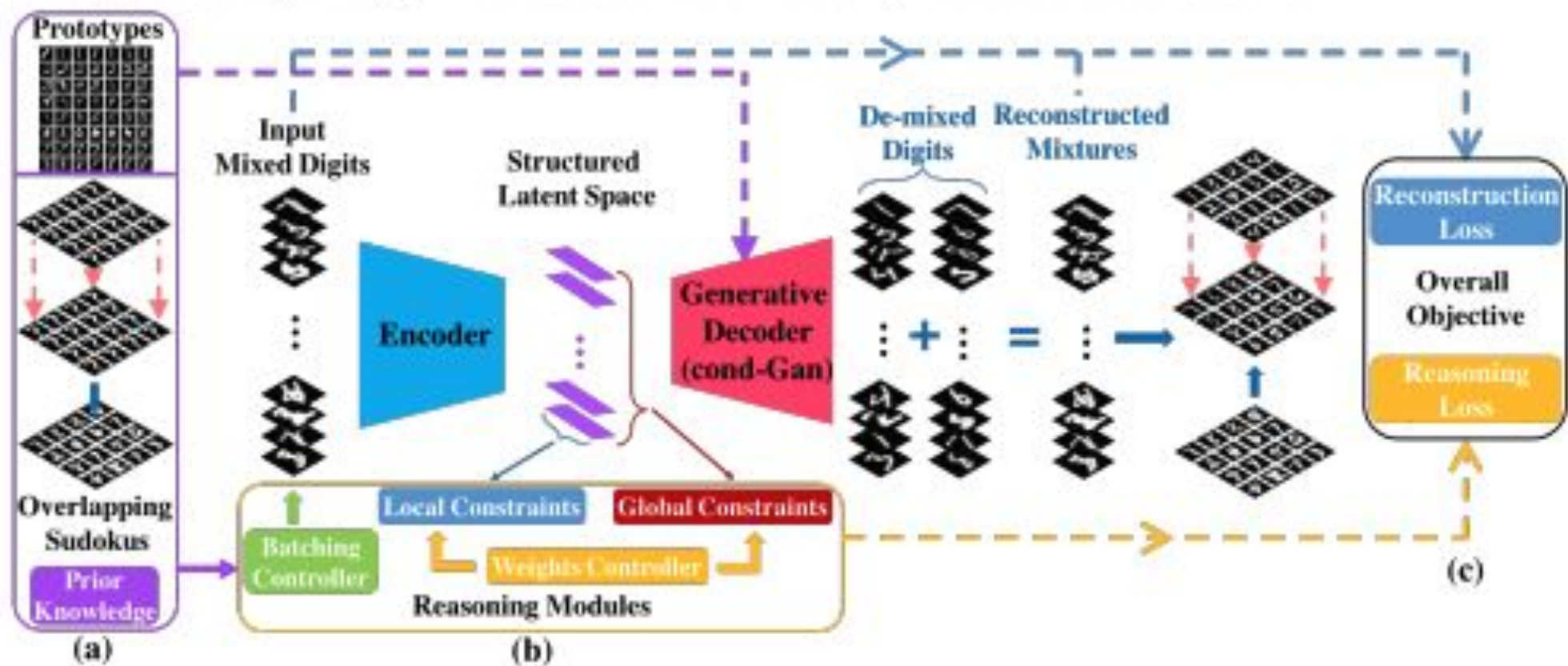
Atención

Efecto de realzar algunos patrones de los datos de entrada y permite transferir información entre redes neuronales



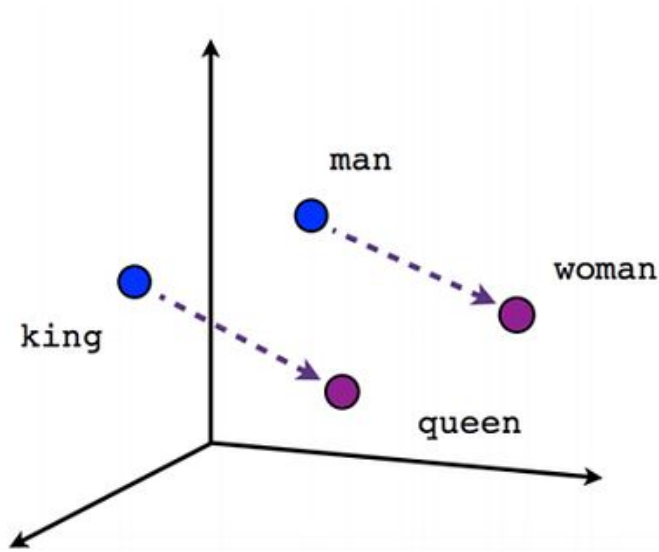
Razonamiento

Capacidad de resolver problemas complejos mediante lógica, inducción o deducción.

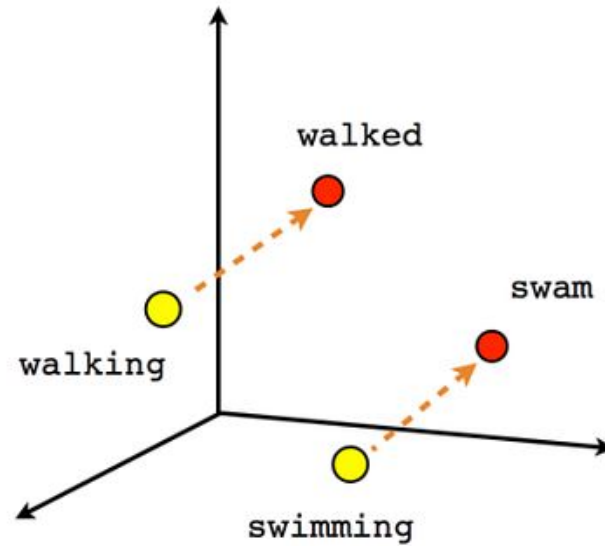


Razonamiento

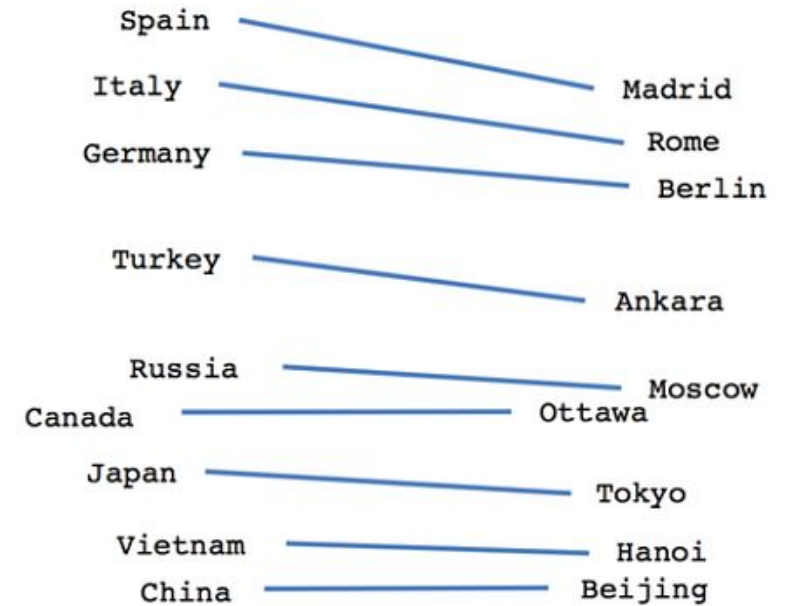
Capacidad de resolver problemas complejos mediante lógica, inducción o deducción.



Male-Female



Verb tense

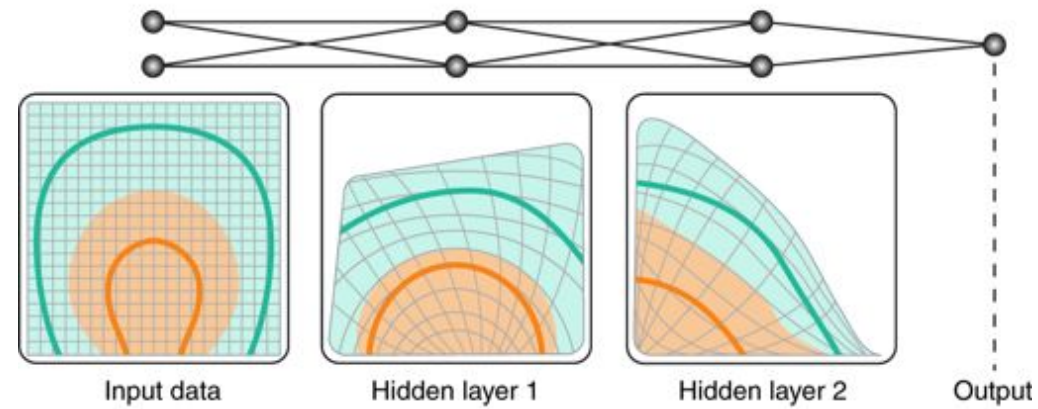


Country-Capital

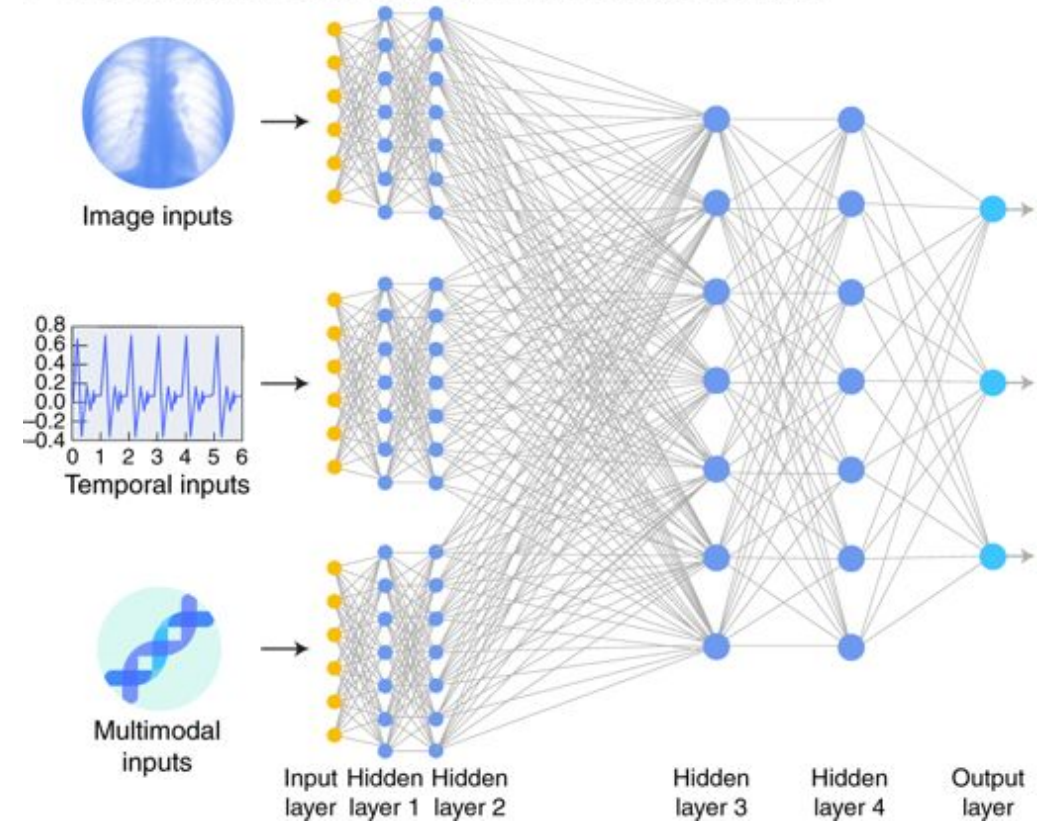
Explicatividad vs predictibilidad

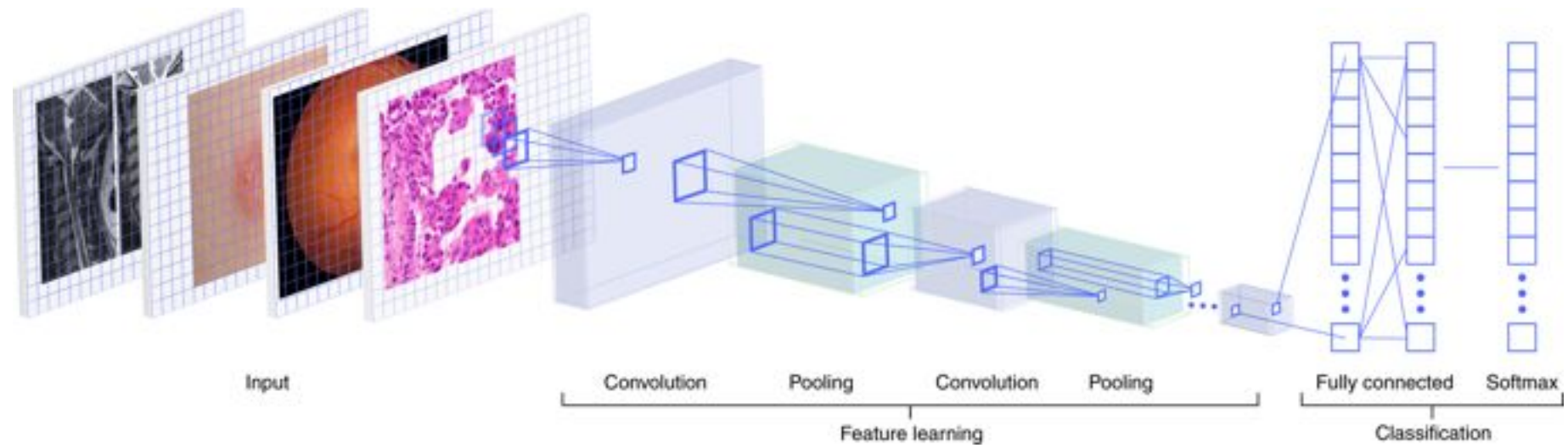
Casos de uso en salud

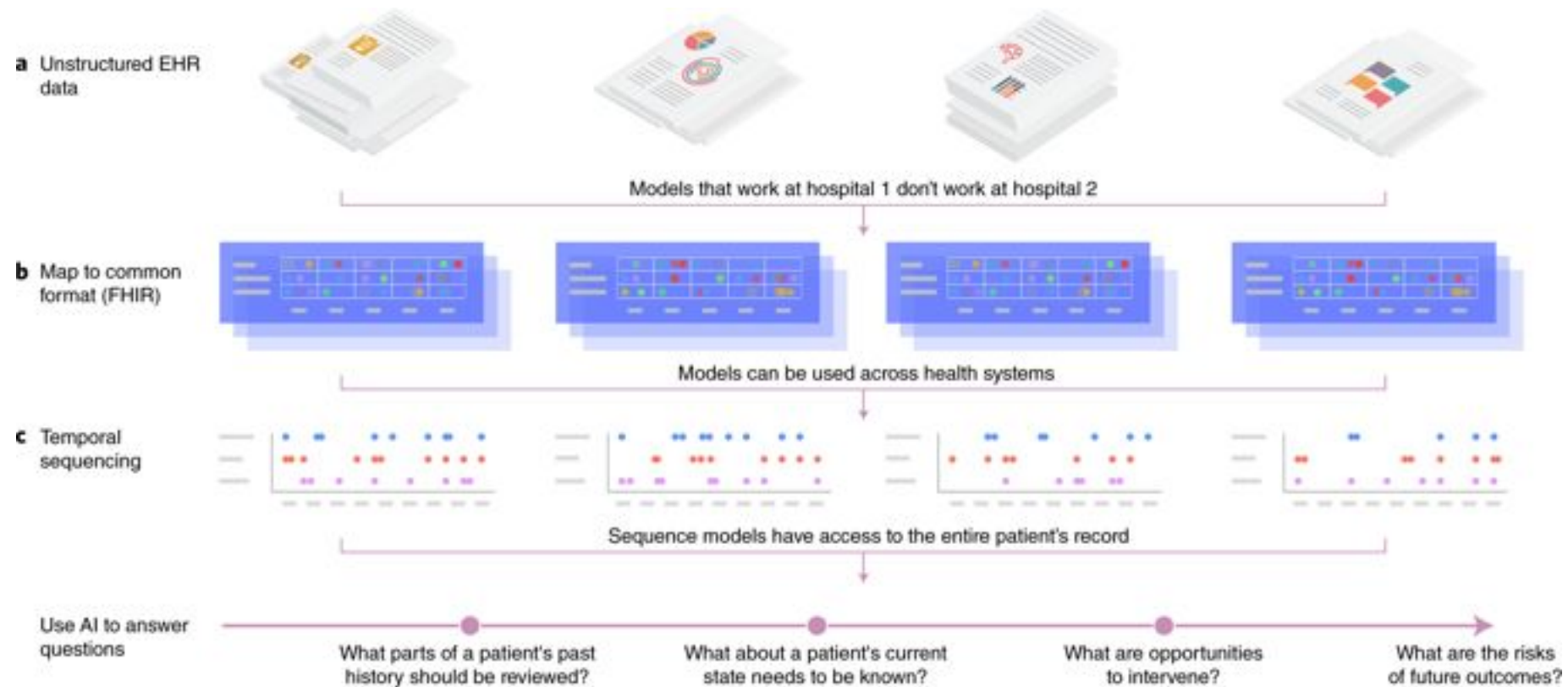
a Neural network layers make data linearly separable

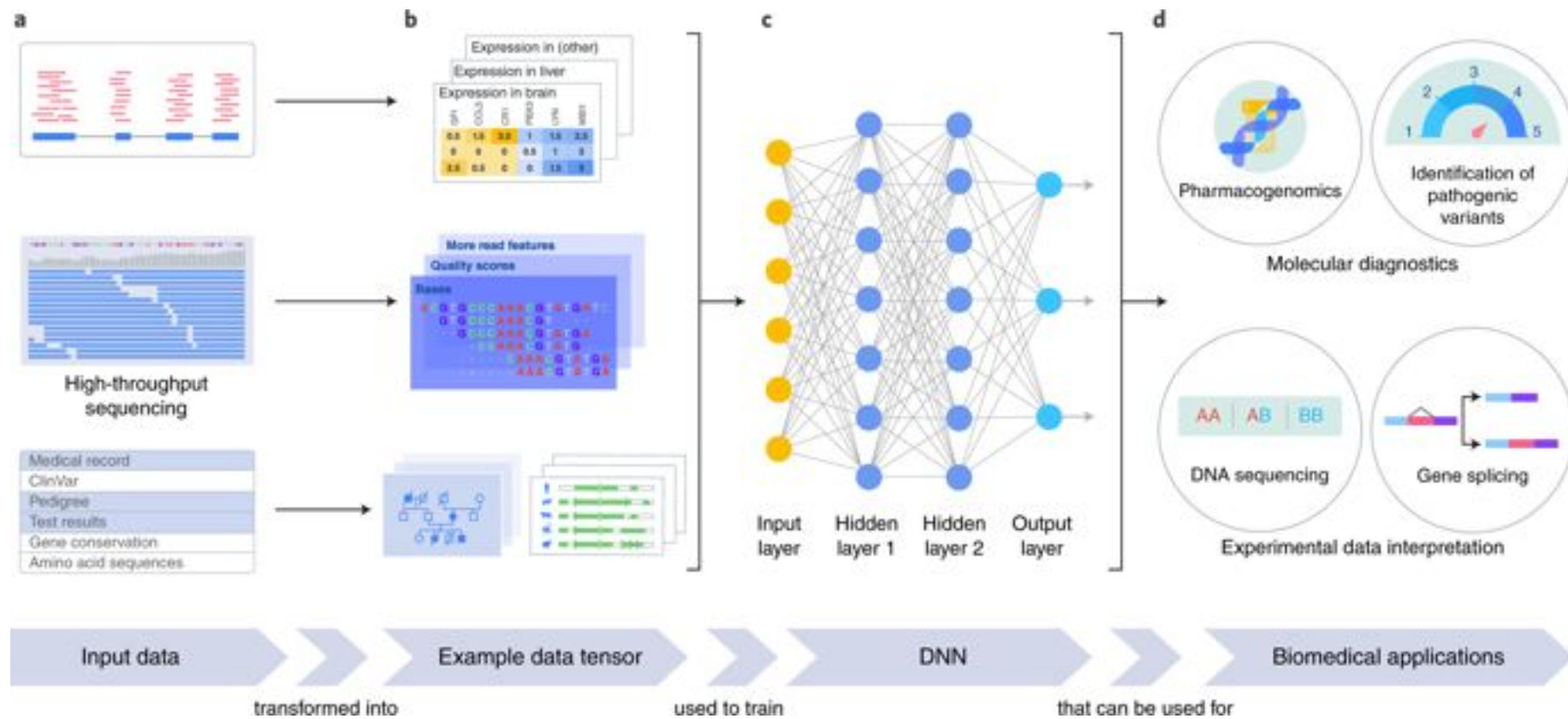


b Deep learning can featurize and learn from a variety of data types

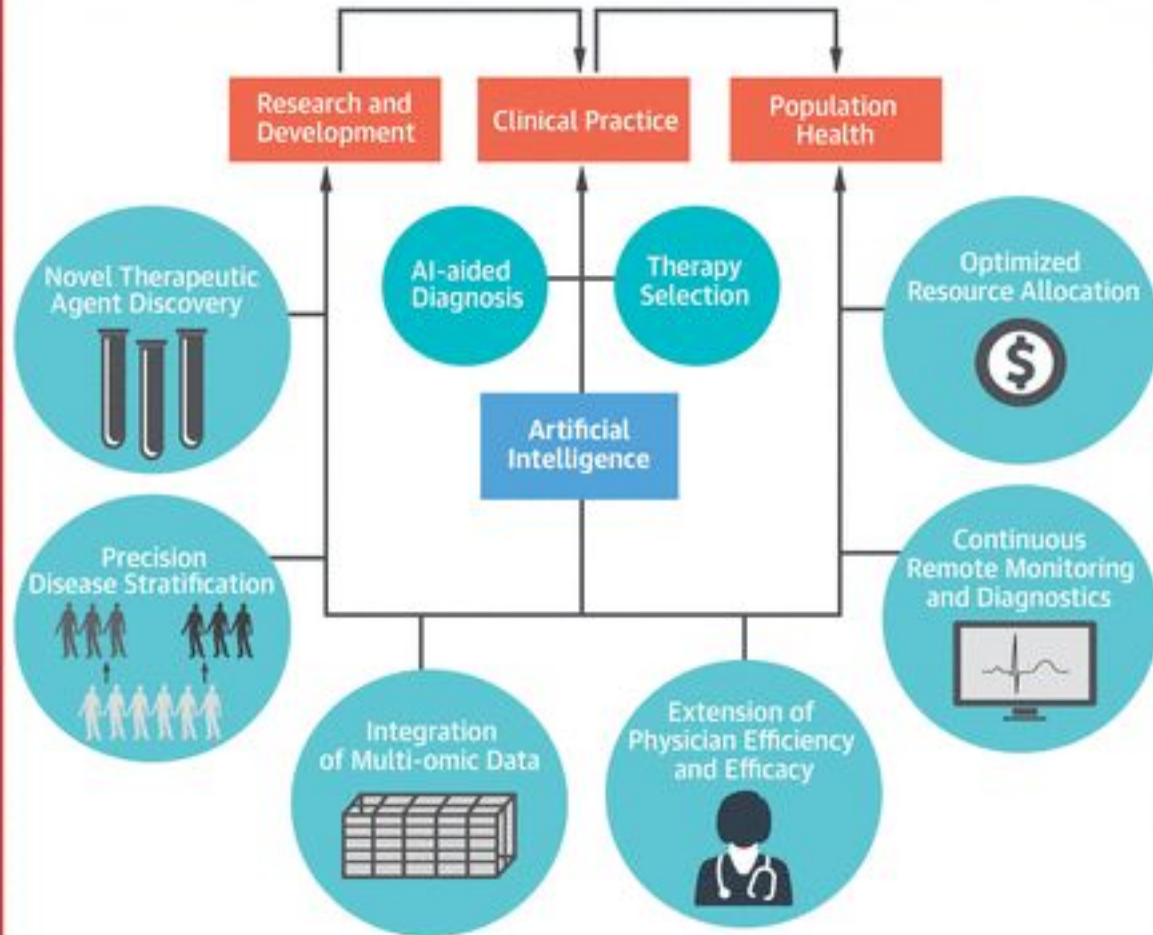






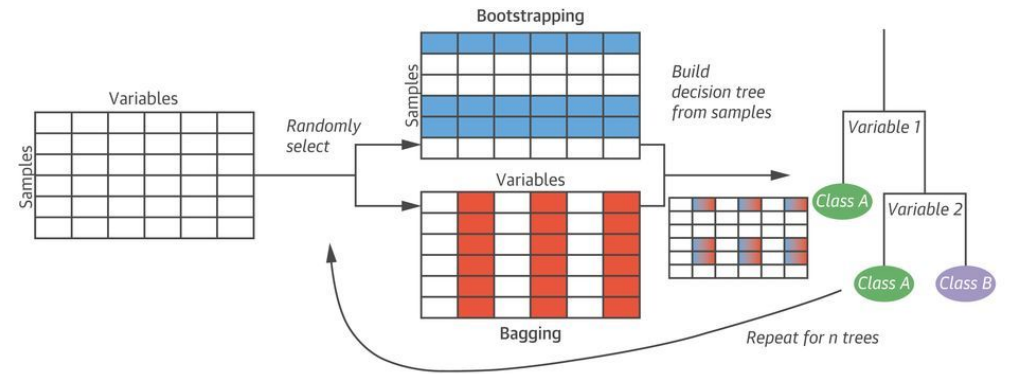


CENTRAL ILLUSTRATION: Role of Artificial Intelligence in Cardiovascular Medicine

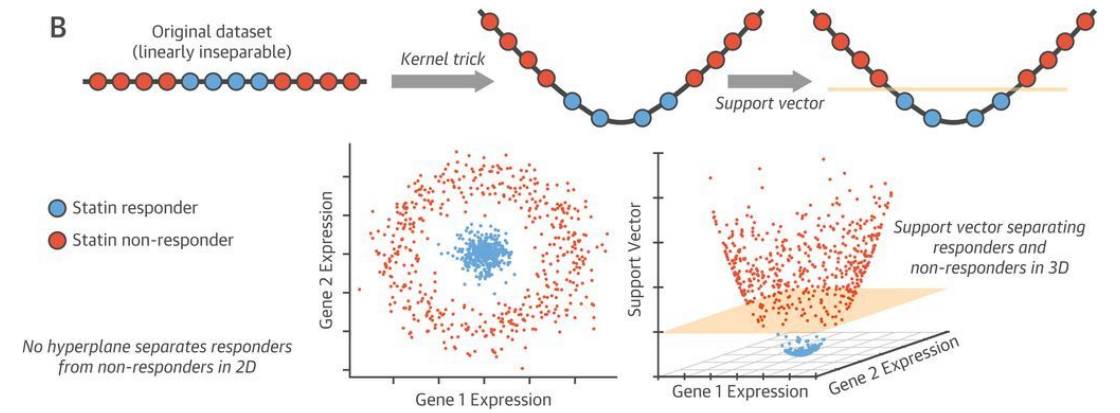


Johnson, K.W. et al. J Am Coll Cardiol. 2018;71(23):2668-79.

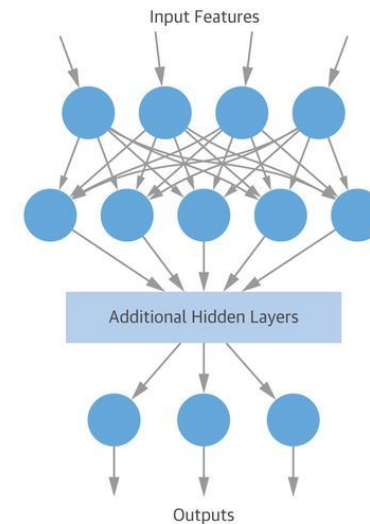
A



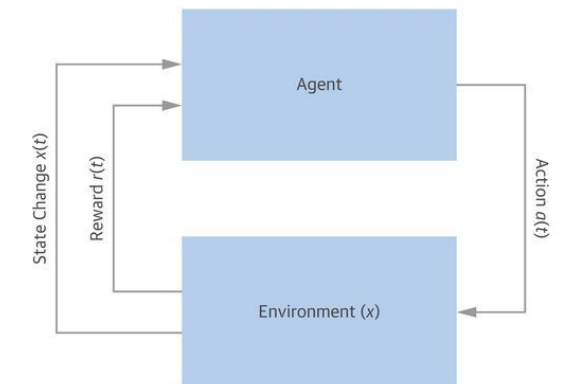
B

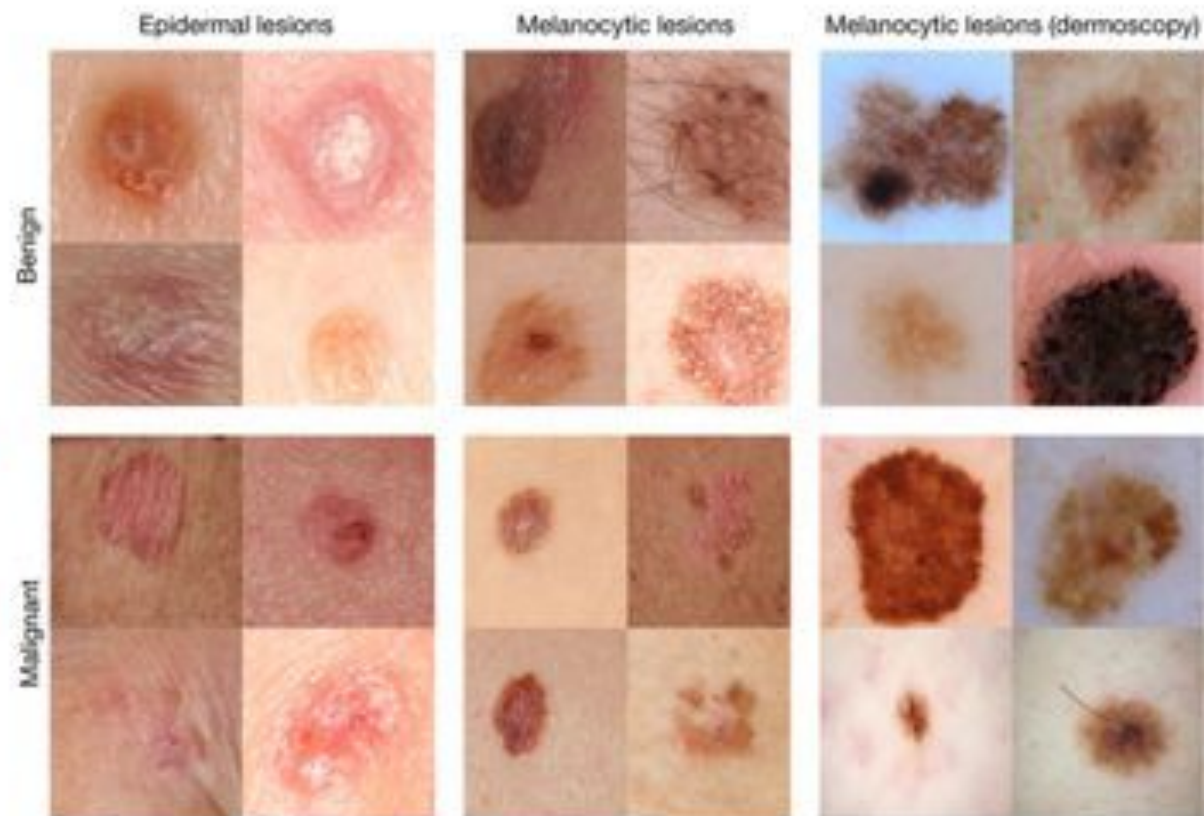


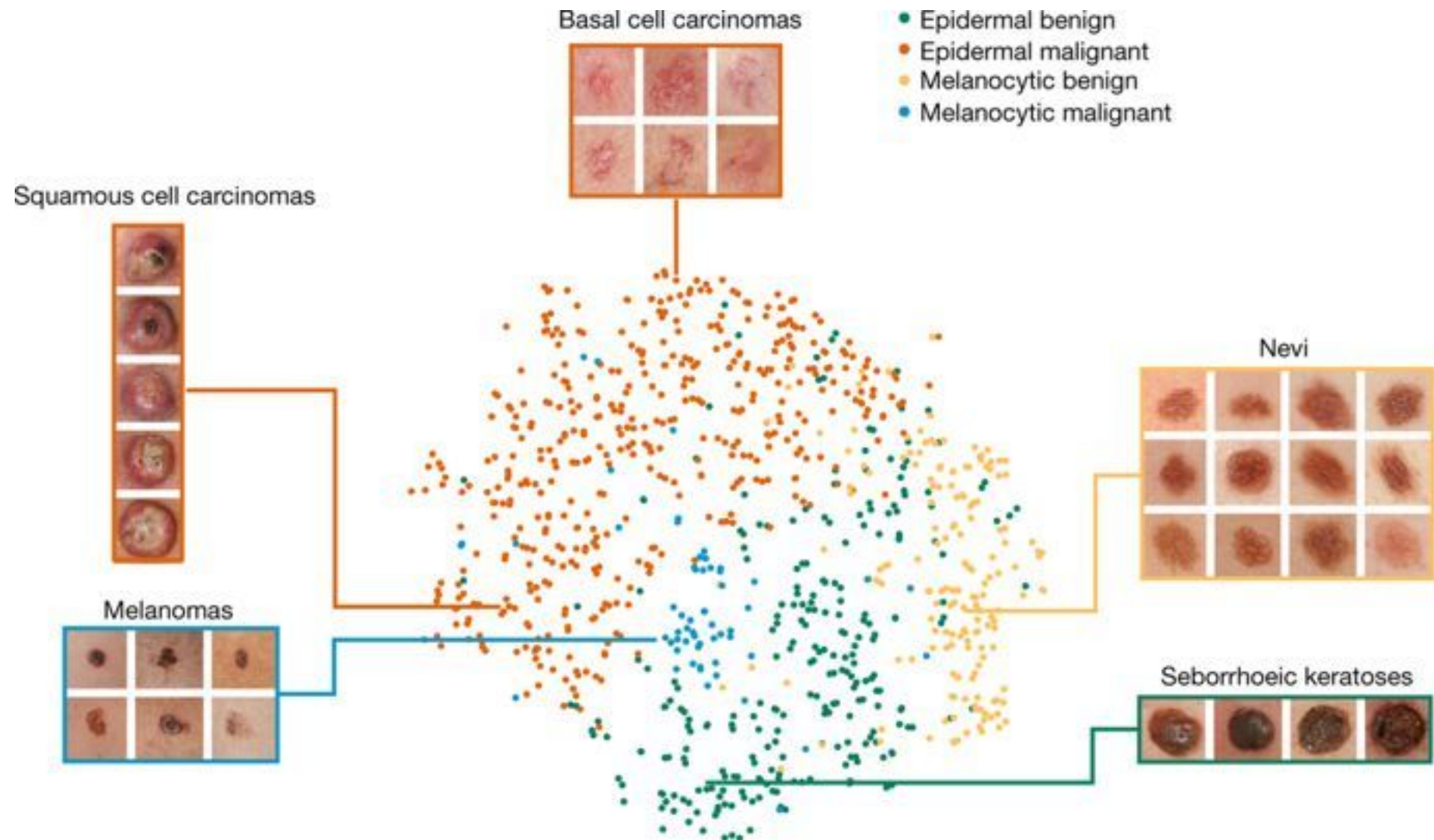
C

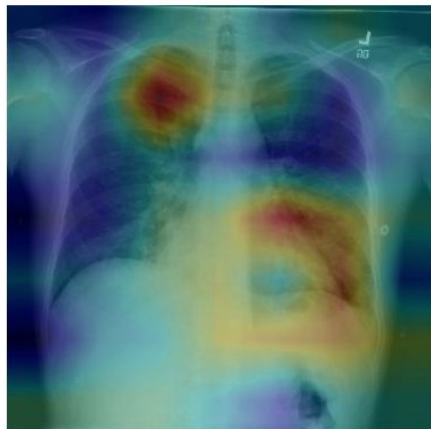


D

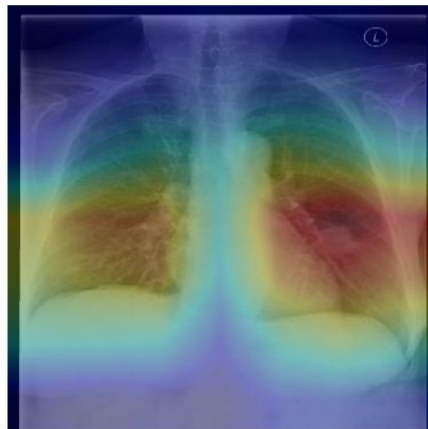


a**b**

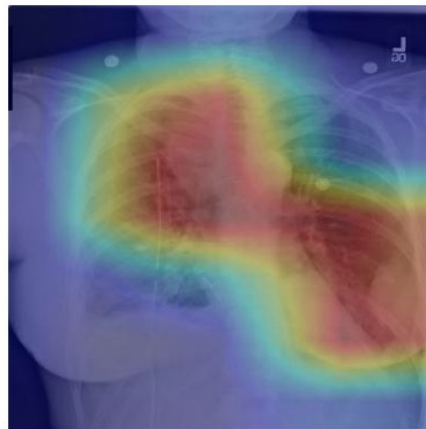




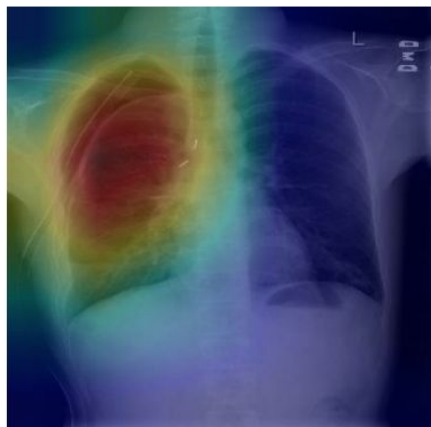
(a) Patient with multifocal community acquired pneumonia. The model correctly detects the airspace disease in the left lower and right upper lobes to arrive at the pneumonia diagnosis.



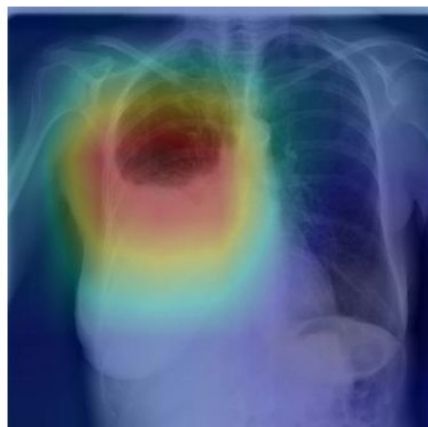
(b) Patient with a left lung nodule. The model identifies the left lower lobe lung nodule and correctly classifies the pathology.



(c) Patient with primary lung malignancy and two large masses, one in the left lower lobe and one in the right upper lobe adjacent to the mediastinum. The model correctly identifies both masses in the X-ray.



(d) Patient with a right-sided pneumothorax and chest tube. The model detects the abnormal lung to correctly predict the presence of pneumothorax (collapsed lung).



(e) Patient with a large right pleural effusion (fluid in the pleural space). The model correctly labels the effusion and focuses on the right lower chest.



(f) Patient with congestive heart failure and cardiomegaly (enlarged heart). The model correctly identifies the enlarged cardiac silhouette.



Input
Chest X-Ray Image

CheXNet
121-layer CNN

Output
Pneumonia Positive (85%)



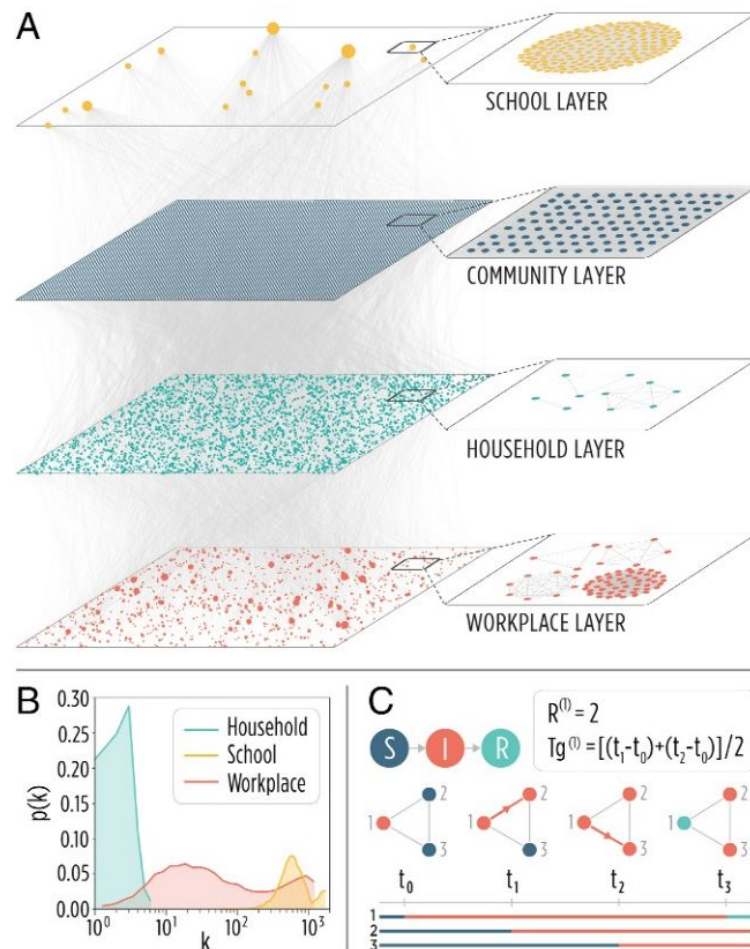


Fig. 1. Model structure. (A) Visualization of the multiplex network representing a subsample of 10,000 individuals of the synthetic population. Note that the community layer is a complete graph, although not all edges are visible for the sake of readability of the illustration. (B) Degree distributions in the school, household, and workplace layers. (C) Schematic representation of the infection transmission model along with examples of the computation of individual reproduction number and generation time for the simulated transmission chains. I, infectious; R, removed; S, susceptible.

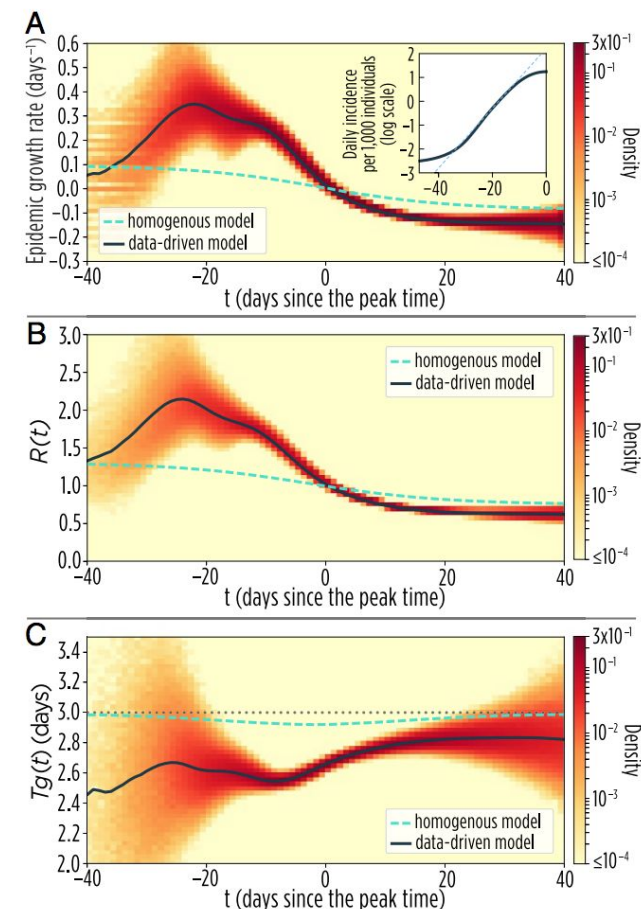
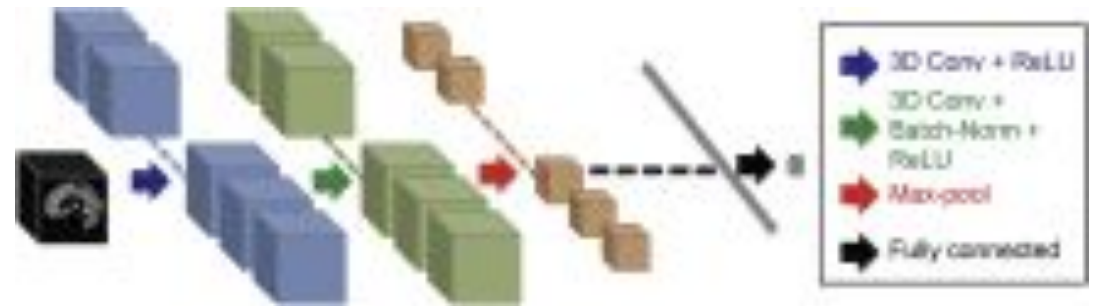
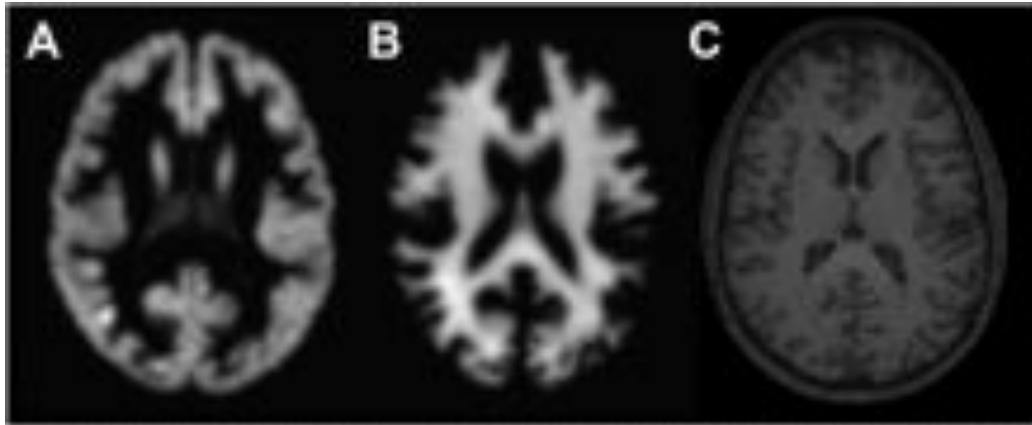
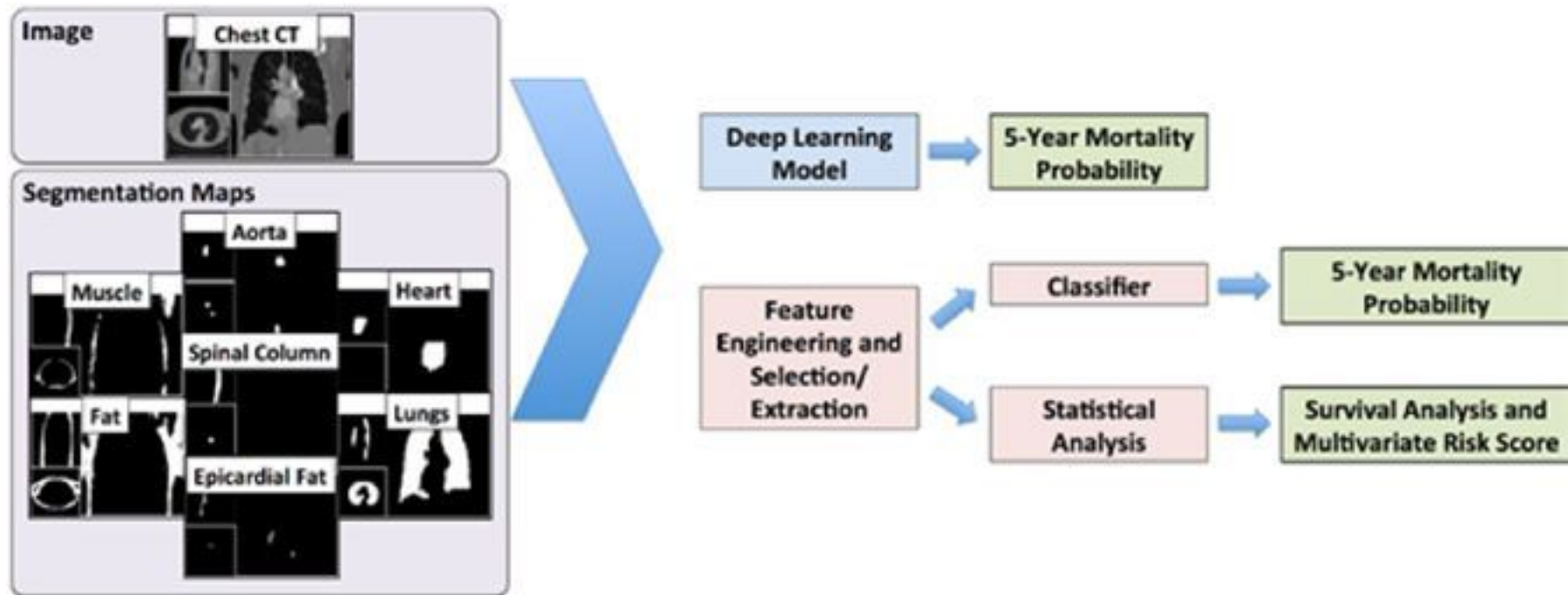


Fig. 2. Fundamental epidemiological indicators. (A) Mean daily exponential epidemic growth rate, r , over time of the data-driven and homogeneous models. The colored area shows the density distribution of $r(t)$ values obtained in the single realizations of the data-driven model. Results are based on 50,000 realizations of each model. Results are aligned at the epidemic peak, which corresponds to time $t = 0$. Inset shows the logarithm of the mean daily incidence of new influenza infections over time, which does not follow a linear trend. (B) Mean $R(t)$ of data-driven and homogeneous models. The colored area shows the density distribution of $R(t)$ values obtained in the single realizations of the data-driven model. (C) The three lines represent the mean $Tg(t)$ of data-driven and homogeneous models. The colored area shows the density distribution of $Tg(t)$ values obtained in the single realizations of the data-driven model. The horizontal dotted





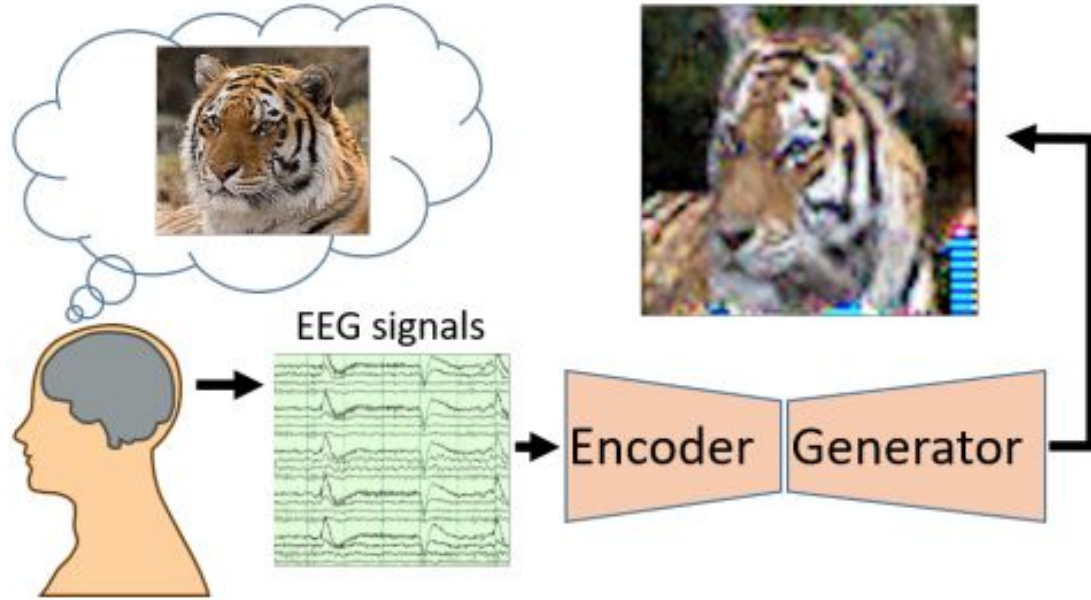
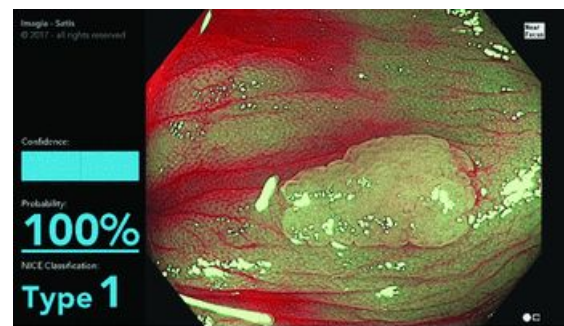
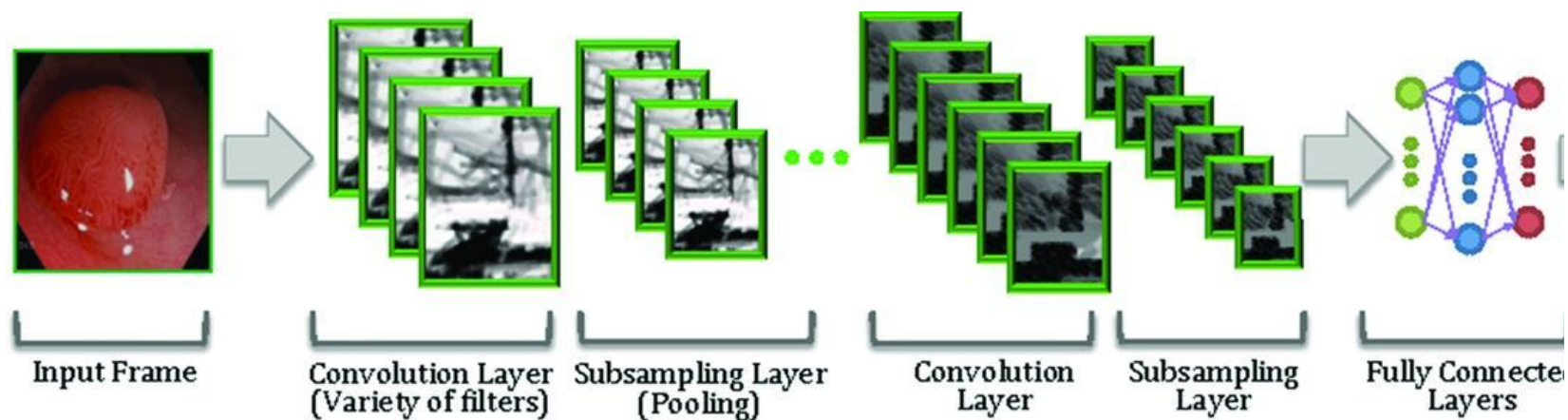
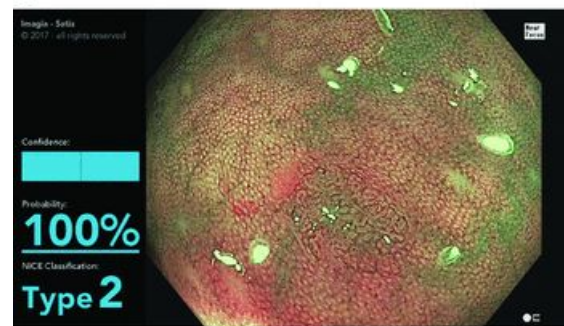


Figure 1: Overview of our proposed method where an EEG signal from brain is sent to the encoder and the encoded signal is used to generate a visualization corresponding to the captured EEG signal.





A



B

El CDC tiene una predicción ensamblada de todos los modelos que se han construido. Un gestor nunca puede confiar en el resultado de un solo modelo.

CDC COVID Data Tracker

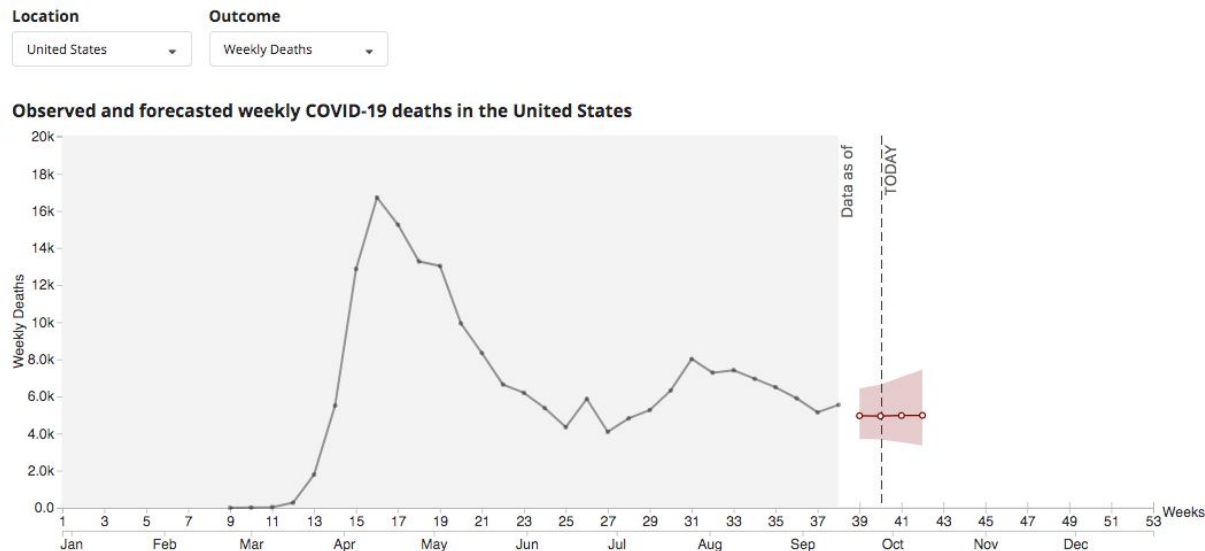
Maps, charts, and data provided by the CDC

Case Trends ▾ Laboratory ▾ Community Impact ▾ Unique Populations ▾ COVID-19 Home

Cases and Deaths by State | US and State Trends | Compare State Trends | Demographics | Trends by Population Factors | Cases and Deaths by County | Forecasting | Trends in ED Visits

United States Forecasting

Updated September 24, 2020. Observed and forecasted cumulative and weekly reported COVID-19 deaths as of September 21, 2020.
Use the interactive tool below to explore observed and forecasted cumulative and weekly COVID-19 deaths in the United States. Click on past weeks to see previous forecasts or use the arrows in the legend. For more information, please visit the CDC COVID-19 Forecasting Page.



Por eso el CDC ha recopilado todas las predicciones colaborando con terceros.

CDC COVID Data Tracker

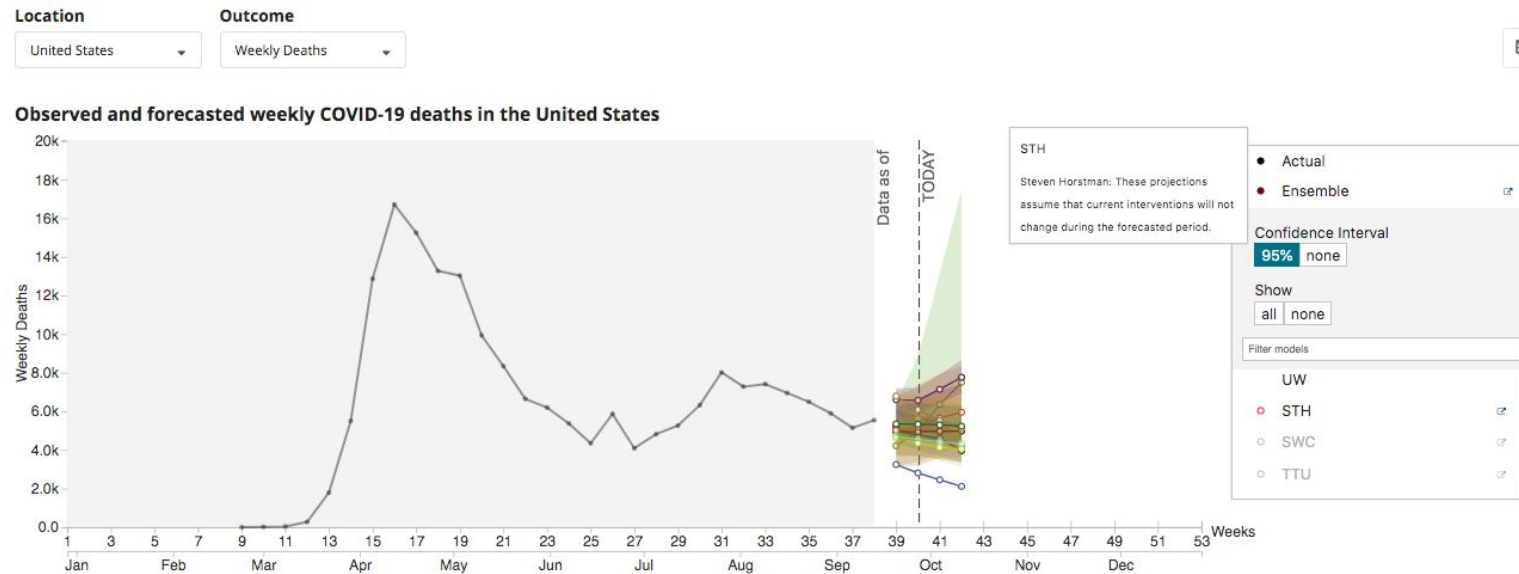
Maps, charts, and data provided by the CDC

Case Trends ▾ Laboratory ▾ Community Impact ▾ Unique Populations ▾ COVID-19 Home

[Cases and Deaths by State](#) | [US and State Trends](#) | [Compare State Trends](#) | [Demographics](#) | [Trends by Population Factors](#) | [Cases and Deaths by County](#) | [Forecasting](#) | [Trends in ED Visits](#)

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Productos basados en datos y
modelos de negocio

¿Productos basados en datos?

Son aquellos en los que el core del negocio son los datos que manejan y permiten.

- ▶ Tomar decisiones
- ▶ Reducción de la incertidumbre.
- ▶ Automatizar o optimizar un proceso

¿Productos basados en datos?

Ibai @IbaiLlanos · 19h
Alguien me explica qué ha sido esto



0:01

1,394 3,194 56.2K 3.4M

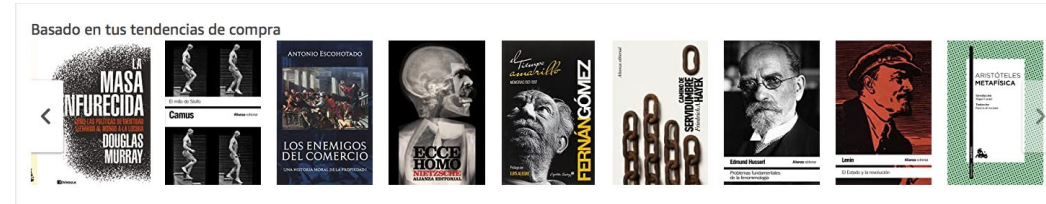
Push notifications
Never miss what's happening on Twitter by enabling push notifications

Turn on notifications

NEAR Protocol @NEARProtocol
200M in total transactions on #NEAR. Probably nothing

Follow us!

NEAR Protocol @NEARProtocol
NEAR is the Blockchain Operating System.
<https://t.co/RUXPdKERJf>



	↑ 26° ↓ 13°	↑ 30° ↓ 16°	↑ 32° ↓ 19°	↑ 29° ↓ 19°	↑ 27° ↓ 16°	↑ 29° ↓ 15°	↑ 31° ↓ 17°
08:00	14°	17°	19°	19°	16°	16°	17°
14:00	26°	30°	32°	29°	27°	29°	31°
20:00	26°	30°	30°	27°	25°	28°	30°
Lluvias	0 mm	0 mm	0 mm	0 mm	0 mm	0 mm	0 mm
Viento	15 km/h	13 km/h	19 km/h	17 km/h	11 km/h	9 km/h	9 km/h
	06:46	06:46	06:47	06:47	06:48	06:48	06:49
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13:06

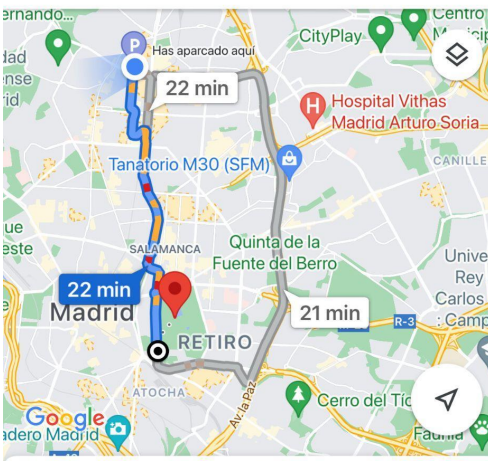
Tu ubicación

Parque de El Retiro

22 min 34 min 1 h 3 22 min

Evitando peajes

Se requiere mascarilla en algunos espacios públicos debido al COVID-19 [Ver más](#)



22 min (6,4 km) Dificil

Actualmente es la mejor ruta debido a las condiciones del tráfico

Pasos Iniciar Fijar

¿Productos basados en datos?



¿Productos basados en datos?



zebra

PROJECTS DATA SEARCH

Docs Hello Tom Goldberg

Reset search SEARCH

http://searchapp.zebra-med.com/?s

186,221 studies

CREATE DATASET

STUDY PARAMETERS

Modality
CT - Computed Tomography,

Body part
Chest,

Male (94,251) Female (94,226)

Age 35 - 110

FINDINGS

- ☐ Abdominal Aortic Aneurysm 7,127
- ☐ Aortic Stentgraft 6,422
- ☐ Appendicitis 30
- ☐ Benign appearing Nodules 565

My Tab

Volume Referencing Flip Selection

Layout Studies Series Sort Tile Mode

ROBIN BATES
F 923472412
Pos: 195.00 mm
SI: 1
Acq#: ZEBRA
Patient Pos: HFS
Study Desc: PET-CT SCAN
Series Desc: Plain Chest 2.5mm Inspiration
← 2 - 1 →

Current
MEDICAL SYSTEMS Discovery STE
38% Pixel
DFOV 146.6 x 60.1 cm

Viewer
10 cm

C 50
W 500

Link 1

Current
MEDICAL SYSTEMS Discovery STE
120kV, 50mAs
SC 500.00 mm

Current
GE MEDICAL SYSTEMS Discovery STE
120kV, 30mAs
SC 500.00 mm

Lung Nodules
No lung nodule has been detected.

Not Detected Detected

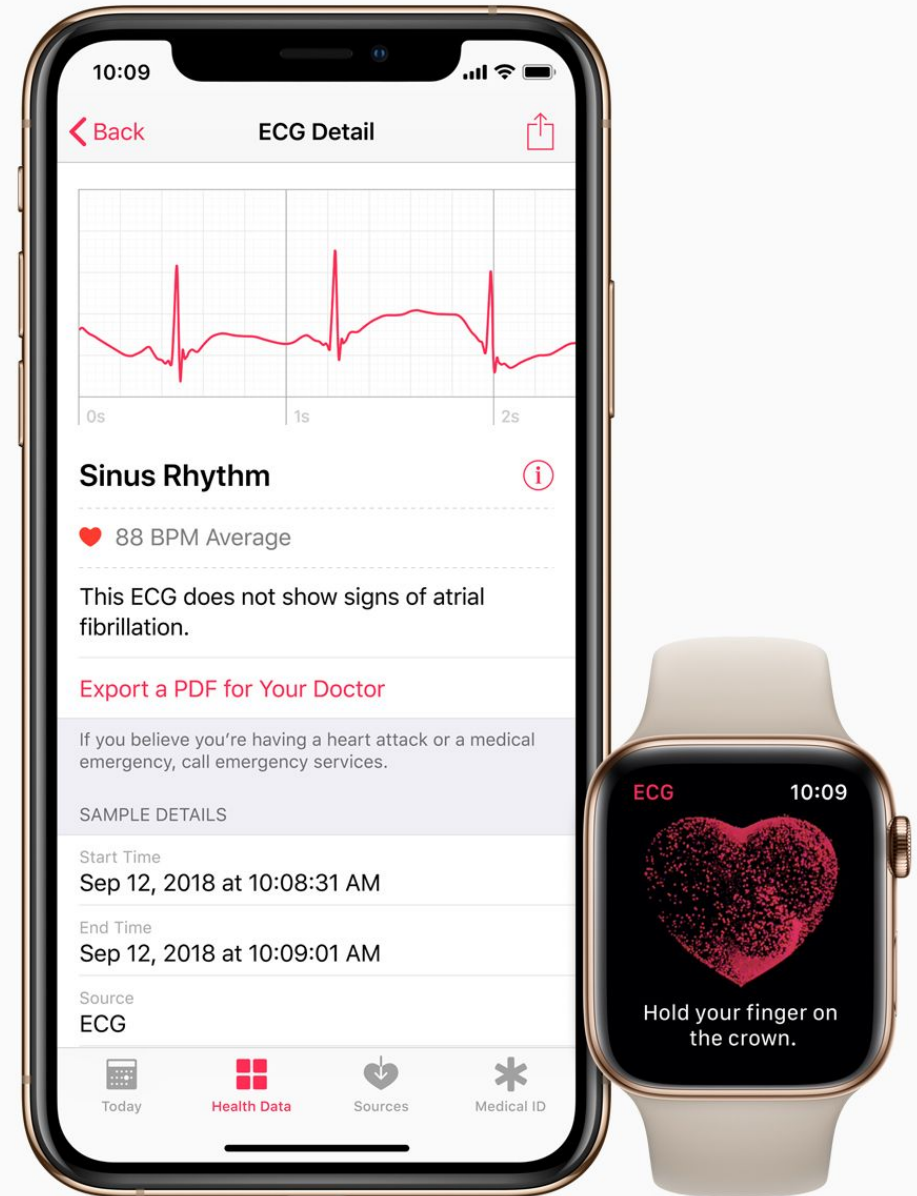
Copy To Clipboard

¿Productos basados en datos?

kaia



¿Productos basados en datos?

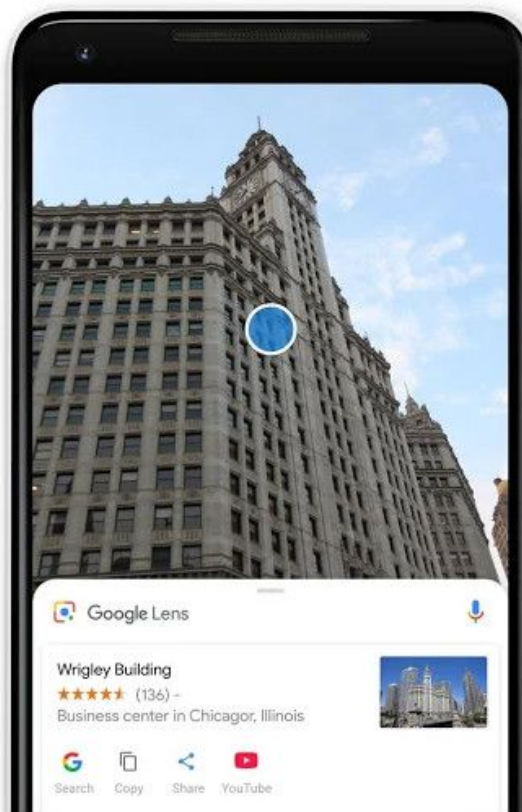


¿Productos basados en datos?

Take action on text



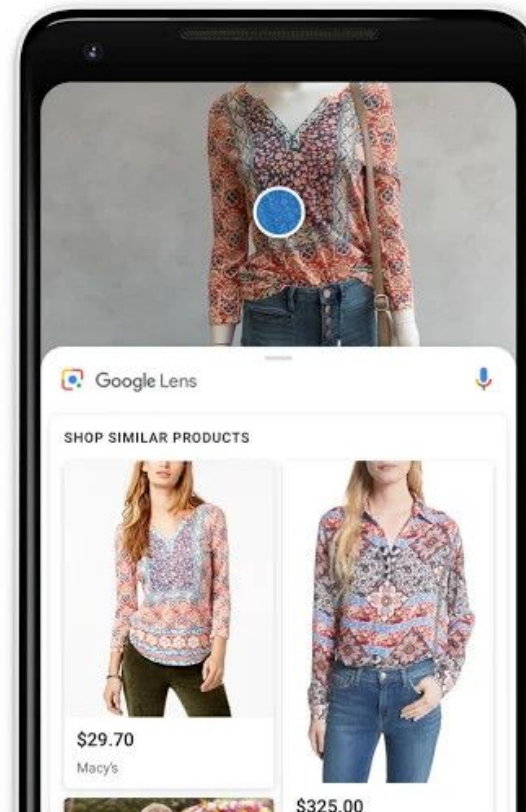
Learn more about the world



Identify plants and animals



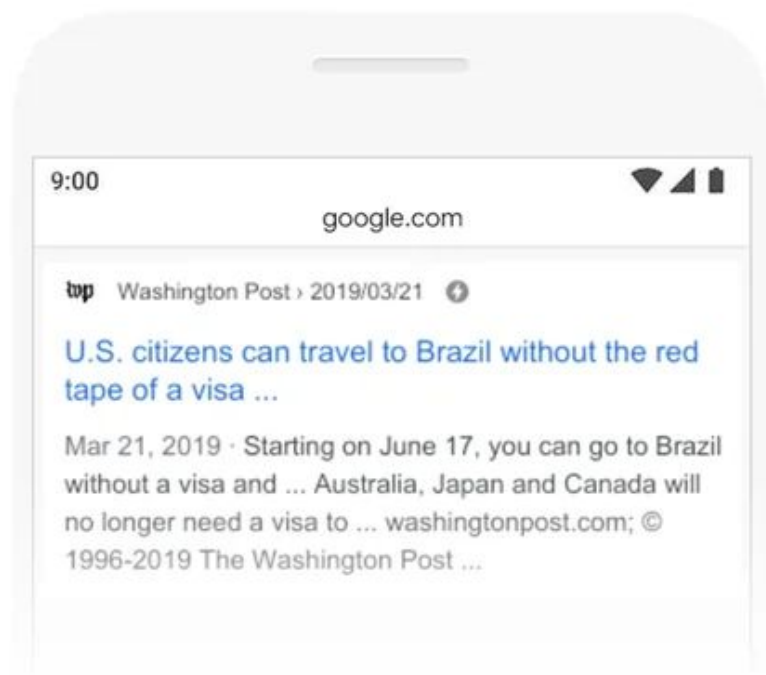
Find a look you like



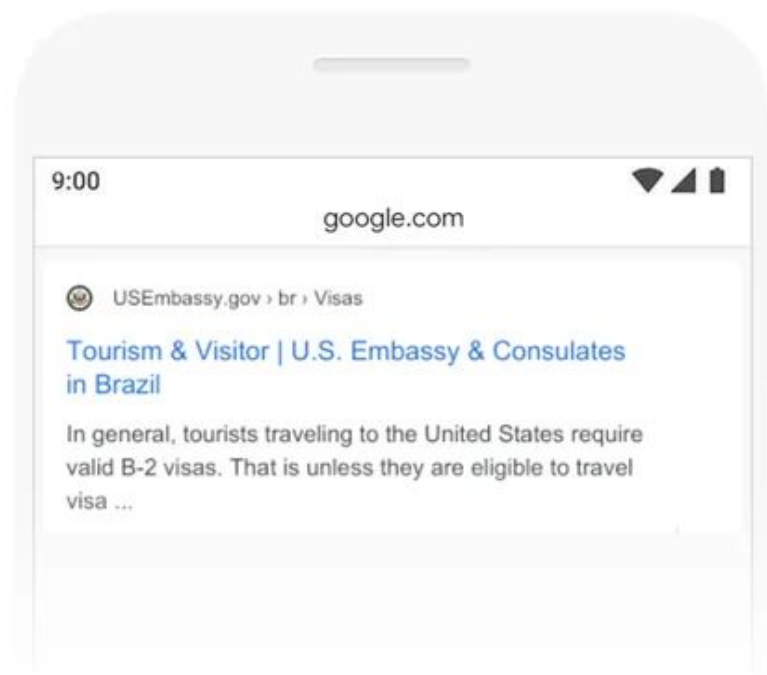
¿Productos basados en datos?

🔍 2019 brazil traveler to usa need a visa

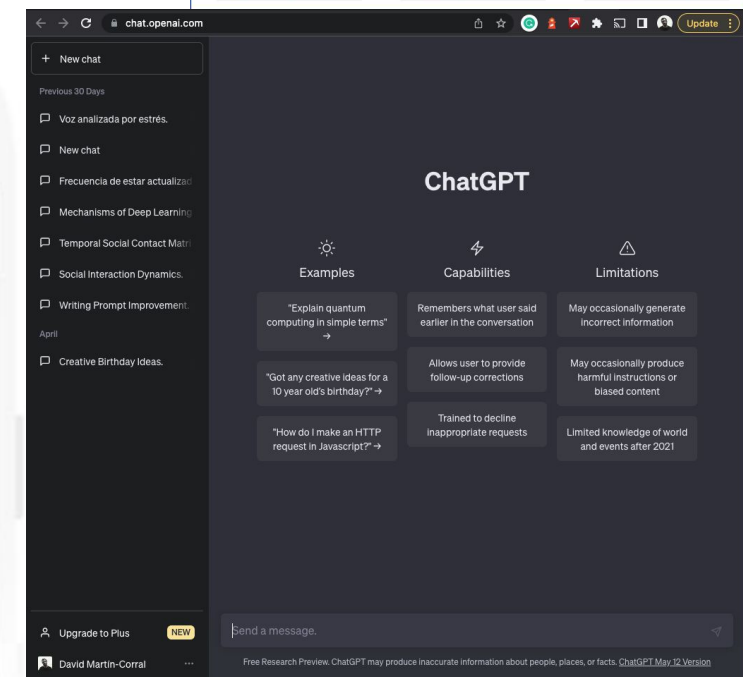
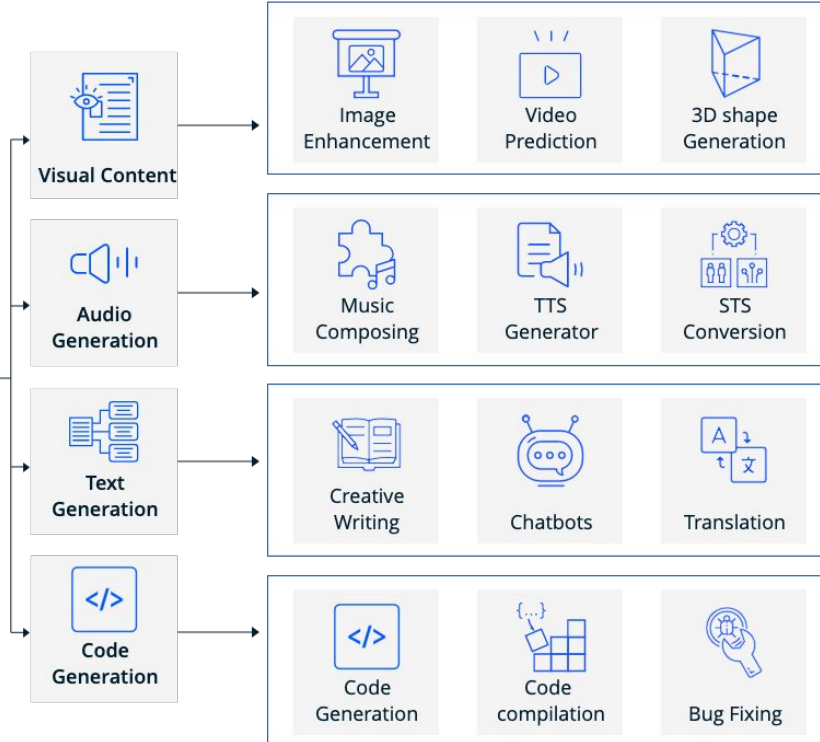
BEFORE



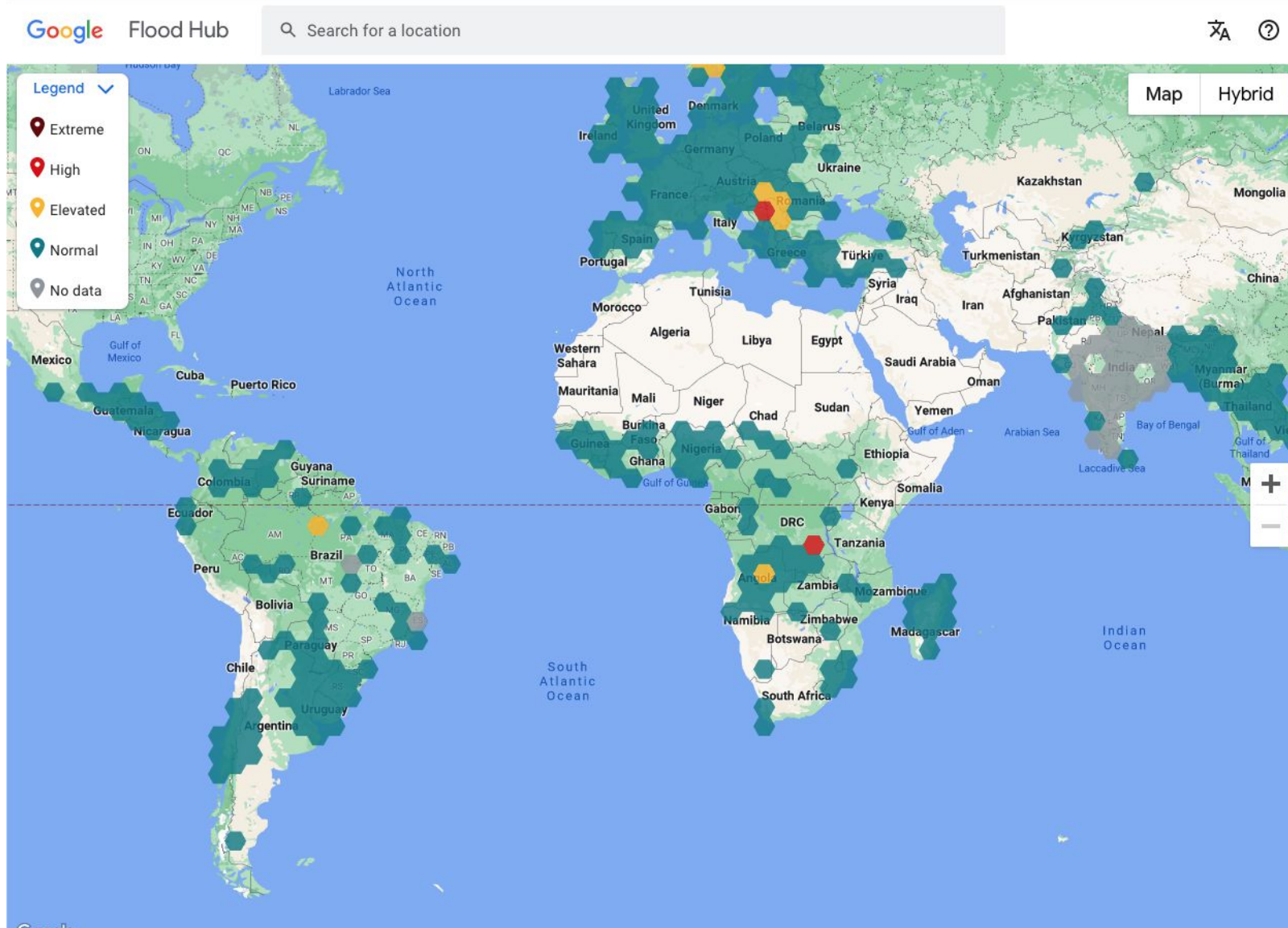
AFTER



Generative AI use cases



¿Productos basados en datos?



Modelos de negocio

Posibles modelos para monetizar los datos en el entorno actual.

- ▶ **Usuarios de datos:** Empresas que utilizan datos para elaborar estrategias y crear mejores productos.
- ▶ **Proveedores de datos:** Empresas que se dedican principalmente al comercio de datos.
- ▶ **Redes de distribución:** Empresas con un modelo de negocio publicitario.
- ▶ **Facilitadores de infraestructura:** Empresas que participan en el suministro de infraestructura de datos, análisis y consultoría.